US 20250280821A1

(19) **United States**

(12) **Patent Application Publication**

(10) **Pub. No.: US 2025/0280821 A1**

(43) **Pub. Date: Sep. 11, 2025**

Vrselja

(54) **PERFUSION AND METABOLIC RESTORATION OF ORGANS**

(71) Applicant: **Bexorg, Inc.**, New Haven, CT (US)

(72) Inventor: **Zvonimir Vrselja**, New Haven, CT (US)

(21) Appl. No.: **19/070,309**

(22) Filed: **Mar. 4, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/562,094, filed on Mar. 6, 2024.

Publication Classification

(51) **Int. Cl.**

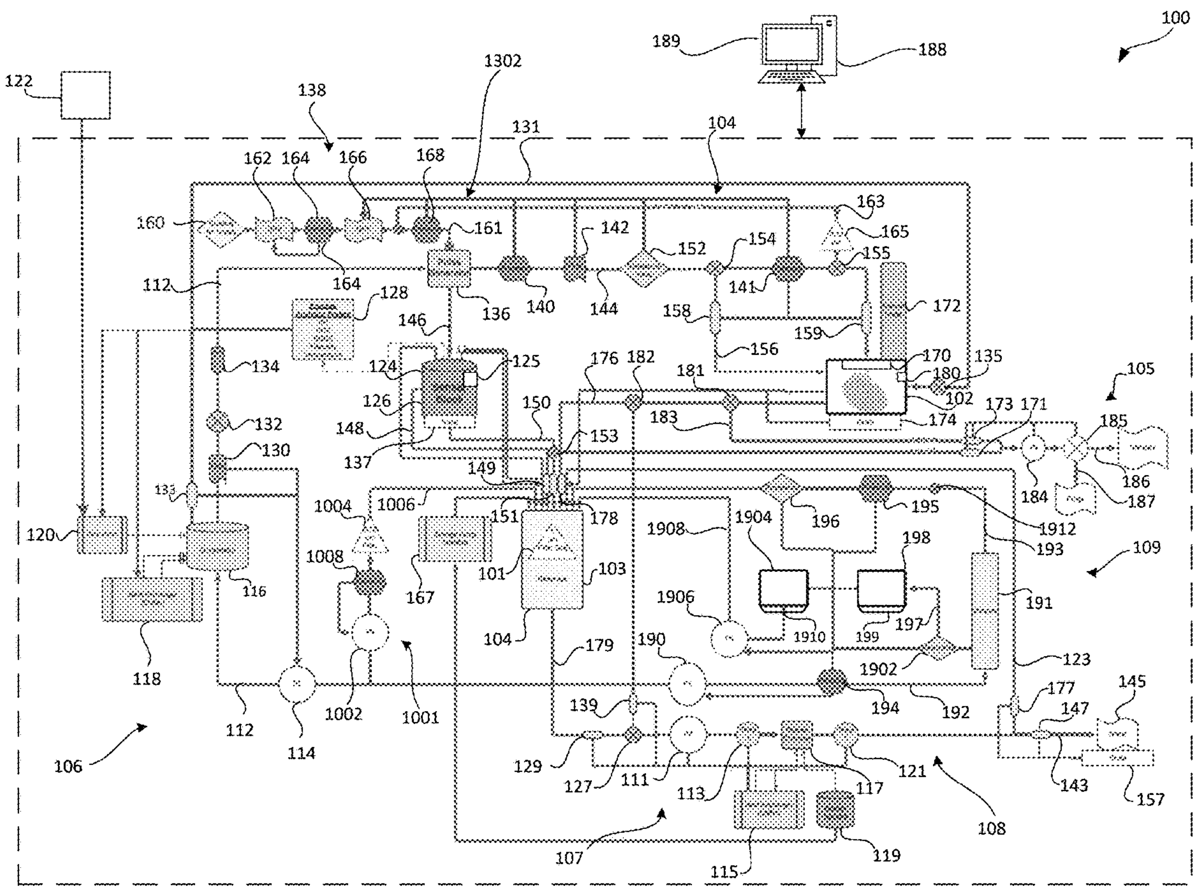
A01N 1/143	(2025.01)
G01N 21/65	(2006.01)
G01N 33/50	(2006.01)

(52) **U.S. Cl.**

CPC ***A01N 1/143*** (2025.01); ***G01N 21/65*** (2013.01); ***G01N 33/5082*** (2013.01)

(57) **ABSTRACT**

An ex-vivo brain perfusion system for perfusion of a brain includes a reservoir configured to receive perfusate, a housing configured to contain the brain, an arterial circuit configured to fluidly couple the reservoir to the brain, a venous circuit fluidly coupling the housing to the reservoir, and a syringe pump fluidly coupled to the reservoir and configured to introduce a therapeutic agent to the perfusate in the reservoir. The ex-vivo brain perfusion system is configured to perfuse the brain with the therapeutic agent to test at least one property of the therapeutic agent.



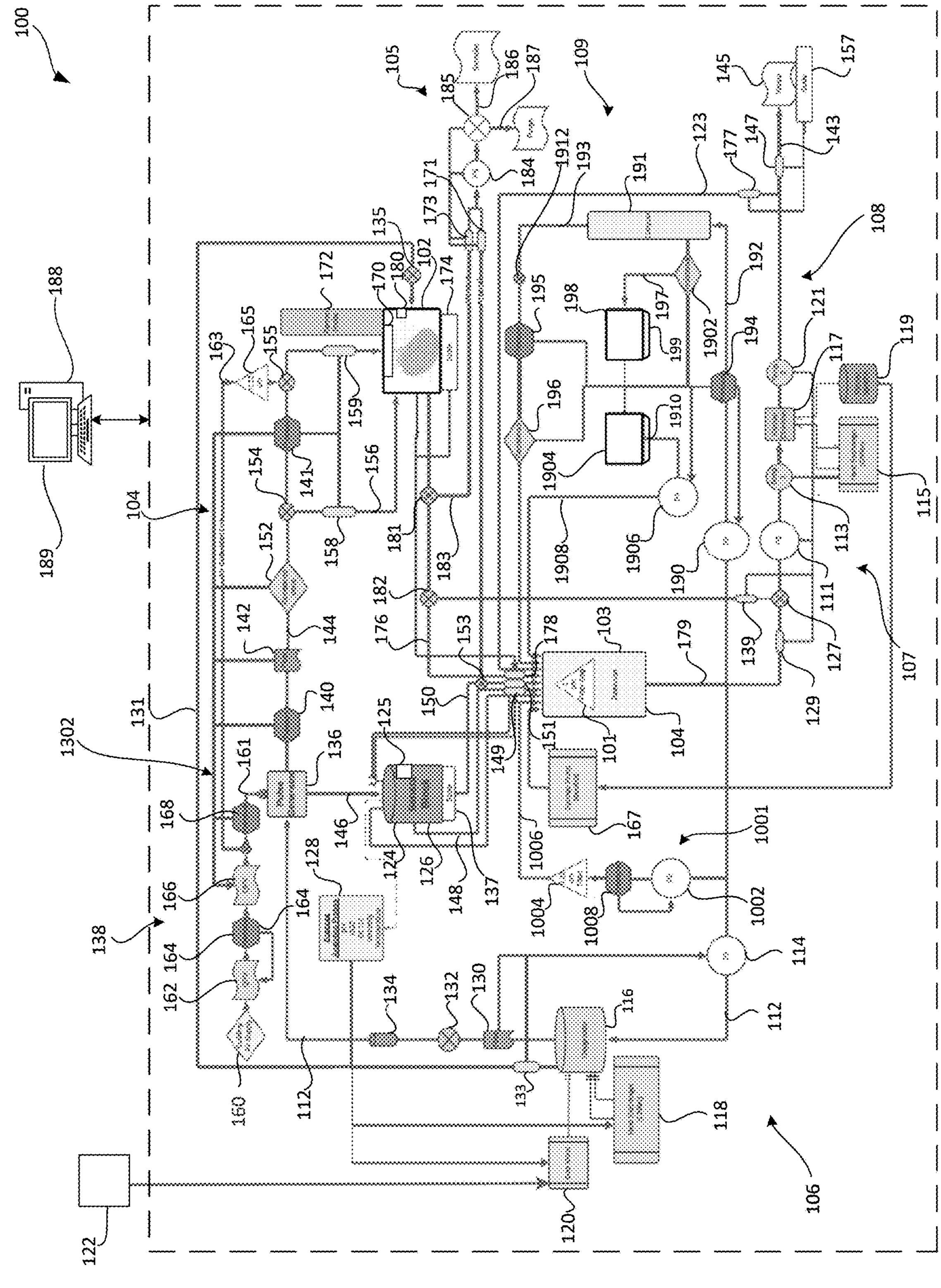


FIG. 1

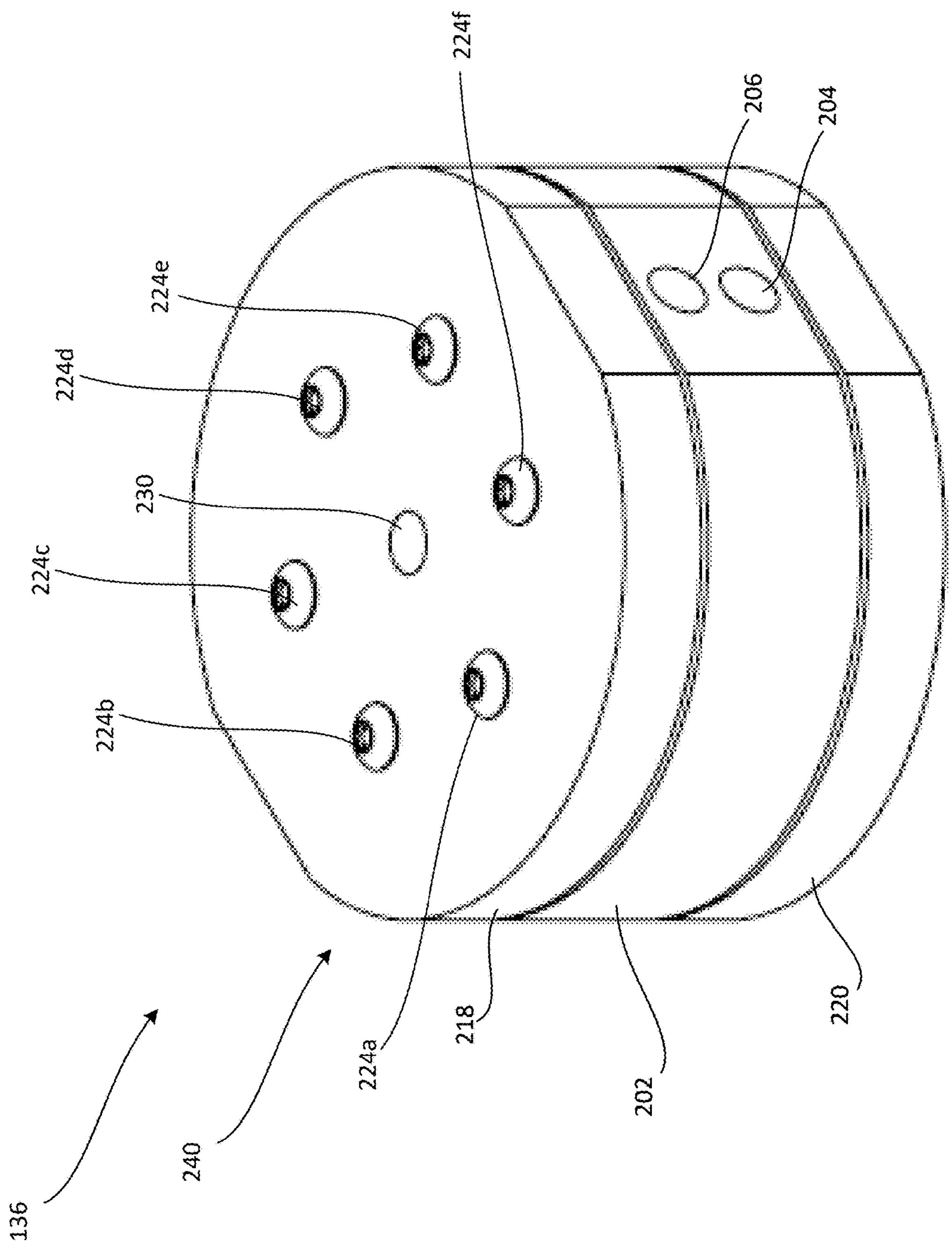


FIG. 2A

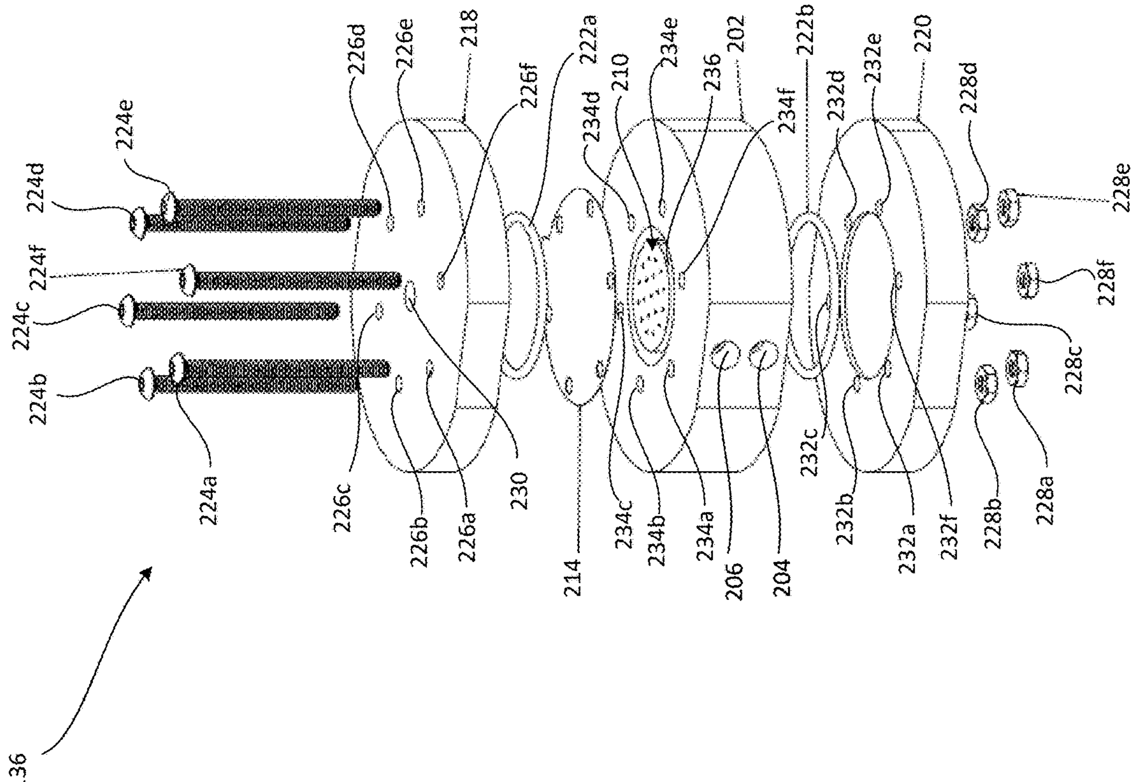

136
224a
224b
224c
224d
224e
224f
226a
226b
226c
226d
226e
226f
218
230
222a
214
234a
234b
234c
234d
234e
234f
210
236
202
206
204
222b
232a
232b
232c
232d
232e
232f
220
228a
228b
228c
228d
228e
228f

FIG. 2B

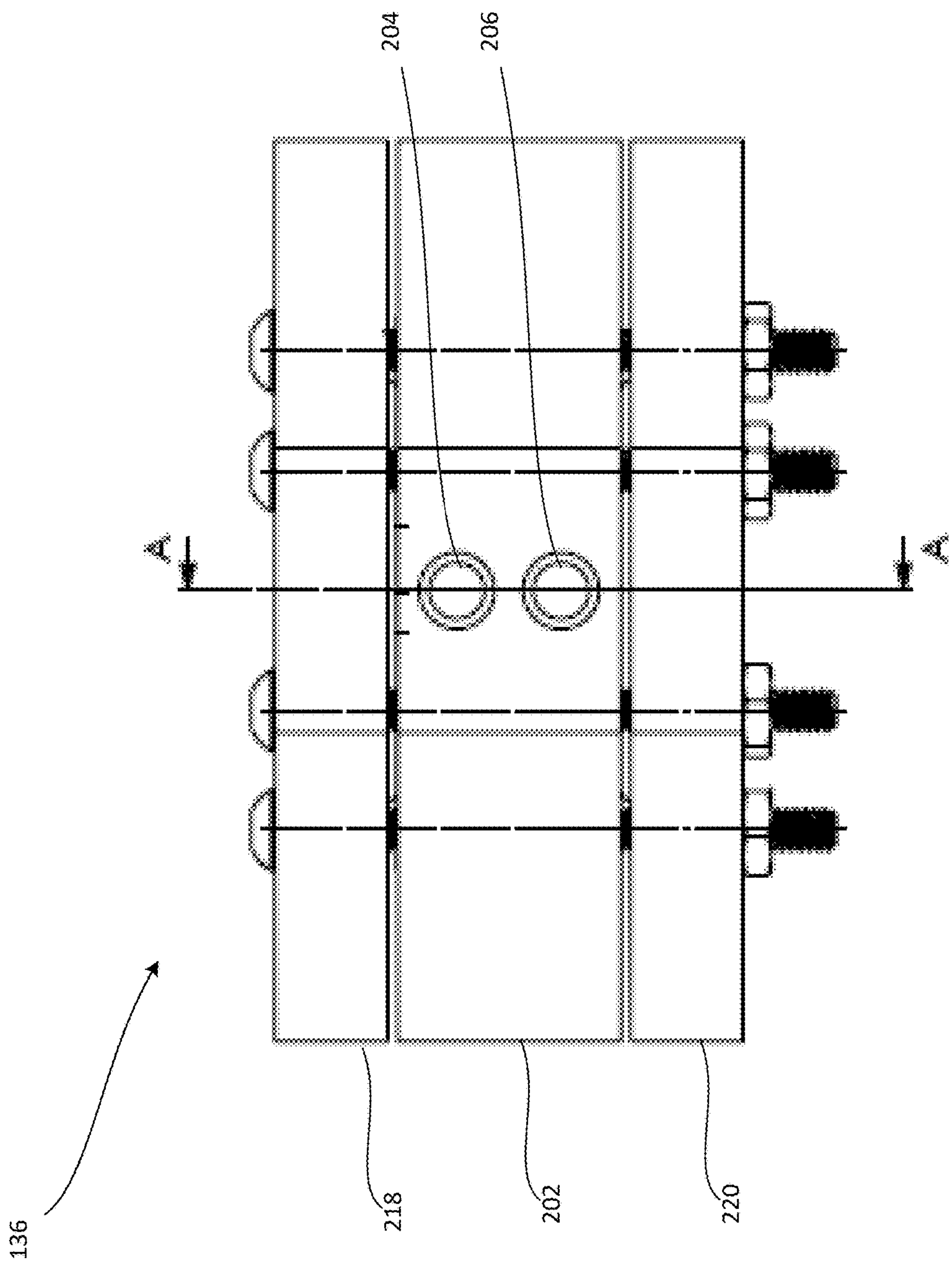


FIG. 2C

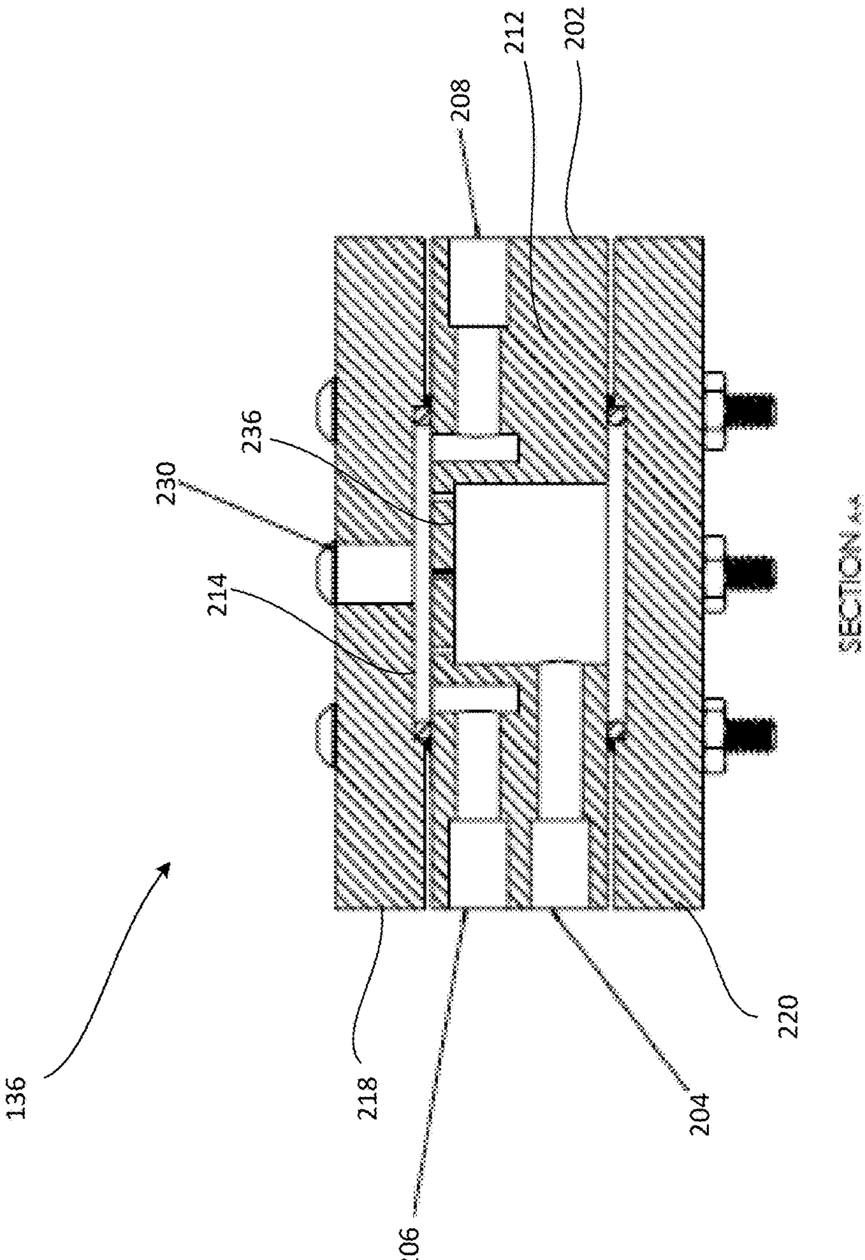


FIG. 2D

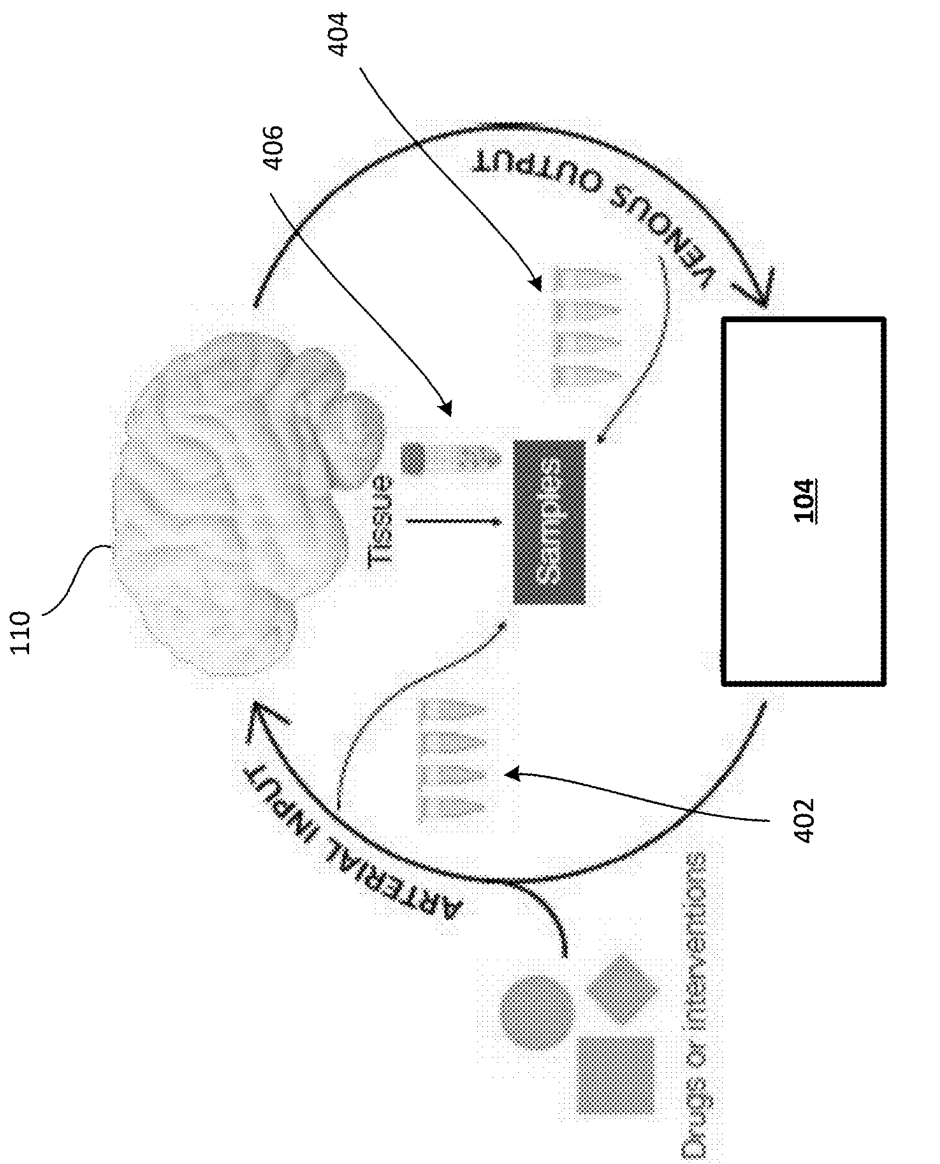

110
406
404
VENOUS OUTPUT
Tissue
Samples
104
402
ARTERIAL INPUT
Drugs or interventions

FIG. 3

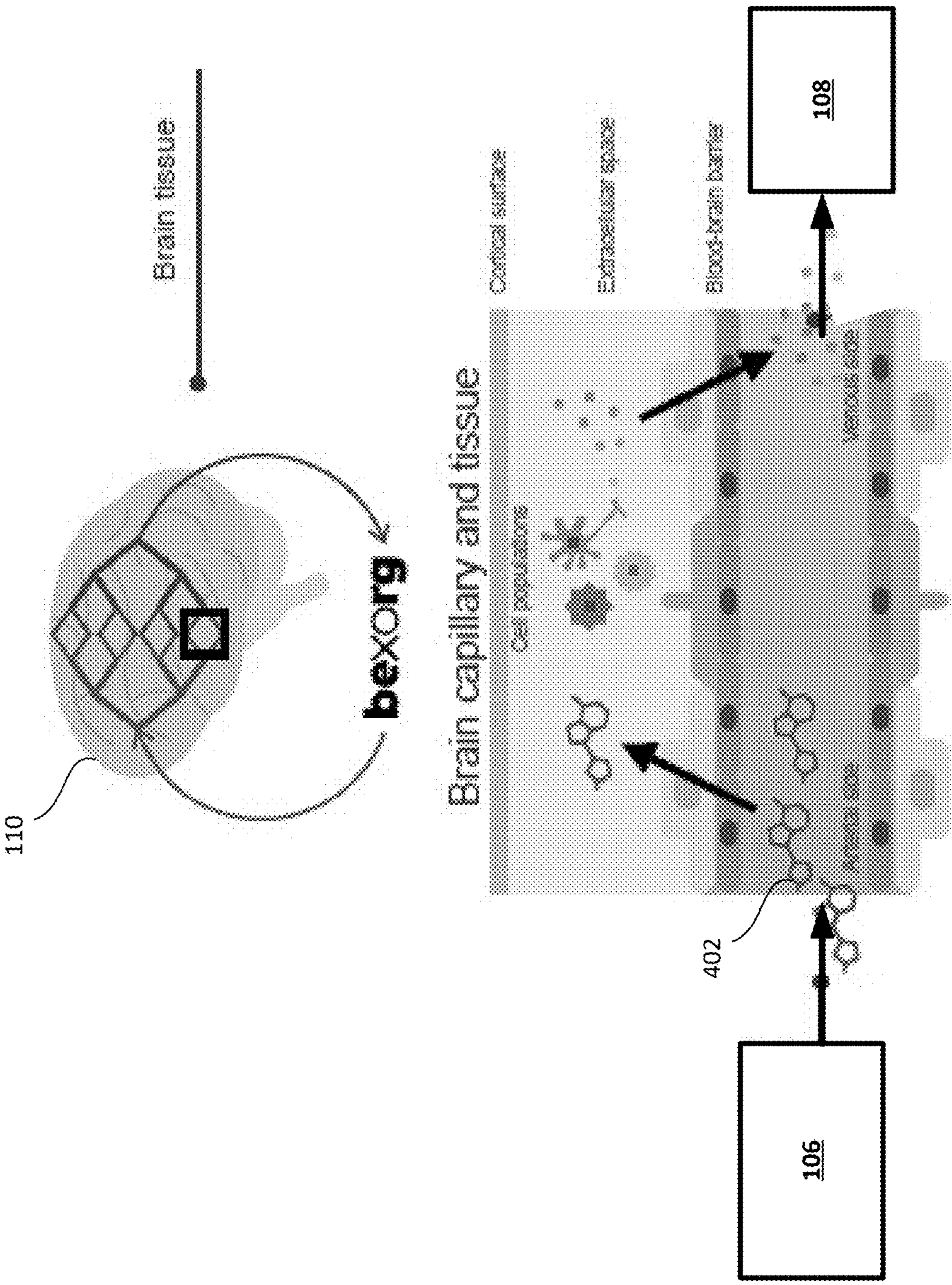

110
bexorg
Brain tissue
Brain capillary and tissue
Cortical surface
Extracellular space
Blood-brain barrier
Cell populations
402
106
108

FIG. 4

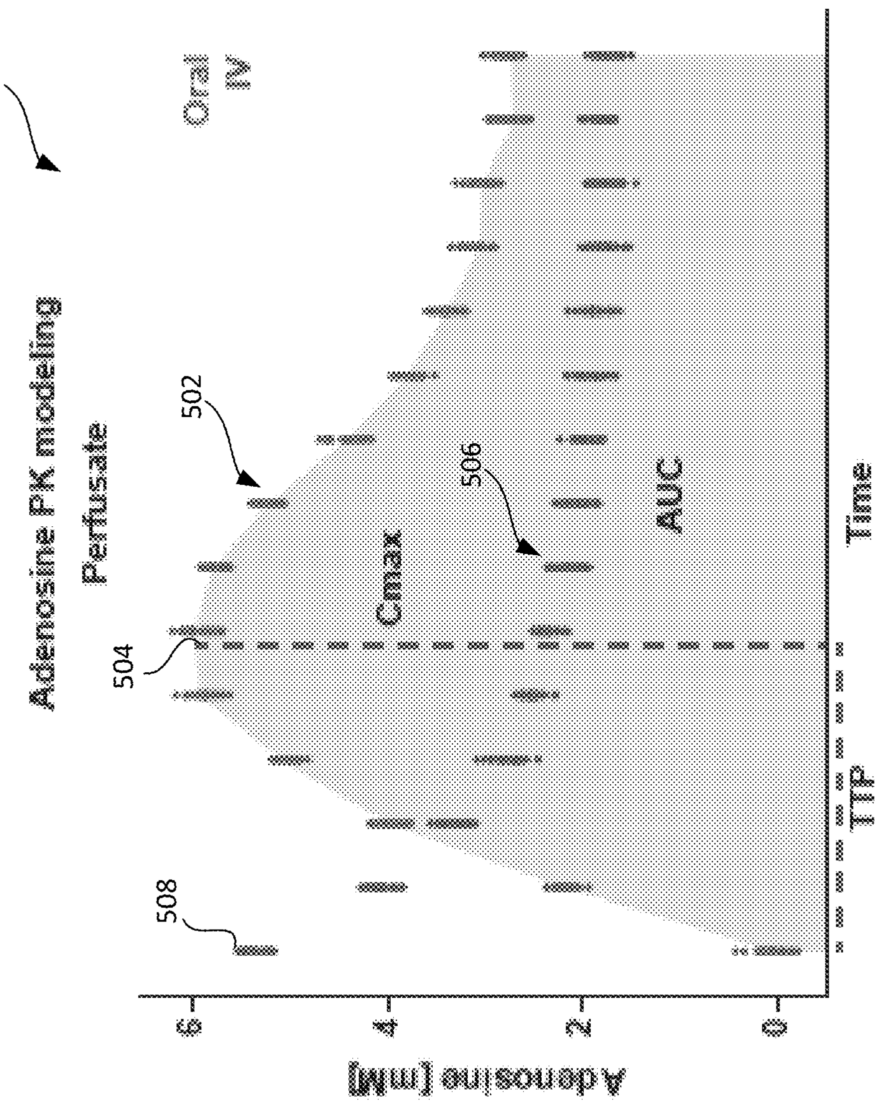

500
Adenosine PK modeling
Perfusate
Oral
IV
502
504
506
508
Cmax
AUC
TTP
Time
Adenosine [mM]
0
2
4
6

FIG. 5

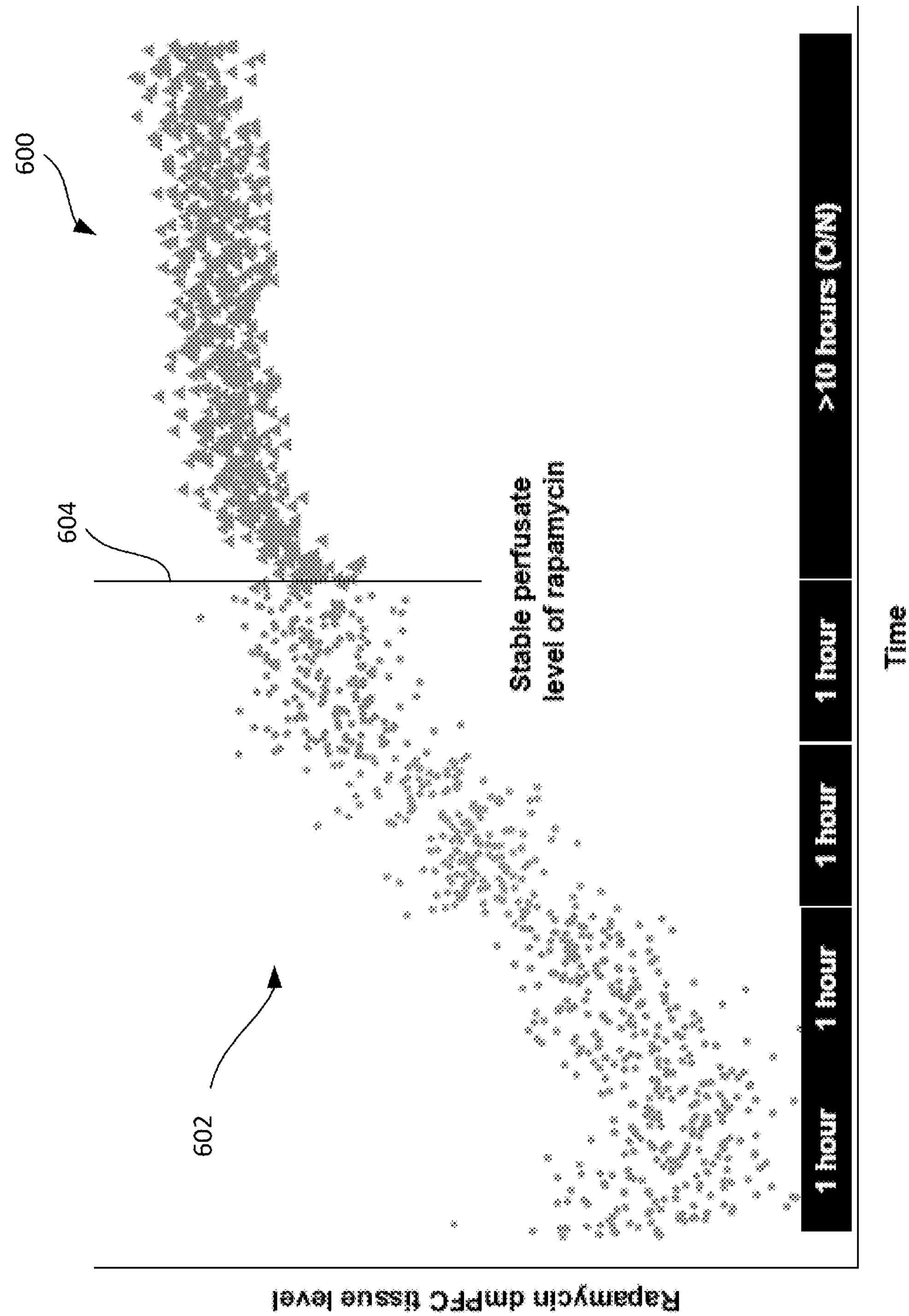
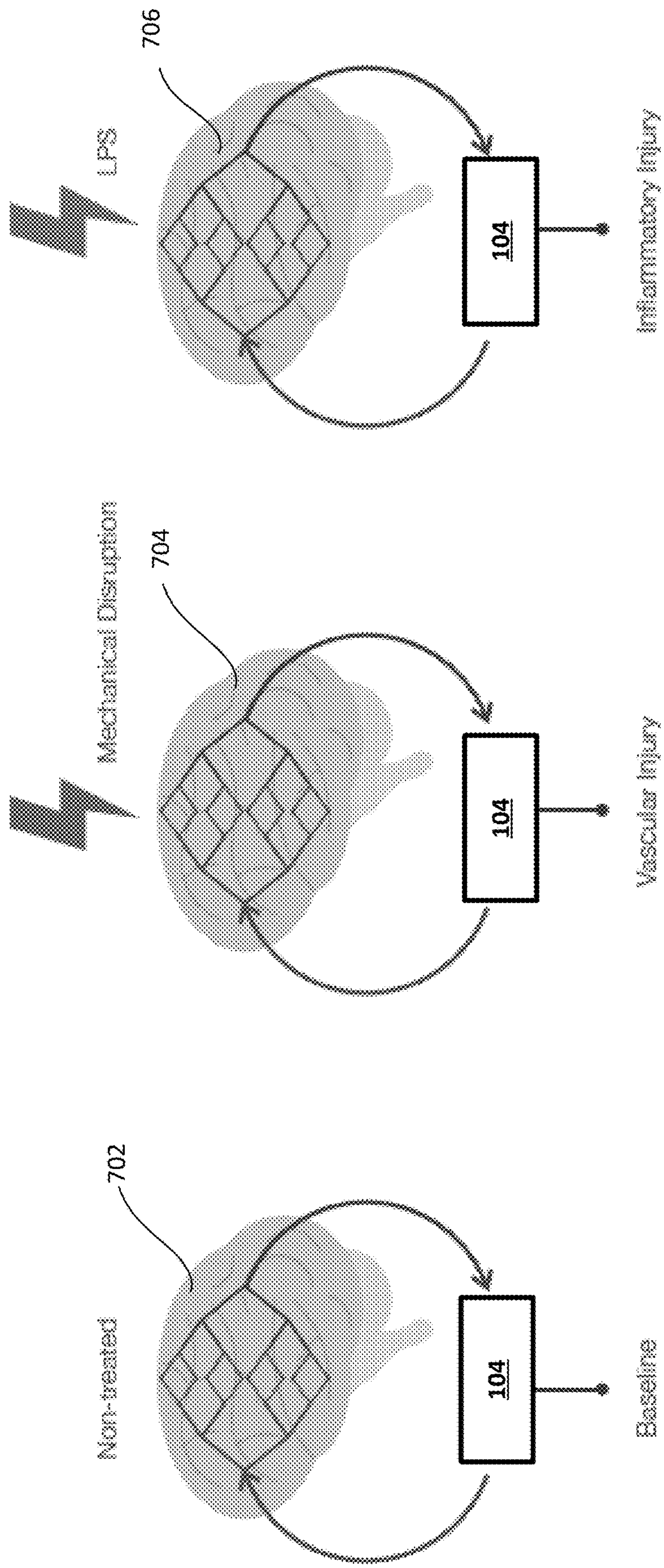
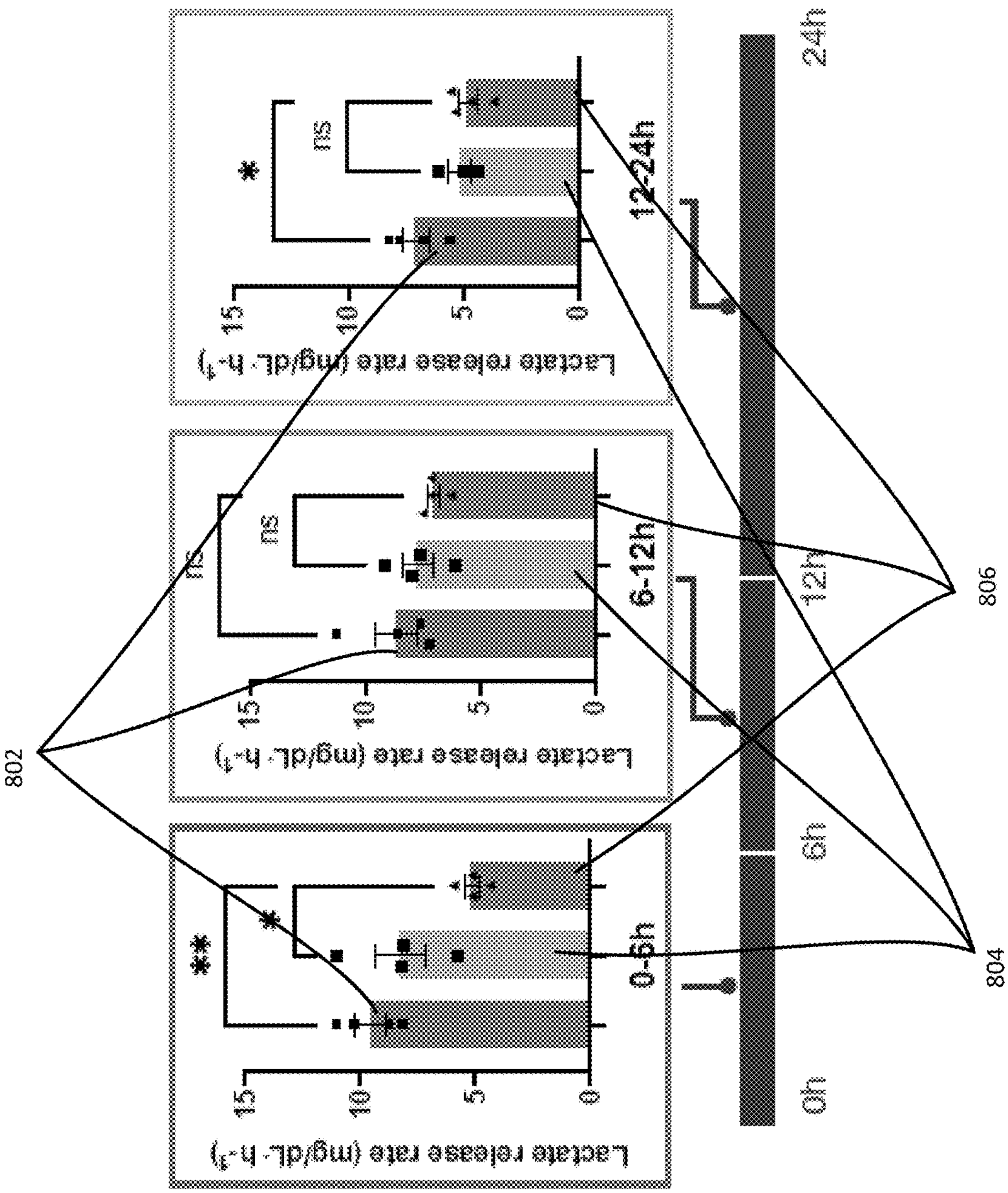

600
604
602
Stable perfusate
level of rapamycin
1 hour
1 hour
1 hour
1 hour
>10 hours (O/N)
Time
Rapamycin dmPFC tissue level

FIG. 6



Non-treated
702
104
Baseline
Mechanical Disruption
704
104
Vascular Injury
LPS
706
104
Inflammatory Injury

FIG. 7



Lactate release rate (mg/dL· h-1)
0-6h
6-12h
12-24h
0h
6h
12h
24h
ns
802
804
806

FIG. 8

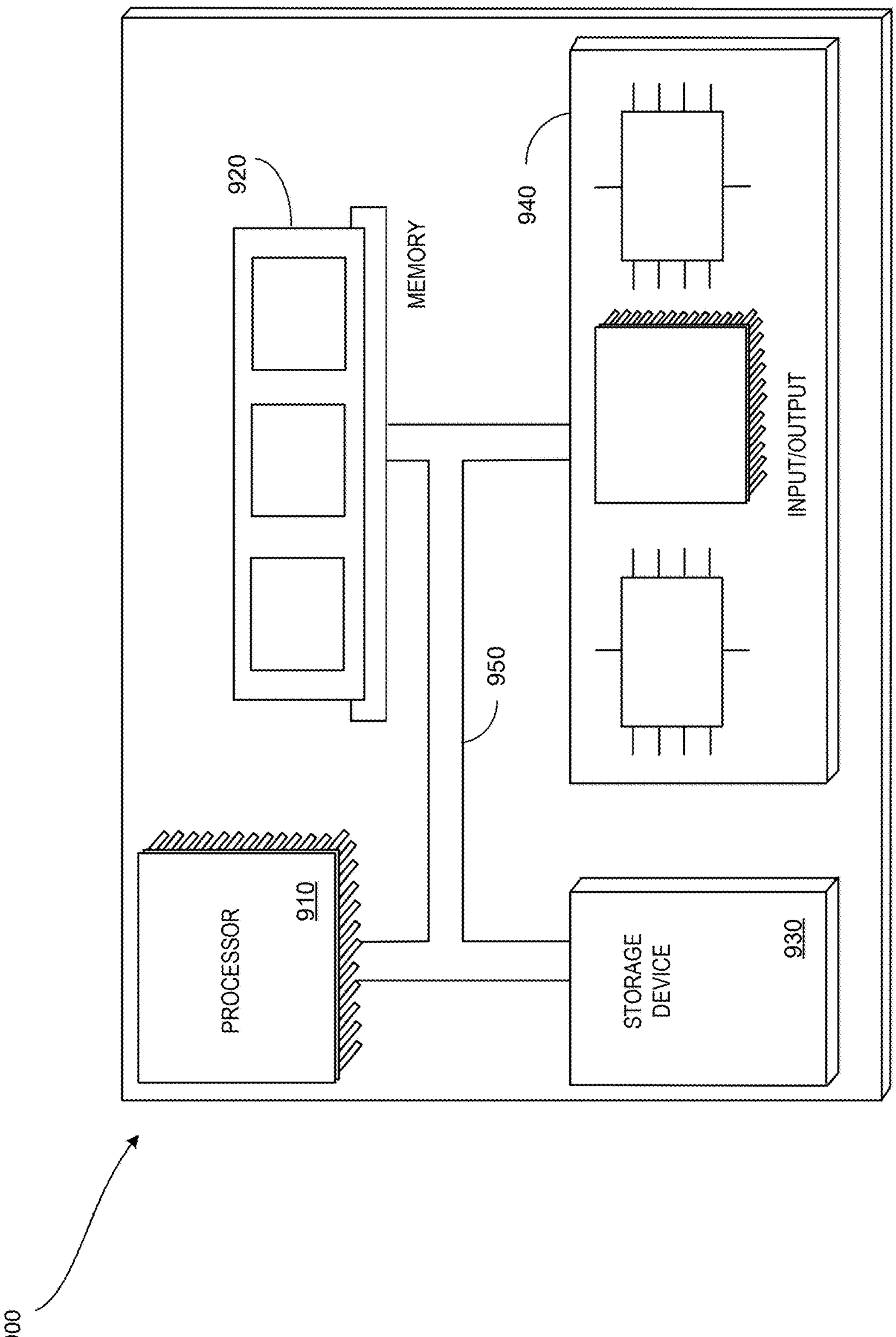

900
920
MEMORY
940
INPUT/OUTPUT
950
PROCESSOR
910
STORAGE
DEVICE
930

FIG. 9

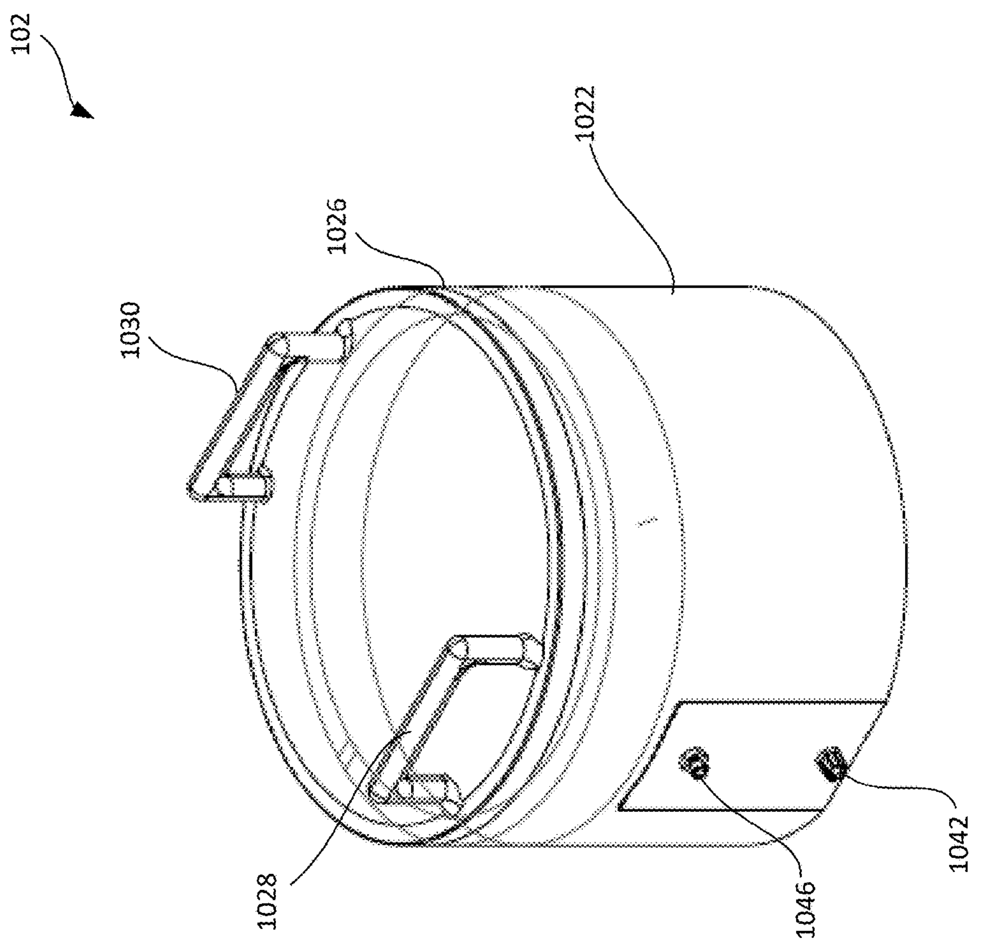


FIG. 10A

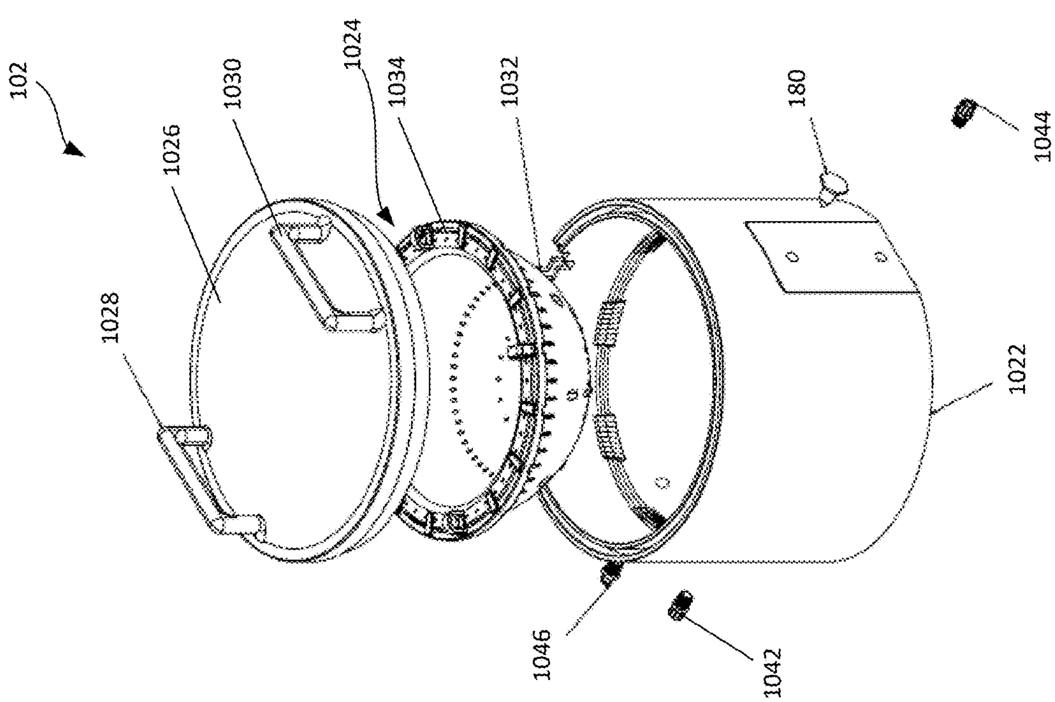


FIG. 10B

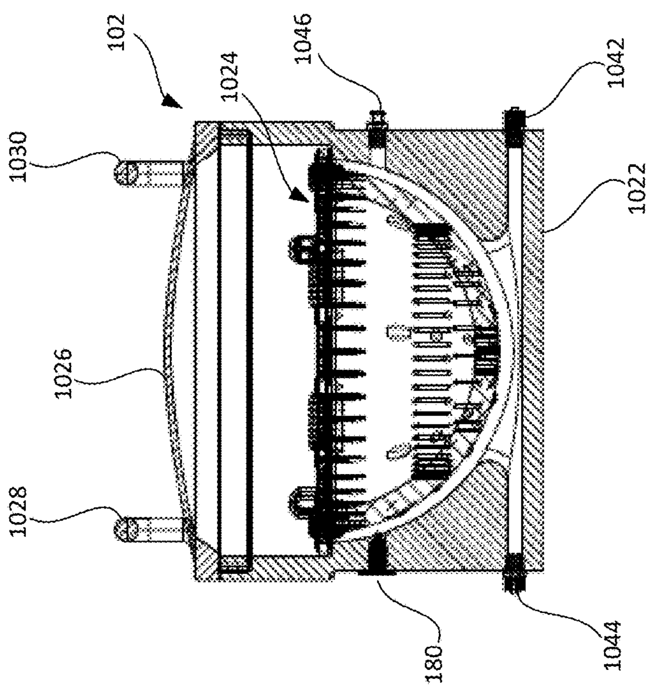


FIG. 10C

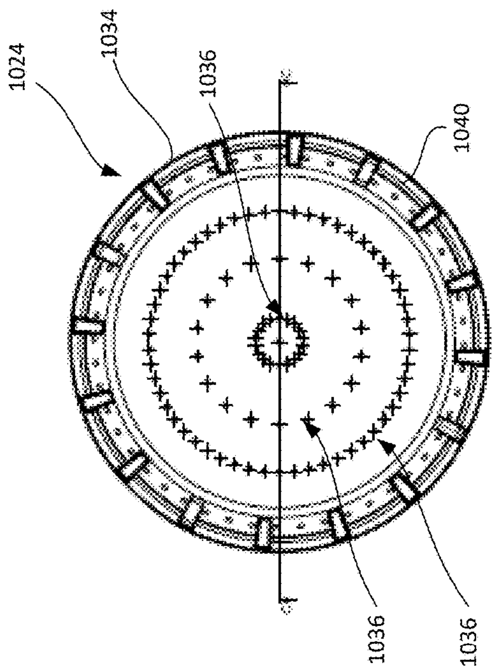


FIG. 10E

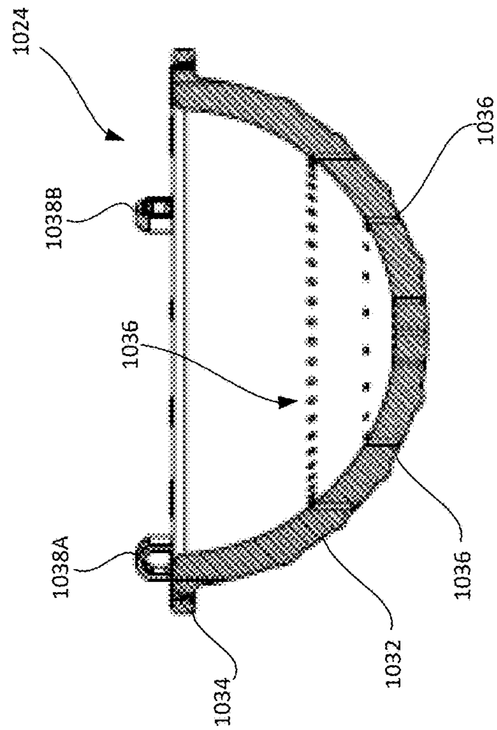


FIG. 10F

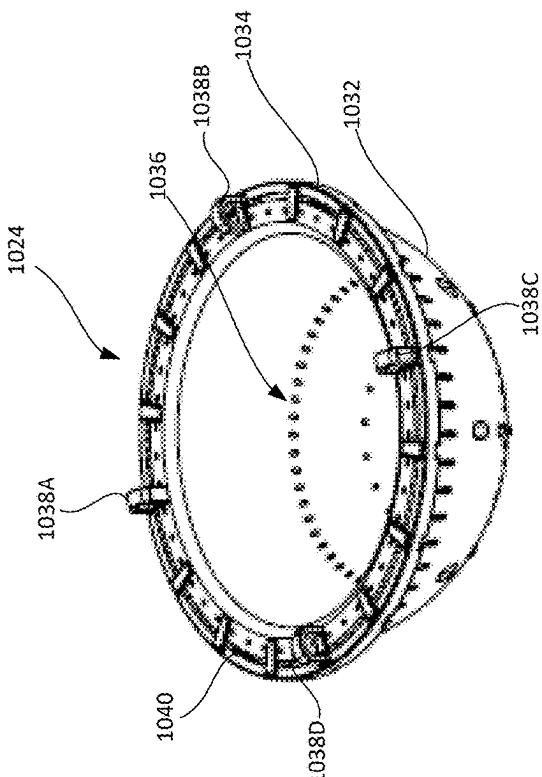


FIG. 10D

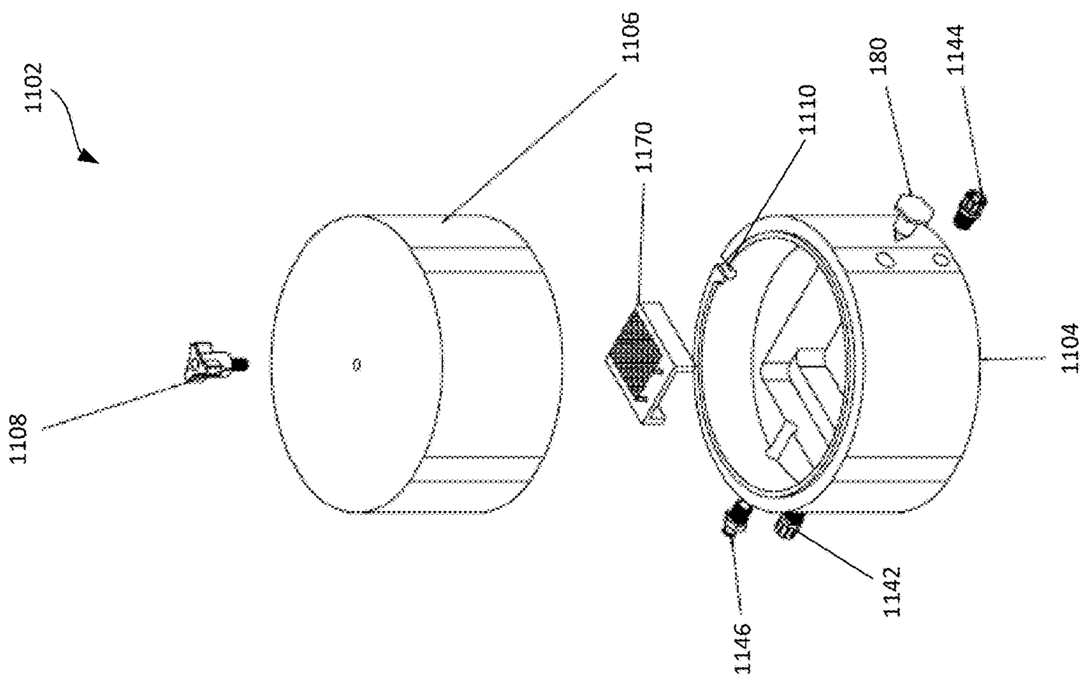


FIG. 11B

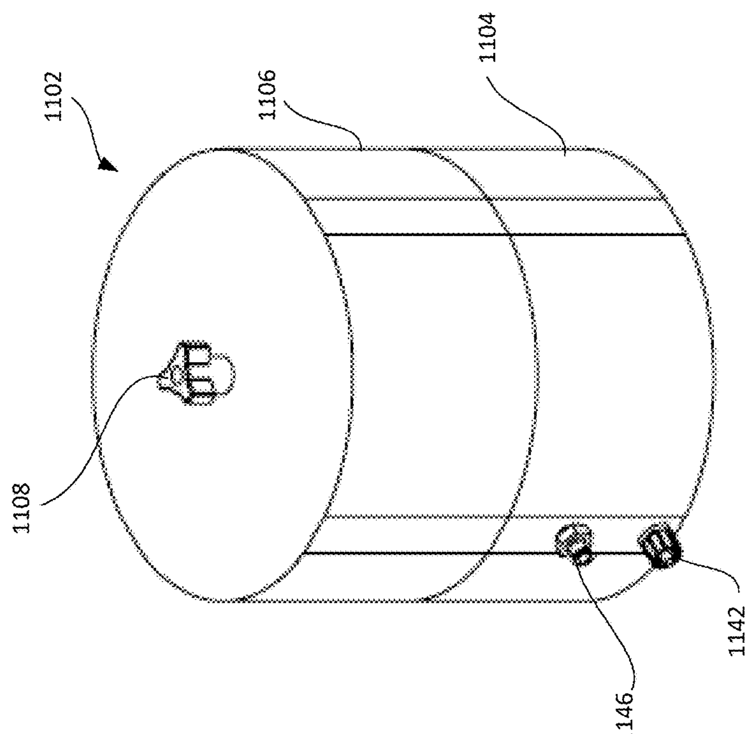


FIG. 11A

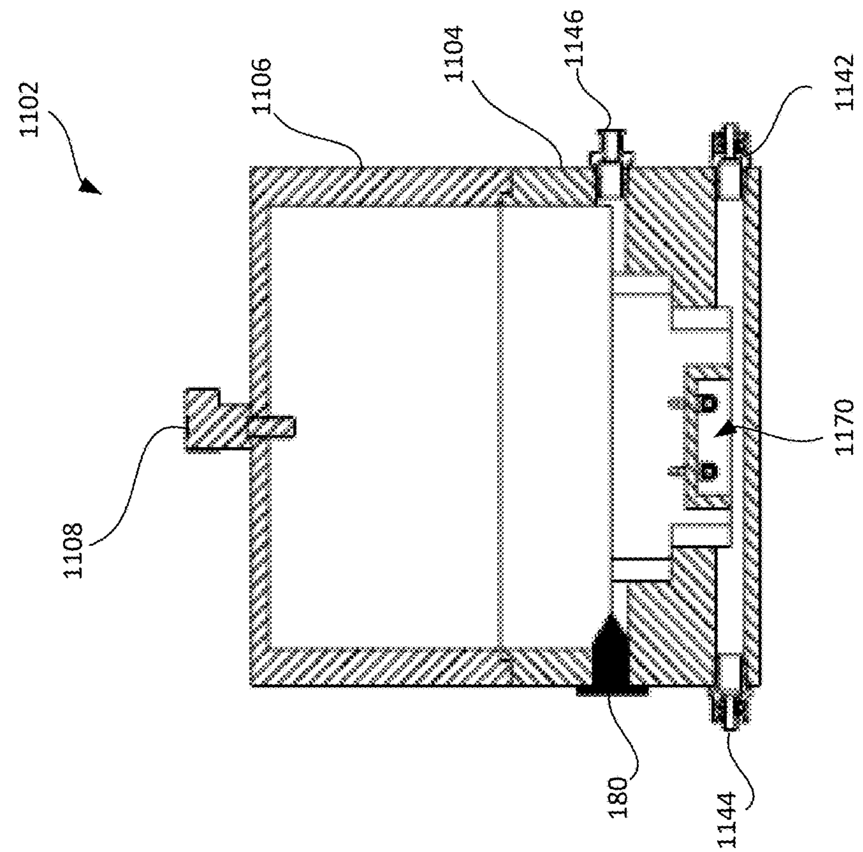


FIG. 11D

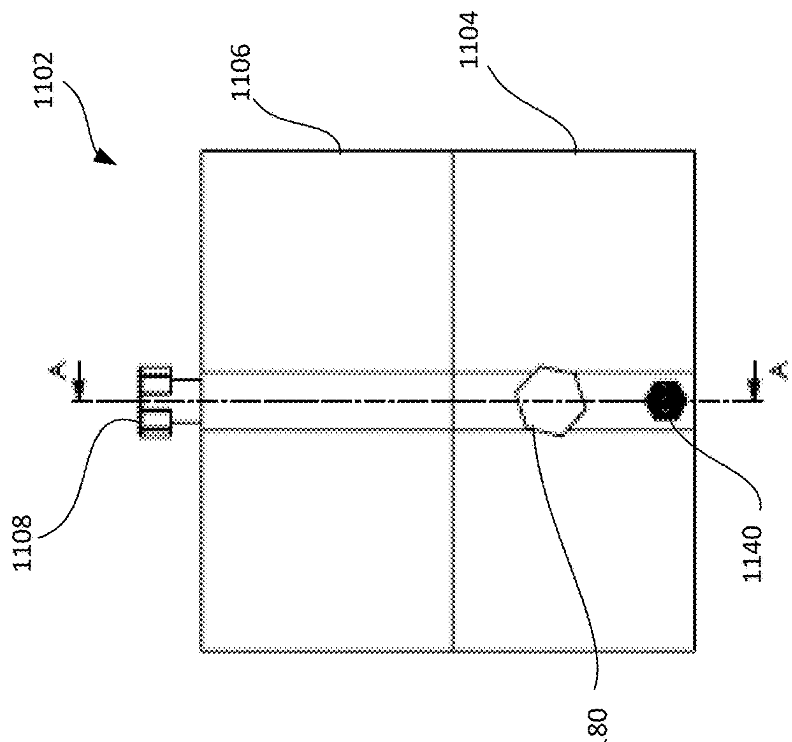


FIG. 11C

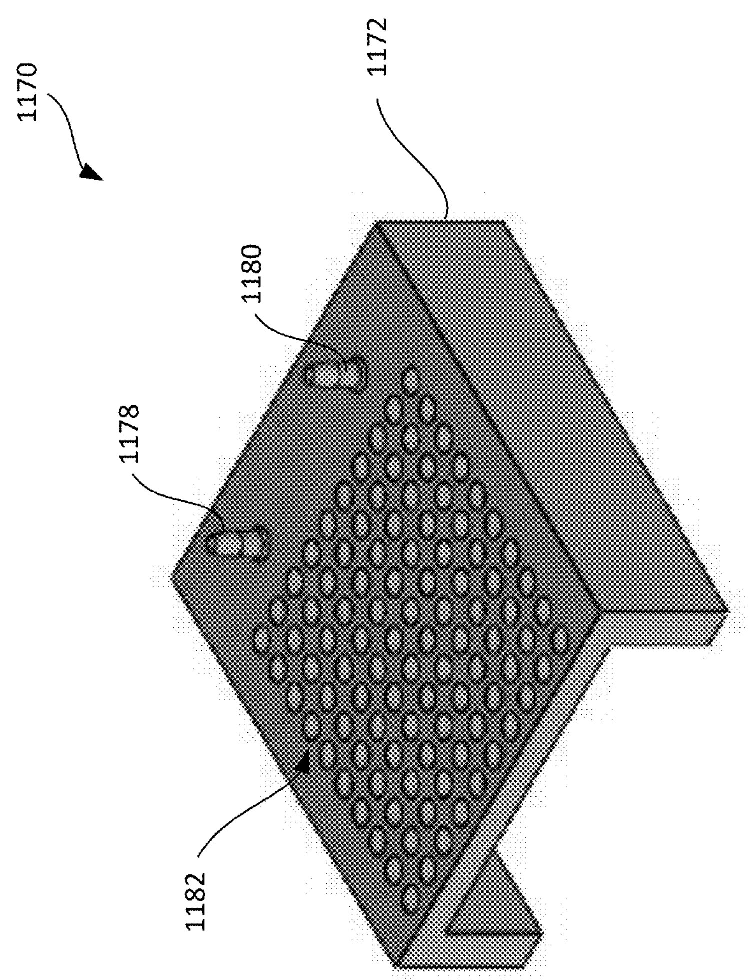


FIG. 12A

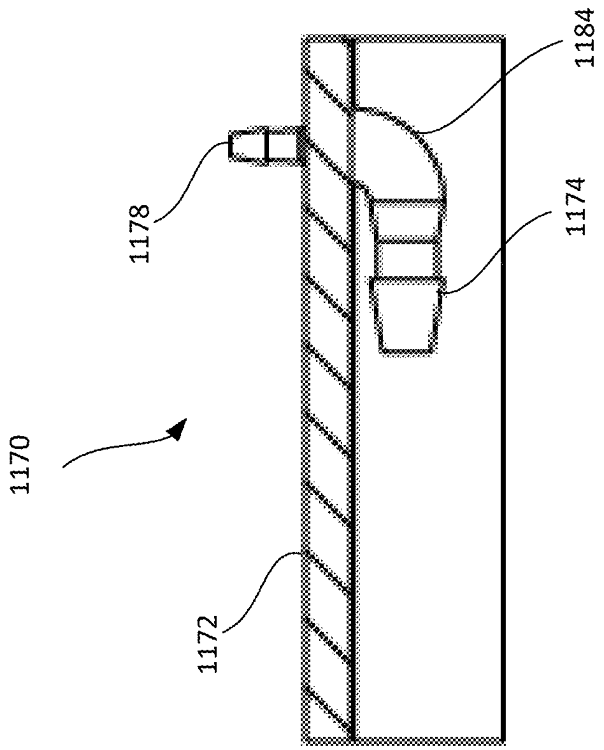


FIG. 12C

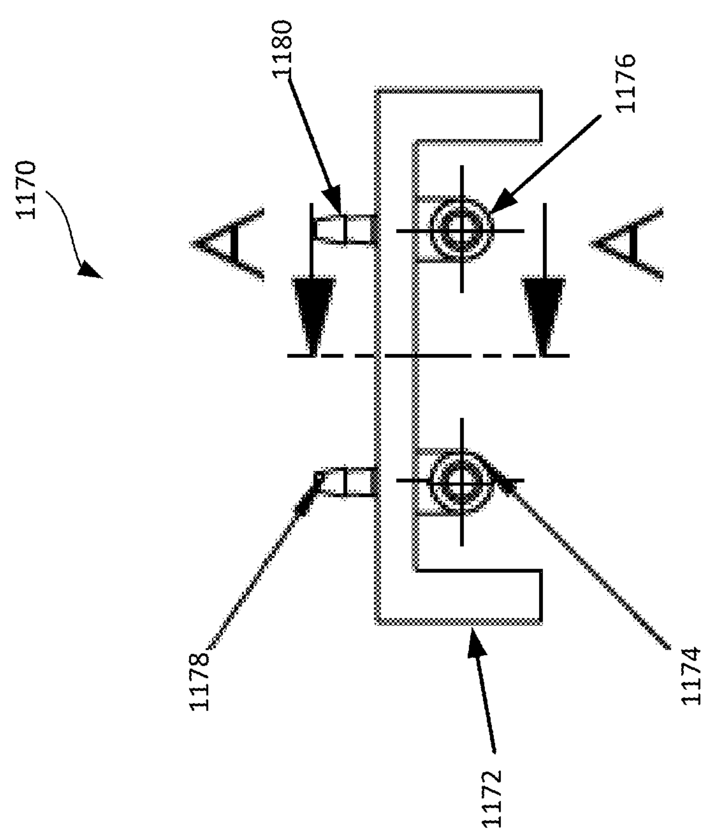


FIG. 12B

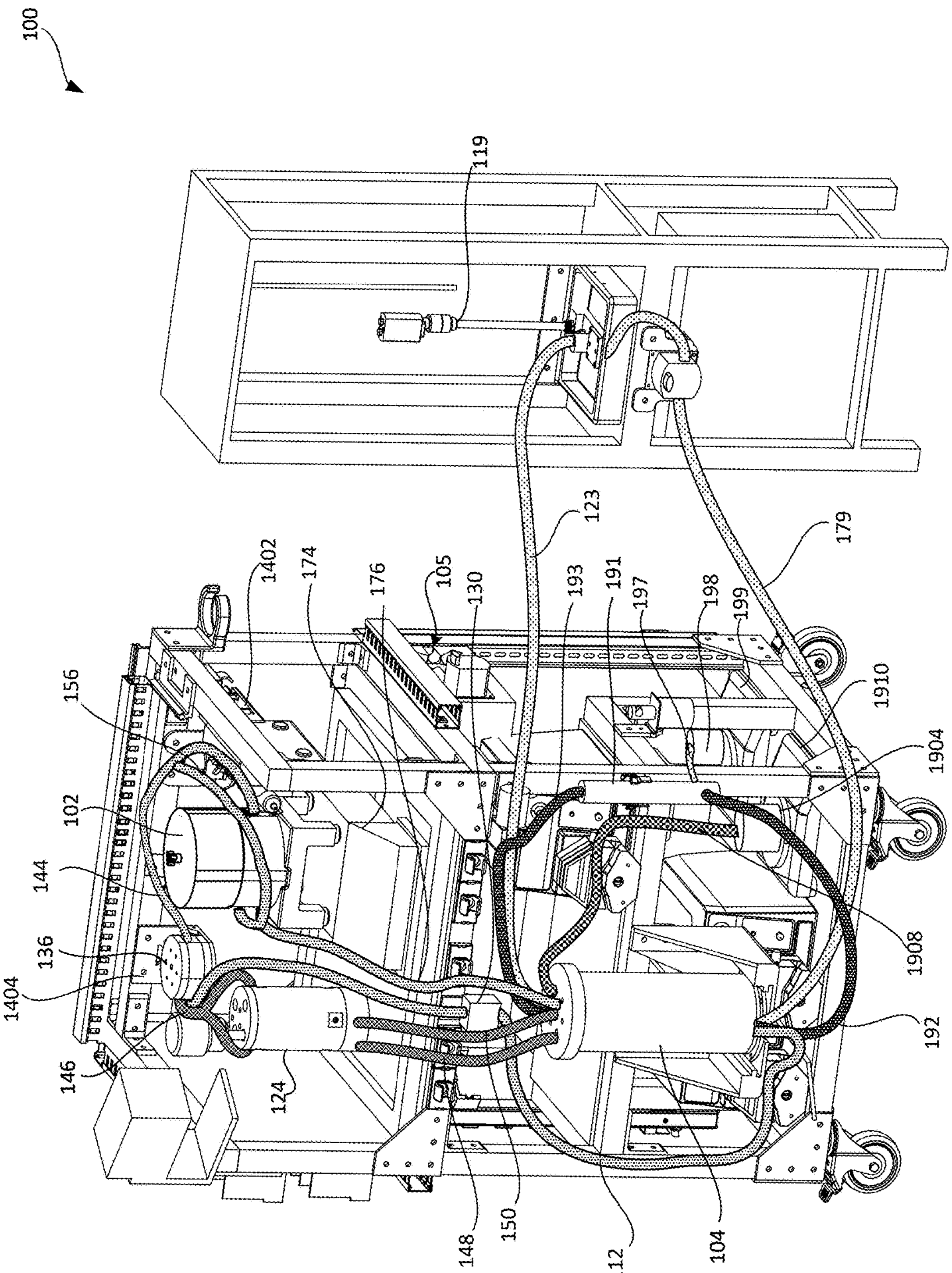

100
119
123
179
1402
174
176
105
130
193
191
197
198
199
1910
1904
156
102
144
136
1404
146
124
148
150
112
104
1908
192

FIG. 13

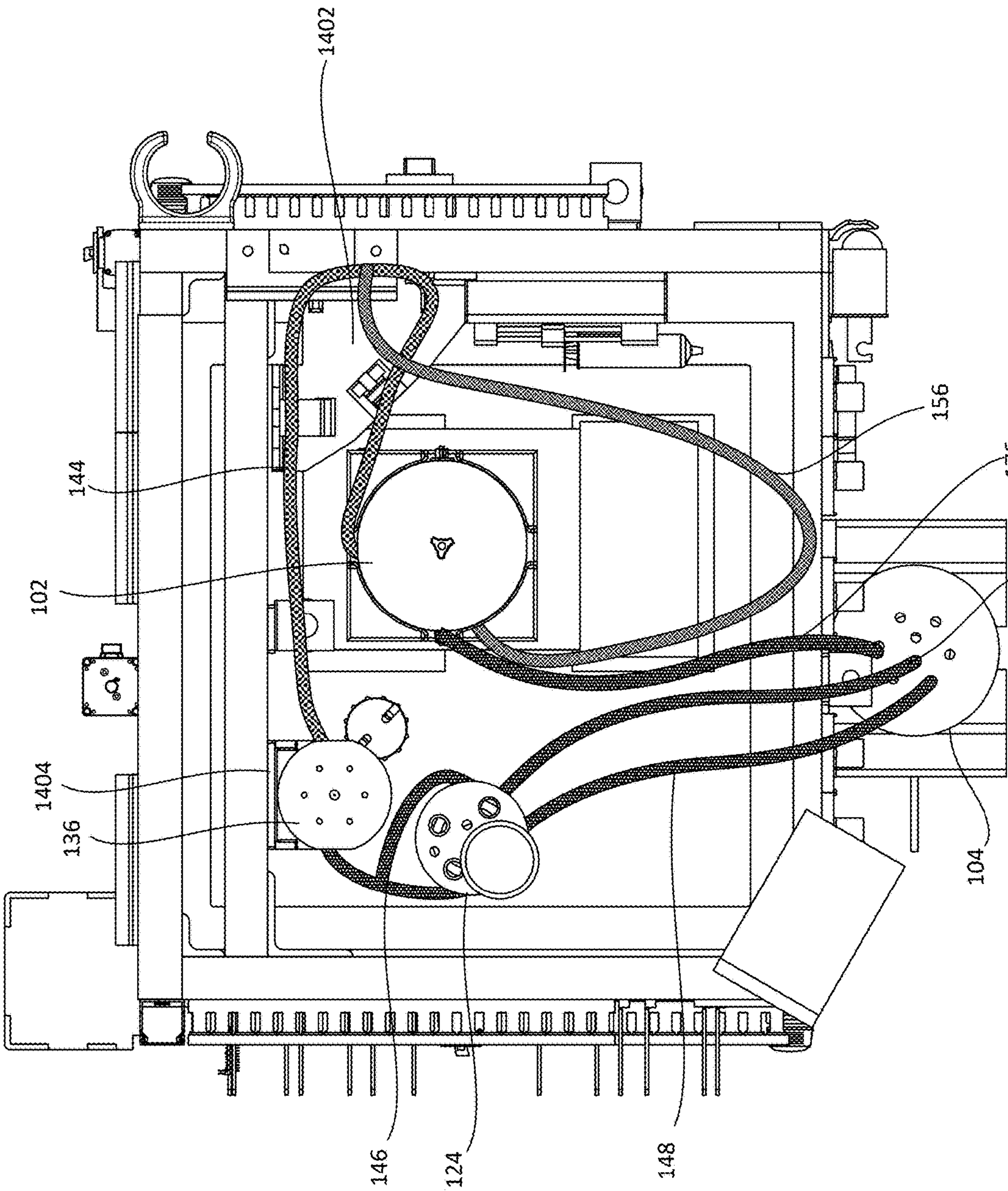


FIG. 14

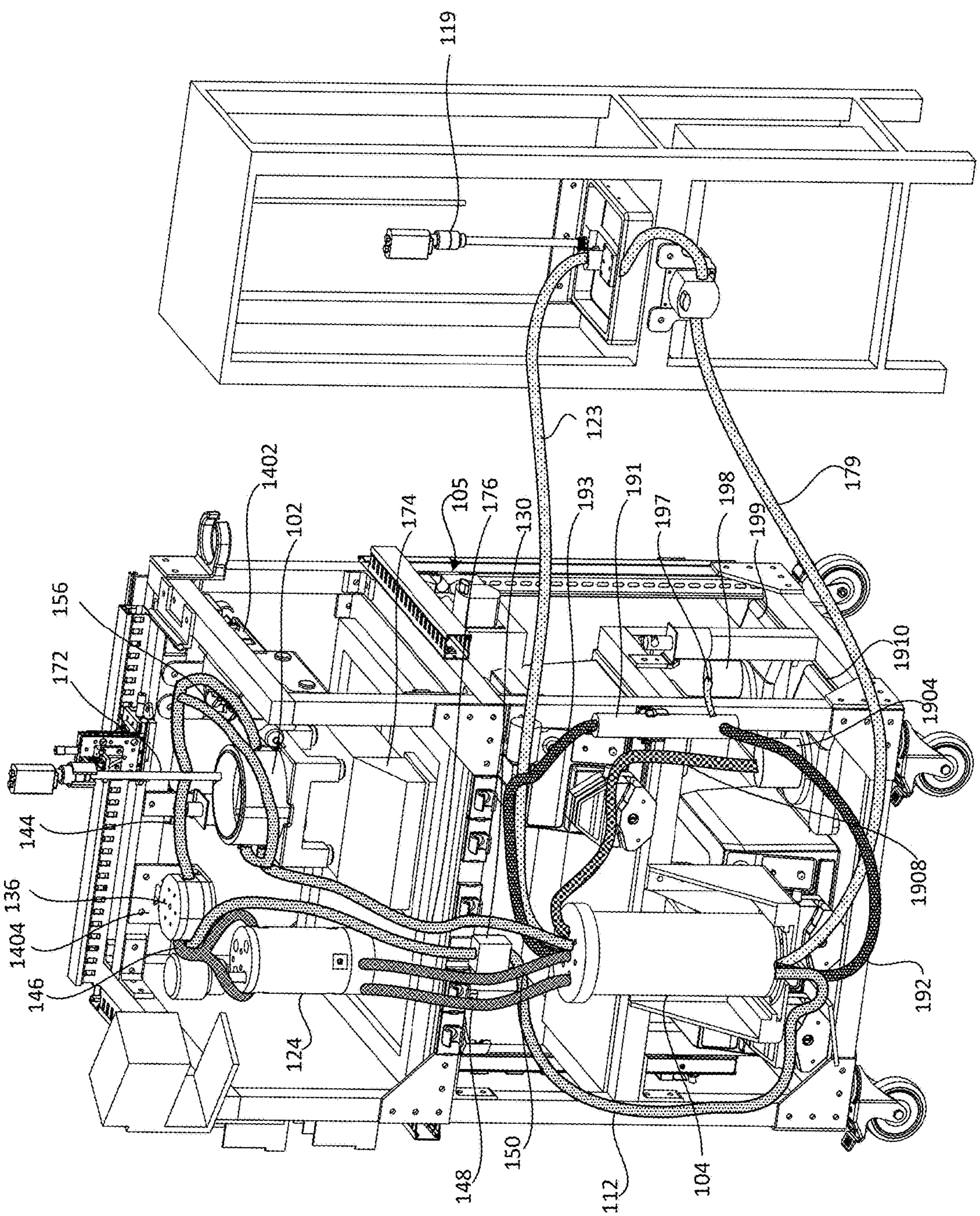

119
123
179
1402
102
174
105
176
130
193
191
197
198
199
156
172
144
136
1404
146
124
148
150
112
104
192
1908
1904
1910

FIG. 15

PERFUSION AND METABOLIC RESTORATION OF ORGANS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/562,094, filed on Mar. 6, 2024. The disclosure of the prior application is considered part of the disclosure of this application and is incorporated in its entirety into this application.

TECHNICAL FIELD

[0002] This specification relates to methods, systems, and compositions for perfusion and metabolic restoration of organs, in particular mammalian brains.

BACKGROUND

[0003] Perfusion in mammalian brains refers to the process of delivering perfusate (e.g., blood or an artificial perfusate) to the brain tissue to supply inputs, such as oxygen, glucose, and nutrients, while removing outputs, such as carbon dioxide and metabolic wastes. The process of perfusion is facilitated by a dense network of blood vessels, including arteries, capillaries, and veins, which collectively carry the cerebral blood flow.

SUMMARY

[0004] In one aspect, an ex-vivo brain perfusion system for perfusion of a brain includes a reservoir configured to receive perfusate, a housing configured to contain the brain, an arterial circuit configured to fluidly couple the reservoir to the brain, a venous circuit fluidly coupling the housing to the reservoir, and a syringe pump fluidly coupled to the reservoir and configured to introduce a therapeutic agent to the perfusate in the reservoir. The ex-vivo brain perfusion system is configured to perfuse the brain with the therapeutic agent to test at least one property of the therapeutic agent.

[0005] Implementations can include one or more of the following features in any combination.

[0006] In some implementations, the arterial circuit includes a fluid line fluidly coupling the reservoir to the brain, a pressure sensor configured to measure pressure along the fluid line, a flow sensor configured to measure a flow rate along the fluid line, and a pulse generation system configured to generate pulsatile flow of perfusate from the reservoir to the brain along the fluid line based on one or more signals generated by at least one of the pressure sensor or the flow sensor.

[0007] In certain implementations, the pulse generation system includes a pulse generator and an air supply system. The pulse generator includes a housing, a housing inlet, and a flexible diaphragm. The air supply system is fluidly coupled to the housing inlet of the pulse generator. The air supply system is configured to provide pressurized air into the housing inlet of the pulse generator based on one or more signals generated by at least one of the pressure sensor or the flow sensor,

[0008] In some implementations, the pulse generation system is controlled based on an internal resistance of the brain detected by the pressure sensor.

[0009] In certain implementations, the syringe pump is configured to introduce the therapeutic agent into the reservoir according to a predefined release profile for the therapeutic agent.

[0010] In some implementations, the system includes a filtration system configured to filter the perfusate. The filtration system includes a dialyzer that includes an inlet port and an outlet port, an inlet line fluidly coupling the reservoir to the inlet port, an outlet line fluidly coupling the outlet port to the reservoir, and a pump configured to pump perfusate from the reservoir through the dialyzer. The dialyzer is configured to filter the perfusate.

[0011] In certain implementations, the system includes an in-line spectroscopy system configured to perform spectroscopy on the perfusate in real-time during a perfusion process.

[0012] In some implementations, the in-line spectroscopy system includes a spectroscopy chamber configured to receive perfusate from the reservoir or the venous circuit, a heat exchanger fluidly coupled to the spectroscopy chamber and configured to control a temperature of perfusate provided to the spectroscopy chamber, and a spectrometer configured to perform spectroscopy on perfusate in the spectroscopy chamber.

[0013] In certain implementations, the in-line spectroscopy system is configured to perform spectroscopy on one or more perfusate samples from the reservoir, and the syringe pump is configured to introduce the therapeutic agent into the reservoir according to a predefined release profile for the therapeutic agent based on spectroscopy data generated by performing spectroscopy on one or more perfusate samples from the reservoir using the in-line spectroscopy system.

[0014] In some implementations, the system further includes a filtration system configured to filter the therapeutic agent out of the perfusate, and the filtration system is configured to remove the therapeutic agent from the perfusate according to a predefined release profile for the therapeutic agent based on spectroscopy data generated by performing spectroscopy on one or more perfusate samples from the reservoir using the in-line spectroscopy system.

[0015] In certain implementations, the in-line spectroscopy system is configured to perform spectroscopy on one or more perfusate samples collected from the venous circuit to test the at least one property of the therapeutic agent.

[0016] In some implementations, the at least one property of the therapeutic agent includes whether the therapeutic agent passes through a blood-brain barrier of the brain.

[0017] In certain implementations, the in-line spectroscopy system is configured to perform Raman spectroscopy on the perfusate.

[0018] In certain implementations, the in-line spectroscopy system is configured to perform surface-enhanced Raman spectroscopy (SERS) on the perfusate.

[0019] In some implementations, the system includes a spectroscopy probe coupled to the housing, and the spectroscopy probe is configured to perform spectroscopy on tissue of the brain in the housing.

[0020] In certain implementations, the at least one property of the therapeutic agent is determined based on spectroscopy data generated by performing spectroscopy on the brain using the spectroscopy probe.

[0021] In some implementations, the spectroscopy probe is configured to perform Raman spectroscopy on the tissue of the brain.

[0022] In certain implementations, the spectroscopy probe is configured to perform surface-enhanced Raman spectroscopy (SERS) on the tissue of the brain.

[0023] In certain implementations, the at least one property of the therapeutic agent is a pharmacodynamic property of the therapeutic agent or pharmacokinetic property of the therapeutic agent.

[0024] In certain implementations, the at least one property of the therapeutic agent is an ability of the therapeutic agent to cross the blood-brain barrier of the brain.

[0025] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0026] In another aspect, a method of testing at least one property of a therapeutic agent includes perfusing a brain with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, while perfusing the brain, performing spectroscopy on tissue of the brain in real-time using the ex-vivo perfusion system to generate spectroscopy data, and determining, based on spectroscopy data, the at least one property of the therapeutic agent.

[0027] Implementations can include one or more of the following features in any combination.

[0028] In some implementations, performing spectroscopy on tissue of the brain in real-time while perfusing the brain using the ex-vivo perfusion system includes performing spectroscopy on tissue of the brain using a spectroscopy probe coupled to a brain housing of the ex-vivo perfusion system.

[0029] In certain implementations, determining, based on the spectroscopy data, the at least one property of the therapeutic agent includes determining whether the therapeutic agent has crossed a blood-brain barrier of the brain.

[0030] In some implementations, determining whether the therapeutic agent has crossed the blood-brain barrier of the brain includes analyzing the spectroscopy data to determine whether the therapeutic agent is present in the tissue of the brain, and in response to determining that the therapeutic agent is present in the tissue of the brain, determining that the therapeutic agent has crossed the blood-brain barrier of the brain.

[0031] In certain implementations, determining, based on the spectroscopy data, the at least one property of the therapeutic agent includes determining one or more molecules produced by the brain in response to the presence of the therapeutic agent.

[0032] In some implementations, determining one or more molecules produced by the brain in response to the presence of the therapeutic agent includes prior to introducing the therapeutic agent into the perfusate, performing spectroscopy on tissue of the brain to generate baseline spectroscopy data, and comparing the spectroscopy data generated after introducing the therapeutic agent with the baseline spectroscopy data.

[0033] In certain implementations, determining, based on the spectroscopy data, the at least one property of the therapeutic agent includes determining, based on the spectroscopy data, a concentration of the therapeutic agent in the tissue of the brain.

[0034] In some implementations, the method includes determining, based on the spectroscopy data, a release profile for the therapeutic agent that provides a desired uptake of the therapeutic agent by the brain.

[0035] In certain implementations, introducing the therapeutic agent into the perfusate includes introducing the therapeutic agent into the perfusate according to a first release profile, performing spectroscopy on the tissue of the brain in real-time while perfusing the brain using the ex-vivo perfusion system includes performing spectroscopy on the tissue of the brain while introducing the therapeutic agent into the perfusate according to the first release profile to generate first spectroscopy data; and the method further includes introducing the therapeutic agent into the perfusate according to a second release profile, perfusing a second brain while introducing the therapeutic agent into the perfusate according to the second release profile, performing spectroscopy on tissue of the second brain while introducing the therapeutic agent into the perfusate according to the second release profile to generate second spectroscopy data, and determining a release profile for the therapeutic agent by comparing the first spectroscopy data and the second spectroscopy data

[0036] In some implementations, determining the release profile for the therapeutic agent includes comparing a first concentration of the therapeutic agent detected in the tissue of the brain based on the first spectroscopy data and a second concentration of the therapeutic agent detected in the tissue of the second brain based on the second spectroscopy data.

[0037] In certain implementations, the determined release profile for the therapeutic agent provides a desired uptake of the therapeutic agent by the brain.

[0038] In some implementations, performing spectroscopy on the tissue of the brain includes performing Raman spectroscopy on the tissue of the brain, and the spectroscopy data is Raman spectroscopy data.

[0039] In certain implementations, performing spectroscopy on the tissue of the brain includes performing surface-enhanced Raman spectroscopy (SERS) on the tissue of the brain, and the spectroscopy data is SERS data.

[0040] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0041] In certain implementations, the at least one property of the therapeutic agent is a pharmacodynamic property of the therapeutic agent or pharmacokinetic property of the therapeutic agent.

[0042] In a further aspect, a method of testing at least one property of a therapeutic agent includes perfusing a brain with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, while perfusing the brain, performing spectroscopy on the perfusate using an in-line spectroscopy system of the ex-vivo perfusion system to generate spectroscopy data, and determining, based on the spectroscopy data, the at least one property of the therapeutic agent.

[0043] Implementations can include one or more of the following features in any combination.

[0044] In some implementations, performing spectroscopy on the perfusate using an in-line spectroscopy system of the ex-vivo perfusion system includes operating a pump to deliver perfusate into a spectroscopy chamber of the in-line spectroscopy system, and controlling a spectrometer to perform spectroscopy on perfusate inside the spectroscopy chamber.

[0045] In certain implementations, performing spectroscopy on the perfusate using an in-line spectroscopy system of the ex-vivo perfusion system includes prior to delivering the perfusate into the spectroscopy chamber, adjusting a temperature of the perfusate to a predetermined temperature using a heat exchanger of the ex-vivo perfusion system.

[0046] In some implementations, performing spectroscopy on the perfusate using an in-line spectroscopy system of the ex-vivo perfusion system includes performing spectroscopy on used perfusate flowing out of the brain.

[0047] In certain implementations, determining, based on the spectroscopy data, the at least one property of the therapeutic agent includes determining whether the therapeutic agent has crossed a blood-brain barrier of the brain.

[0048] In some implementations, determining whether the therapeutic agent has crossed the blood-brain barrier of the brain includes analyzing the spectroscopy data generated by the in-line spectroscopy system to determine a concentration of the therapeutic agent in the used perfusate flowing out of the brain, comparing the concentration of the therapeutic agent in the used perfusate flowing out of the brain to a concentration of the therapeutic agent in fresh perfusate flowing into the brain, and in response to determining that the concentration of the therapeutic agent in the used perfusate flowing out of the brain is less than the concentration of the therapeutic agent in the fresh perfusate, determining that the therapeutic agent has crossed the blood-brain barrier of the brain.

[0049] In certain implementations, the method includes performing spectroscopy on the fresh perfusate using the in-line spectroscopy system to determine the concentration of the therapeutic agent in the fresh perfusate.

[0050] In some implementations, determining, based on the spectroscopy data, the at least one property of the therapeutic agent includes detecting one or more molecules produced by the brain in response to the presence of the therapeutic agent.

[0051] In certain implementations, detecting one or more molecules produced by the brain in response to the presence of the therapeutic agent includes prior to introducing the therapeutic agent into the perfusate, performing, using the in-line spectroscopy system, spectroscopy on fresh perfusate flowing to the brain to generate baseline spectroscopy data, and comparing the spectroscopy data generated by performing spectroscopy on the used perfusate with the baseline spectroscopy data.

[0052] In some implementations, introducing the therapeutic agent includes performing spectroscopy on fresh perfusate flowing into the brain using the in-line spectroscopy system to determine a concentration of the therapeutic agent in the fresh perfusate; and controlling the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate or remove the therapeutic agent from the fresh perfusate at a particular rate based on the concentration of the therapeutic agent in the fresh perfusate.

[0053] In certain implementations, controlling the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate or remove the therapeutic agent from the fresh perfusate at a particular rate based on the concentration of the therapeutic agent in the fresh perfusate includes controlling a syringe pump of the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate at the particular rate determined based on the concentration of the therapeutic agent in the fresh perfusate.

[0054] In some implementations, controlling the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate or remove the therapeutic agent from the fresh perfusate at a particular rate based on the concentration of the therapeutic agent in the fresh perfusate includes controlling a filtration system of the ex-vivo perfusion system to remove the therapeutic agent from the fresh perfusate at the particular rate determined based on the concentration of the therapeutic agent in the fresh perfusate.

[0055] In certain implementations, controlling the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate or remove the therapeutic agent from the fresh perfusate at a particular rate based on the concentration of the therapeutic agent in the fresh perfusate includes controlling the ex-vivo perfusion system to introduce the therapeutic agent into the fresh perfusate or remove the therapeutic agent from the fresh perfusate according to a predefined release profile.

[0056] In some implementations, the method includes determining, based on the spectroscopy data generated by the in-line spectroscopy system, a release profile for the therapeutic agent that provides a desired uptake of the therapeutic agent by the brain.

[0057] In certain implementations, introducing the therapeutic agent into the perfusate includes introducing the therapeutic agent into the perfusate according to a first release profile; performing spectroscopy on perfusate solution while perfusing the brain includes performing spectroscopy on used perfusate flowing out of the brain while introducing the therapeutic agent into the perfusate according to the first release profile to generate first spectroscopy data; and the method further includes introducing the therapeutic agent into the perfusate according to a second release profile, perfusing a second brain while introducing the therapeutic agent into the perfusate according to the second release profile, performing spectroscopy on used perfusate flowing out of the second brain using the in-line spectroscopy system while introducing the therapeutic agent into the perfusate according to the second release profile to generate second spectroscopy data, and determining a release profile for the therapeutic agent by comparing the first spectroscopy data and the second spectroscopy data.

[0058] In some implementations, determining a release profile for the therapeutic agent includes comparing a first concentration of the therapeutic agent detected based on the first spectroscopy data and a second concentration of the therapeutic agent detected based on the second spectroscopy data.

[0059] In certain implementations, the determined release profile for the therapeutic agent provides a desired uptake of the therapeutic agent by the brain.

[0060] In some implementations, the at least one property of the therapeutic agent is a pharmacodynamic property of the therapeutic agent or pharmacokinetic property of the therapeutic agent.

[0061] In certain implementations, performing spectroscopy on the perfusate while perfusing the brain using the in-line spectroscopy system includes performing Raman spectroscopy on the perfusate using the in-line spectroscopy system; and the spectroscopy data is Raman spectroscopy data.

[0062] In certain implementations, performing spectroscopy on the perfusate includes performing surface-enhanced

Raman spectroscopy (SERS) on the perfusate, and the spectroscopy data is SERS data.

[0063] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0064] In a further aspect, a method of perfusing a brain using an ex-vivo perfusion system to reduce internal resistance of the brain includes fluidly coupling the brain to a fluid line of the ex-vivo perfusion system, controlling a pulse generation system to provide pulsatile flow of the perfusate to the brain along the fluid line at an initial pressure, adjusting the pulsatile flow of the perfusate to increase a pressure along the fluid line until a flow rate of perfusate along the fluid line corresponds to a threshold flow rate, and in response to detecting that the flow rate of perfusate along the fluid line corresponds to the threshold flow rate, controlling the pulse generation system to maintain the threshold flow rate of perfusate along the fluid line.

[0065] Implementations can include one or more of the following features in any combination.

[0066] In some implementations, the pulse generation system includes a pulse generator and an air supply system fluidly coupled to the pulse generator, and adjusting the pulsatile flow of the perfusate to increase the pressure along the fluid line until a flow rate of perfusate along the fluid line corresponds to a threshold flow rate includes controlling the pulse generation system to increase the pressure along the fluid line.

[0067] In certain implementations, controlling the pulse generation system to increase the pressure along the fluid line includes controlling the air supply system to supply pressurized air to pulse generator at a particular frequency determined based on signals generated by a pressure sensor positioned along the fluid line.

[0068] In some implementations, detecting that the flow rate of perfusate along the fluid line corresponds to the threshold flow rate includes measuring the flow rate of perfusate along the fluid line using a flow sensor positioned along the fluid line.

[0069] In certain implementations, the method includes controlling a resistance valve positioned along the fluid line based on the pressure detected along the fluid line.

[0070] In some implementations, the method includes gradually increasing a temperature of the perfusate to a threshold temperature using a heat exchanger of the ex-vivo perfusion system.

[0071] In certain implementations, the method includes gradually increasing a concentration of oxygen in the perfusate to a threshold concentration using a gas mixer of the ex-vivo perfusion system.

[0072] In a further aspect, a pulse generation system configured to provide pulsatile flow of perfusate to a brain coupled to an ex-vivo perfusion system includes a pulse generator and an air supply system. The pulse generator includes a housing configured to receive perfusate from the ex-vivo perfusion system, an air inlet, and a flexible diaphragm. The air supply system is fluidly coupled the air inlet of the pulse generator. The air supply system is configured to provide pressurized air into the air inlet of the pulse generator based on one or more signals generated by at least one of a pressure sensor or a flow sensor downstream of the pulse generator.

[0073] Implementations can include one or more of the following features in any combination.

[0074] In some implementations, the pulse generator includes a perfusate inlet configured to be fluidly connected to a fluid line of the ex-vivo perfusion system upstream of the brain, a sensor block outlet configured to be fluidly coupled to a sensor block of the ex-vivo perfusion system, and an organ line outlet configured to be fluidly coupled to the brain.

[0075] In certain implementations, the pulse generator is configured to prevent flow of perfusate through the organ line outlet when the flexible diaphragm is in an unflexed position.

[0076] In some implementations, the pulse generator is configured to direct all of the perfusate in the housing to flow through the sensor block outlet when the flexible diaphragm is in an unflexed position.

[0077] In certain implementations, the pulse generator is configured to force a portion of the perfusate in the housing through the organ line outlet when the flexible diaphragm is in a flexed position.

[0078] In some implementations, the air supply system is configured to supply pressurized air to pulse generator at a particular frequency determined based on signals generated by a pressure sensor positioned along a fluid line extending between the pulse generator and the brain.

[0079] In some implementations, the air supply system includes an air source, a first electronic pressure regulator fluidly coupled to the air source, and a second electronic pressure regulator fluidly coupled to the first electronic pressure regulator. The first electronic pressure regulator is configured to regulate a pressure of a first stream of air provided by the air source. The second electronic pressure regulator is configured to regulate a pressure of a second stream of air provided by the first electronic pressure regulator.

[0080] In certain implementations, the second electronic pressure regulator has a smaller tolerance range than the first electronic pressure regulator.

[0081] In some implementations, the system includes a first pressure sensor coupled to a fluid line downstream of the first electronic pressure regulator and upstream of the second electronic pressure regulator, and the first electronic pressure regulator is controlled based on one or more signals generated by the first pressure sensor.

[0082] In certain implementations, the system includes a second pressure sensor coupled to the fluid line downstream of the second electronic pressure regulator, and the second electronic pressure regulator is controlled based on one or more signals generated by the second pressure sensor.

[0083] In a further aspect, an ex-vivo perfusion system for perfusion of a mammalian organ includes a reservoir configured to receive perfusate, a housing configured to contain the mammalian organ, an arterial circuit configured to fluidly couple the reservoir to the mammalian organ, a venous circuit fluidly coupling the housing to the reservoir, and a syringe pump fluidly coupled to the reservoir and configured to introduce a therapeutic agent to the perfusate in the reservoir. The ex-vivo perfusion system is configured to perfuse the mammalian organ with the therapeutic agent to test at least one property of the therapeutic agent.

[0084] Implementations can include one or more of the following features in any combination.

[0085] In some implementations, the mammalian organ is a human organ.

[0086] In some implementations, the mammalian organ is a brain.

[0087] In certain implementations, the mammalian organ is a human brain.

[0088] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0089] In a further aspect, a method of testing at least one property of a therapeutic agent includes perfusing a mammalian organ with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, while perfusing the mammalian organ, performing spectroscopy on tissue of the mammalian organ in real-time using the ex-vivo perfusion system to generate spectroscopy data, and determining, based on spectroscopy data generated by the spectroscopy, the at least one property of the therapeutic agent.

[0090] Implementations can include one or more of the following features in any combination.

[0091] In some implementations, the mammalian organ is a human organ.

[0092] In certain implementations, the mammalian organ is a brain.

[0093] In some implementations, the mammalian organ is a human brain.

[0094] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0095] In a further aspect, a method of testing at least one property of a therapeutic agent includes perfusing a mammalian organ with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, while perfusing the mammalian organ, performing spectroscopy on perfusate solution using an in-line spectroscopy system of the ex-vivo perfusion system to generate spectroscopy data, and determining, based on the spectroscopy data, the at least one property of the therapeutic agent.

[0096] Implementations can include one or more of the following features in any combination.

[0097] In some implementations, the mammalian organ is a human organ.

[0098] In certain implementations, the mammalian organ is a brain.

[0099] In some implementations, the mammalian organ is a human brain.

[0100] In some implementations, the therapeutic agent includes at least one of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0101] In a further aspect, a pulse generation system configured to provide pulsatile flow of perfusate to a mammalian organ coupled to an ex-vivo perfusion system includes a pulse generator and an air supply system. The pulse generator includes a housing configured to receive perfusate from the ex-vivo perfusion system, an air inlet, and a flexible diaphragm. The air supply system is fluidly coupled the air inlet of the pulse generator. The air supply system is configured to provide pressurized air into the air inlet of the pulse generator based on one or more signals generated by at least one of a pressure sensor or a flow sensor downstream of the pulse generator

[0102] Implementations can include one or more of the following features in any combination.

[0103] In some implementations, the mammalian organ is a human organ.

[0104] In certain implementations, the mammalian organ is a brain.

[0105] In some implementations, the mammalian organ is a human brain.

[0106] In another aspect, a method of testing at least one property of a therapeutic agent includes perfusing a brain with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, testing tissue of the brain using the ex-vivo perfusion system to generate test data, and determining, based on the test data, the at least one property of the therapeutic agent.

[0107] Implementations can include one or more of the following features in any combination.

[0108] In certain implementations, testing the tissue of the brain comprises performing one or more of spectroscopy, gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, or metabolomic analysis.

[0109] In some implementations, the therapeutic agent comprises one or more of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0110] In certain implementations, determining, based on the test data, the at least one property of the therapeutic agent comprises determining whether the therapeutic agent has crossed a blood-brain barrier of the brain.

[0111] In a further aspect, a method of testing at least one property of a therapeutic agent includes perfusing a brain with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, testing the perfusate using a test device of the ex-vivo perfusion system to generate test data, and determining, based on the test data, the at least one property of the therapeutic agent.

[0112] Implementations can include one or more of the following features in any combination.

[0113] In certain implementations, the test device comprises one or more of a spectroscopy device, a gene sequencing device, a cytokine analysis device, an anti-body analysis device, a genomic analysis device, a transcriptomic analysis device, a proteomic analysis device, or a metabolomic analysis device.

[0114] In some implementations, the therapeutic agent comprises one or more of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0115] In some implementations, determining, based on the test data, the at least one property of the therapeutic agent comprises determining whether the therapeutic agent has crossed a blood-brain barrier of the brain.

[0116] In an additional aspect, a method of testing at least one property of a therapeutic agent includes perfusing a mammalian organ with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, testing tissue of the mammalian organ using the ex-vivo perfusion system to generate test data, and determining, based on test data, the at least one property of the therapeutic agent.

[0117] Implementations can include one or more of the following features in any combination.

[0118] In certain implementations, testing the tissue of the mammalian organ comprises performing one or more of spectroscopy, gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, or metabolomic analysis.

[0119] In some implementations, the therapeutic agent comprises one or more of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0120] In some implementations, the mammalian organ is a human organ.

[0121] In certain implementations, the mammalian organ is a brain.

[0122] In some implementations, the mammalian organ is a human brain.

[0123] In another aspect, a method of testing at least one property of a therapeutic agent includes perfusing a mammalian organ with a perfusate using an ex-vivo perfusion system, introducing the therapeutic agent into the perfusate, testing the perfusate to generate test data, and determining, based on the test data, the at least one property of the therapeutic agent.

[0124] Implementations can include one or more of the following features in any combination.

[0125] In certain implementations, testing the perfusate comprises performing one or more of spectroscopy, gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, or metabolomic analysis.

[0126] In some implementations, the therapeutic agent comprises one or more of a pharmaceutical compound, an antibody, a biologic, a viral vector, a lipid nanoparticle, or a peptide.

[0127] In some implementations, the mammalian organ is a human organ.

[0128] In certain implementations, the mammalian organ is a brain.

[0129] In some implementations, the mammalian organ is a human brain.

[0130] In a further aspect, a method performed by one or more computers includes receiving test data that is generated by a test device of an ex-vivo brain perfusion system. The test data relates to (i) tissue of a brain being perfused by the ex-vivo brain perfusion system, or (ii) perfusate that has passed through the tissue of the brain being perfused by the ex-vivo brain perfusion system. The method further includes processing a model input that comprises the test data using a machine learning model, in accordance with values of a set of machine learning model parameters, to generate a model output that comprises a prediction characterizing the brain, and outputting the prediction characterizing the brain.

[0131] Implementations can include one or more of the following features in any combination.

[0132] In some implementations, the prediction characterizing the brain includes one or more of: a prediction for whether the brain is currently experiencing swelling; or a prediction for whether the brain is currently infected; or a prediction for a reaction of the brain to a therapeutic agent being provided to the brain by the ex-vivo brain perfusion system.

[0133] In certain implementations, the machine learning model includes one or more of: a neural network, or a random forest, or a support vector machine.

[0134] In some implementations, the model input comprises test data captured at multiple time points in a sequence of time points.

[0135] In another aspect, a system includes one or more computers and one or more storage devices communicatively coupled to the one or more computers, wherein the one or more storage devices store instructions that, when executed by the one or more computers, cause the one or more computers to perform operations comprising: receiving test data that is generated by a test device of an ex-vivo brain perfusion system; processing a model input that comprises the test data using a machine learning model, in accordance with values of a set of machine learning model parameters, to generate a model output that comprises a prediction characterizing the brain, and outputting the prediction characterizing the brain. The test data relates to (i) tissue of a brain being perfused by the ex-vivo brain perfusion system, or (ii) perfusate that has passed through the tissue of the brain being perfused by the ex-vivo brain perfusion system.

[0136] In an additional aspect, one or more non-transitory computer storage media store instructions that when executed by one or more computers cause the one or more computers to perform operations comprising: receiving test data that is generated by a test device of an ex-vivo brain perfusion system; processing a model input that comprises the test data using a machine learning model, in accordance with values of a set of machine learning model parameters, to generate a model output that comprises a prediction characterizing the brain, and outputting the prediction characterizing the brain. The test data relates to (i) tissue of a brain being perfused by the ex-vivo brain perfusion system, or (ii) perfusate that has passed through the tissue of the brain being perfused by the ex-vivo brain perfusion system.

[0137] Implementations can include one or more of the following advantages. The perfusion systems and methods described herein enable restoration and preservation of a human brain, including normal cellular activity of the brain, for up to **24** hours or longer post-mortem. As a result, cellular and metabolic health of the brain can be preserved in the brain.

[0138] In addition, human brains that are perfused using the perfusion systems and methods described herein can be tested to study disease mechanisms within the human brain. For example, brains exhibiting a particular disease or condition can be exposed to certain therapeutic agents (e.g., pharmaceutical compounds, antibodies, biologics, viral vectors, lipid nanoparticles, peptides, etc.) during perfusion using the perfusion systems and methods described herein, and the response of the brains to the therapeutic agents can be studied by performing testing on the brain and/or the perfusate in real-time during the perfusion process. In addition, brains can be mechanically or chemically altered prior to perfusion in order to simulate a particular disease mechanism. The mechanically or chemically altered brains can be exposed to certain therapeutic agents during perfusion using the perfusion systems and methods described herein, and the response of the brains to the therapeutic agents can be studied by performing testing on the brain and/or the perfusate in real-time during the perfusion process. In addition, individual mechanisms associated with particular brain diseases can be studied using the systems and methods described herein. For example, brains exhibiting a particular disease or altered to simulate the disease can be analyzed using the systems and methods described herein to identify and study particular compounds or molecules generated by the brain related to the disease or particular chemical or electrical activity of the brain related to disease.

[0139] The perfusion systems and methods described herein can also be used to determine the pharmacokinetics of therapeutic agents (also referred to herein as "drugs"), the pharmacodynamics of therapeutic agents, the likely efficacy of particular therapeutic agents on treating targeted diseases, and potential side effects that may result from administration of therapeutic agents. The perfusion systems and methods described herein can also be used to determine whether a therapeutic agent is able to cross the blood-brain barrier of the brain. As a result, the safety and efficacy of potential new drugs can be tested using the perfusate systems and methods described herein to accurately determine their likely effectiveness in treating the targeted diseases in humans without requiring administration of the drug to humans. For example, perfusion systems and techniques described herein provide a tightly controlled arterial input and high-dimension data sampling that allows investigators to expose the brain to any therapeutic modality in order to generate real and precise human multi-omic brain response networks required for phenotype-based drug discovery. The perfusion systems and techniques described herein can also provide highly differentiated capacity to determine safety profiles, pharmacological properties, and target engagement validation and deconvolution of desired therapeutics. The systems and methods described herein can be used to study the effects of drugs on the brain using non-diseased brains, diseased brains, or modeled brains.

[0140] The perfusion systems and techniques described herein can also be used to study mechanisms and potential treatments for brain injuries.

[0141] The accuracy of the data related to the brain captured using the systems and methods described herein has a high level of accuracy (e.g., a high signal to noise ratio) due to the isolation of the brain and the lack of influence from peripheral organs. The response of the brain to a particular stimulus can be isolated and determined with high levels of accuracy.

[0142] The details of certain implementations are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

[0143] FIG. **1** is a schematic illustration of a system for ex-vivo perfusion of a mammalian brain.

[0144] FIG. **2**A is a perspective view of an example pulse generator of the system of FIG. **1**.

[0145] FIG. **2**B is an exploded view of the example pulse generator of FIG. **2**A.

[0146] FIG. **2**C is a side view of the example pulse generator of FIG. **2**A.

[0147] FIG. **2**D is a cross-sectional view of the pulse generator of FIG. **2**A along axis A-A depicted in FIG. **2**C.

[0148] FIG. **3** is a schematic illustration of a process of sampling perfusate and brain tissue during perfusion of a brain using the system of FIG. **1**.

[0149] FIG. **4** is a schematic illustration of molecules in perfusate crossing the blood-brain barrier of a brain being perfused by the system of FIG. **1**.

[0150] FIG. **5** depicts example Raman spectroscopy data indicating release profiles of rapamycin provided through perfusate of the perfusion system of FIG. **1**.

[0151] FIG. **6** depicts example Raman spectroscopy data indicating the concentration of rapamycin in the tissue of a brain following introduction of rapamycin into the perfusate of the system of FIG. **1**.

[0152] FIG. **7** depicts a schematic illustration of different mammalian brains perfused and tested using the system of FIG. **1**.

[0153] FIG. **8** depicts example rates of lactate released by the mammalian brains depicted in FIG. **7** during perfusion of the brains using the system of FIG. **1**.

[0154] FIG. **9** is a block diagram of an example computer system by which a computer system of the perfusion system of FIG. **1** can be implemented.

[0155] FIG. **10**A is a perspective view of an example brain housing of the system of FIG. **1**.

[0156] FIG. **10**B is an exploded view of the example brain housing of FIG. **10**A.

[0157] FIG. **10**C is a cross-sectional view of the example brain housing of FIG. **10**A.

[0158] FIG. **10**D is a perspective view of an example hemispherical brain chamber of the example brain housing of FIG. **10**A.

[0159] FIG. **10**E is a top view of the example hemispherical brain chamber of FIG. **10**D.

[0160] FIG. **10**F is a cross-sectional view of the example hemispherical brain chamber of FIG. **10**D.

[0161] FIG. **11**A is a perspective view of another example brain housing.

[0162] FIG. **11**B is an exploded view of the example brain housing of FIG. **11**A.

[0163] FIG. **11**C is a side view of the example brain housing of FIG. **11**A.

[0164] FIG. **11**D is a cross-sectional view of the example brain housing of FIG. **11**A.

[0165] FIG. **12**A is a perspective view of an example brain interface configured to fluidly couple a brain to an organ line of the system of FIG. **1**.

[0166] FIG. **12**B is a side view of the example brain interface of FIG. **12**A.

[0167] FIG. **12**C is cross-sectional view of the example brain interface of FIG. **12**A.

[0168] FIG. **13** is a perspective view of the example system of FIG. **1**.

[0169] FIG. **14** is a top down view of a portion of the example system of FIG. **1**.

[0170] FIG. **15** is a perspective view of the example system of FIG. **1** with an example Raman probe positioned over the brain housing.

[0171] FIG. **16** is an RNAScope image showing GFP and mCherry expressed in the frontal cortex of a brain.

[0172] Like references symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0173] This specification generally describes devices, systems, and methods for ex-vivo perfusion of mammalian organs. In particular, this specification describes devices, systems, and methods for perfusing a human brain for ex-vivo restoration and preservation of the human brain, including normal cellular activity of the brain, for 24 hours or longer post-mortem or after global ischemia. In addition, this specification describes devices, systems, and methods for performing pharmacokinetic modeling and pharmacological testing using a perfused brain.

[0174] An example of an electromechanical perfusion device is described in: Zvonimir Vrselja et al., "Restoration of brain circulation and cellular functions hours postmortem," *Nature,* 2019 April; 568 (7752), which is incorporated by reference herein. Examples of electromechanical perfusion devices and artificial perfusion experiments are also described in U.S. patent application Ser. No. 16/967,925, which is incorporated by reference herein.

[0175] FIGS. **1**, **13**, and **15** depict an example perfusion system **100** for perfusion, restoration, and preservation of a human brain **110**. The perfusion system **100** includes a brain housing **102** that is configured to enclose a human brain **110** and a perfusate reservoir **104** that is configured to contain a perfusate solution (also referred to herein as "perfusate"). As will be described in further detail herein, the perfusate contained within the perfusate reservoir **104** is configured to perfuse and restore a human brain **110** contained inside the brain housing **102**.

[0176] The brain housing **102** is fluidly connected to the perfusate reservoir **104** through an arterial circuit **106** and a venous circuit **108**, which together form a closed-loop fluid circuit to circulate a synthetic acellular perfusate under a pulsatile flow. The arterial circuit **106** includes an arterial fluid line **112** and an organ line **144** fluidly coupling the perfusate reservoir **104** to the brain housing **102**, and the fresh perfusate contained within the perfusate reservoir **104** is provided to brain **110** inside the brain housing **102** along the arterial fluid line **112** and the organ line **144**. The arterial circuit **106** also includes a flow pump **114** positioned along the arterial fluid line **112** between the perfusate reservoir **104** and an oxygenator **116**. The flow pump **114** is a peristaltic pump configured to pump fresh perfusate from the perfusate reservoir **104** to the oxygenator **116**. The peristaltic flow pump **114** is configured to minimize shear of the perfusate as it is pumped along the arterial fluid line **112**. In some implementations, the flow pump **114** is configured to flow the perfusate to the oxygenator **116** at a rate of 400 mL/minute.

[0177] The oxygenator **116** includes a heat exchanger **118** and a gas mixer **120**. The heat exchanger **118** includes one or more metal heat exchanging coils configured to regulate the temperature of the perfusate received by the oxygenator **116**. As will be described in further detail herein, the arterial circuit **106** includes a sensor block **124** with sensor probes **128** configured to measure certain characteristics of the fresh perfusate flowing through the arterial circuit **106**, including the temperature of the fresh perfusate, and the heat exchanger **118** is configured to adjust the temperature of the perfusate flowing through the heat exchanger **118** based on the temperature of the perfusate detected by the sensor block **124**. For example, the heat exchanger **118** is configured to adjust the temperature of the perfusate to be within a predetermined range of temperature based on the temperature signals generated by the sensor block **124**. For example, as will be described in further detail herein, at the beginning of treatment, the temperature of the perfusate in the system **100** is in a range of 16° C. to 20° C. and the heat exchanger **118** can be used to increase the temperature of the perfusate at a controlled rate at the beginning of the perfusion process. In some implementations, the heat exchanger **118** is configured to increase to temperature of the perfusate to be within a range of 35° C. to 37° C.

[0178] The gas mixer **120** of the oxygenator **116** is fluidly coupled to and receives one or more gasses from a gas source **122**, and the gas mixer **120** is configured to dissolve the one or more gasses into the perfusate flowing through the oxygenator **116**. The gas mixer **120** includes three inlets fluidly coupled to the gas source **122**. A first inlet of the gas mixer **120** is configured to be fluidly coupled to a source of O_2, a second inlet of the gas mixer **120** is configured to be coupled to a source of N_2, and a third inlet of the gas mixer **120** is configured to be coupled to a source of CO_2. The gas mixer **120** is controlled based on one or more signals generated by the sensor block **124** indicating the dissolved gas levels of the perfusate within the sensor block **124**. For example, the sensor block **124** can generate one or more signals indicating the dissolved oxygen concentration of the perfusate within the sensor block **124**, the dissolved nitrogen concentration of the perfusate within the sensor block **124**, and the dissolved carbon dioxide concentration of the perfusate within the sensor block **124** and, based on the signals, the gas mixer **120** adjusts the amount of oxygen, nitrogen, and/or carbon dioxide provided to the perfusate within the oxygenator **116** in order to control the dissolved oxygen concentration, the dissolved nitrogen concentration, and the dissolved carbon dioxide concentration of the perfusate to be within a predetermined range. For example, based on the signals generated by the sensor block **124**, the gas mixer **120** generates commands to control internal gas valves of the gas mixer **120** to control the flow rate of gas into each respective inlet of the gas mixer **120** in order to control the amount and rate that each gas (O_2, N_2, and CO_2) are provided to the oxygenator **116**.

[0179] In addition, the gas mixer **120** monitors the pressure and flow rate of gas provided from the gas mixer **120** to the oxygenator **116**. In some implementations, the gas mixer **120** maintains the flow rate of gas from the gas mixer **120** to the oxygenator **116** at approximately **700** mL/minute. As described herein, in some implementations, the flow pump **114** is configured to flow the perfusate to the oxygenator **116** at a rate of 400 mL/minute. By flowing the gas into the oxygenator **116** at a rate of 700 mL/minute and simultaneously flowing perfusate into the oxygenator **116** at a rate of 400 mL/minute, a predetermined ventilation/perfusion coefficient of 700/400 can be maintained throughout the perfusion process. By adjusting the flow rate of each of the gasses flowing into the gas mixer **120** while maintaining the rate of gas flowing out of the gas mixer **120** into oxygenator **116**, the$_2$ gas mixer **120** can control and adjust the concentration of each of the respective gasses within the perfusate while maintaining the predetermined ventilation/perfusion coefficient.

[0180] The concentration of the CO_2, N_2, and O_2 within the perfusate are maintained within physiological limits, as determined based on the concentration detected using the sensor probes **128** in the sensor block **124**. In some implementations, the gas mixer **120** and oxygenator **116** are controlled based on an alpha stat approach to maintain the blood gas concentrations within normal, physiological limits. In some implementations, the partial pressure of CO_2 within the partial pressure of the CO_2 within the perfusate is maintained at a partial pressure of approximately 400 mmHg during the perfusion process. In some implementations, the partial pressure of O_2 in the perfusate is maintained at 450 mmHg during the perfusion process. As will be described in further detail herein, at the beginning of a perfusion procedure, the temperature of the perfusate is gradually increased by the heat exchanger **118** and the O_2 concentration of the perfusate is gradually increased by the gas mixer **120** until a target temperature and O_2 concentration for the perfusate are reached, as detected based on the sensor probes **128** in the sensor block **124**. For example, at the beginning of perfusion, the partial pressure of O_2 in the perfusate is 200 mmHg and the gas mixer **120** and oxygenator **116** are controlled to gradually increase the partial pressure of O_2 in the perfusate to 450 mmHg. By gradually increasing the temperature and the O_2 concentration of the perfusate, the internal resistance of the brain **110** is decreased, which results in improved perfusion of the brain **110** and reduced risk of injury to the brain **110**.

[0181] A shunt line **131** is coupled to and extends from the oxygenator **116** and fluidly couples the oxygenator **116** to the brain housing **102**. The shunt line **131** is configured to carry fresh perfusate out of the oxygenator **116** directly to the brain housing **102** without passing through a pulse generator **136** of the perfusate system **100**. The shunt line **131** can be used in preparation stages to quality check the device prior to the connection of the brain. The shunt line **131** can be used to provide perfusate to the brain housing **102** during the quality check period but is generally not used during the primary application of brain perfusion. A stopcock **135** is positioned along the shunt line **131** downstream of the oxygenator **116**. The stopcock **135** can be used to fluidly coupled the shunt line **131** to the brain housing **102** and to sample the perfusate flowing out of the oxygenator **116** prior to providing the perfusate to the brain **110**, for example, in order to conduct tests on the perfusate following oxygenation of the perfusate by the oxygenator **116**.

[0182] A solenoid valve **133** is positioned along the shunt line **131** and is configured to control the flow of perfusate between the oxygenator **116** and the brain housing **102** along the shunt line **131**. For example, during the initial set up of the perfusate system **100** for a perfusate procedure, the stopcock **135** is controlled (e.g., using a servo-motor) to fluidly couple the shunt line **131** to the brain housing **102**, the solenoid valve **133** along the shunt line **131** is opened and the flow pump **114** is operated to gradually increase the flow of perfusate through the oxygenator **116** to 400 mL/minute, and perfusate flows through the oxygenator **116** along the shunt line **131** and into the brain housing **102**. As perfusate flows through the oxygenator **116** and the shunt line **131**, any air in the fibers of the oxygenator **116** is forced out of the oxygenator **116** along the shunt line **131** to the brain housing **102**, which allow for gas exchange into the perfusate in the oxygenator **116**. Once all of the air has been flushed out of the oxygenator **116**, the solenoid valve **133** is closed to prevent flow of perfusate from the oxygenator **116** to the brain housing **102** along the shunt line **131**.

[0183] The arterial circuit **106** includes a flow sensor **130** positioned along the arterial fluid line **112** downstream of the oxygenator **116**. When the solenoid valve **133** along the shunt line **131** is closed and the flow pump **114** is operated, fresh perfusate exiting the oxygenator **116** flows along the arterial fluid line **112** through the flow sensor **130**, and the flow pump **114** is controlled based on signals generated by the flow sensor **130**. In some implementations, the pump speed of the flow pump **114** is adjusted based on one or more signals received from the flow sensor **130** indicating the flow rate of the perfusate exiting the oxygenator **116** in order to control the flow rate of the perfusate exiting the oxygenator **116** to be within a predetermined range. In some implementations, the flow pump **114** is controlled to maintain the flow rate of perfusate exiting the oxygenator **116** to be 400 mL/minute. As previously discussed, the flow rate of perfusate into the oxygenator **116** and the flow rate of gas into the oxygenator **116** can each be controlled to maintain a predetermined ventilation/perfusion coefficient. Thus, in some implementations, the flow pump **114** and the gas mixer **120** are each controlled to adjust the flow rates of perfusate and gas, respectively, into the oxygenator **116** in order to maintain the predetermined ventilation/perfusion coefficient.

[0184] Still referring to FIG. **1**, the arterial circuit **106** includes a stopcock **132** positioned along the arterial fluid line **112** downstream of the oxygenator **116**. The stopcock **132** can be used to sample the perfusate flowing out of the oxygenator **116**, for example, in order to conduct tests on the perfusate following oxygenation of the perfusate by the oxygenator **116**.

[0185] The arterial circuit **106** also includes a one-way valve **134** positioned downstream of the oxygenator **116** and the flow sensor **130**. The one-way valve **134** is configured to prevent perfusate in upstream portions of the arterial fluid line **112** from flowing back into the oxygenator **116**.

[0186] The arterial circuit **106** also includes a pulse generation system **1302** configured to convert laminar flow of perfusate along the arterial fluid line **112** to pulsatile flow along the organ line **144**. The pulse generation system **1302** includes a pulse generator device **136** (also referred to herein as "pulse generator **136**") and an air supply system **138** fluidly coupled to the pulse generator **136**. After flowing out of the oxygenator **116**, the perfusate flows along the arterial fluid line **112** to the pulse generator **136**. The pulse generator **136** is configured to convert the steady state flow of the fresh perfusate along the arterial fluid line **112** to pulsatile flow. As a result, fresh perfusate is provided to the brain **110** as a pulsatile flow that more closely mimics the anatomical flow of blood to the human brain, which results in improved perfusion of the brain **110**. As will be described in detail herein, the pulsatile flow of perfusate generated by the pulse generation system **1302** is dynamically adjusted throughout the perfusion process to optimize the flow rate and fluid pressure of perfusate provided to the brain **110** responsive to the level of resistance of the brain **110**, which improves longevity of the cellular and metabolic activity and tissue of the brain **110** being perfused by the system **100**.

[0187] FIGS. **2A-2D** depict the example pulse generator device **136**. The pulse generator **136** includes a body **240** that defines a perfusate inlet **204**, a sensor block outlet **206**, and an organ line outlet **208**. Referring to FIGS. **1** and **2A-2D**, the perfusate inlet **204** is fluidly coupled to a fluid chamber **236** of the pulse generator **136**. The fluid chamber **236** is configured to receive perfusate flowing into the pulse generator **136** from the arterial fluid line **112** through the perfusate inlet **204**. The fluid chamber **236** includes a plurality of openings **210** through an upper surface of the fluid chamber **236**, and perfusate received by the fluid chamber **236** from the perfusate inlet **204** is configured to flow through the openings **210** into the interior **212** of a central housing **202** of the pulse generator **136**. Fluid contained within the interior **212** of the central housing **202** flows through the sensor block outlet **206** and/or the organ line outlet **208**, as will be described herein. In some implementations, the diaphragm **214** is a back pressure regulator membrane manufactured by EQUILIBAR®.

[0188] Referring to FIG. 2B, the pulse generator **136** further includes a lid **218** and a base **220**. A flexible diaphragm **214** is positioned over the fluid chamber **236** between the lid **218** and the central housing **202** of the pulse generator **136**. The lid **218** is coupled to an upper surface of the central housing **202** and the base **220** is coupled to a lower surface of the central housing **202**. O-rings **222***a,* **222***b* are positioned between the lid **218** and the central housing **202** and between the base **220** and the central housing **202**, respectively, in order to prevent perfusate contained within the interior **212** of the central housing **202** from leaking outside of the central housing **202**. The lid **218** and the base **220** are coupled to the central housing **202** by a plurality of bolts **224***a,* **224***b,* **224***c,* **224***d,* **224***e,* **224***f* (collectively referred to herein as bolts **224**) inserted through a series of respective openings **226***a,* **226***b,* **226***c,* **226***d,* **226***e,* **226***f* through the lid **218**, respective openings **234***a,* **234***b,* **234***c,* **234***d,* **234***e,* **234***f* through the central housing **202**, and respective openings **232***a,* **232***b,* **232***c,* **232***d,* **232***e,* **232***f* through the base **220**. A respective plurality of nuts **228***a,* **228***b,* **228***c,* **228***d,* **228***e,* **228***f* are to the bolts **224**.

[0189] The lid **28** of the pulse generator **136** defines an air inlet **230** fluidly coupled to and configured to receive pressurized air from the air supply system **138**. The air supply system **138** is configured to provided pressurized air to the pulse generator **136** in order to control the pulsatile flow of perfusate received by pulse generator **136**. For example, the diaphragm **214** flexes downwards towards the fluid chamber **236** in response to pressurized air being provided by the air supply system **138** through the air inlet **230** of the pulse generator **136**. The flexure of the diaphragm **214** causes the openings **210** in the fluid chamber **236** to be covered by the diaphragm **214** and increases the pressure within the interior **212** of the central housing **202**, which results in increased flow through the organ line outlet **208** of the pulse generator **136** to the organ line **144** that fluidly couples the pulse generator to the brain **110**.

[0190] Referring to FIG. **1**, the air supply system **138** includes an air source **160**, a first electronic pressure regulator **162**, a first pressure sensor **164** positioned downstream of the first electronic pressure regulator **162**, a second electronic pressure regulator **166** positioned downstream of the first electronic pressure regulator **162**, and a second pressure sensor **168** positioned downstream of the second electronic pressure regulator **166**. The air source **160**, the first electronic pressure regulator **162**, the second electronic pressure regulator **166**, and the pulse generator **136** are each fluidly coupled along an air supply line **161**.

[0191] The first electronic pressure regulator **162** is controlled based on pressure signals received from the first pressure sensor **164** in order to provide a pressurized stream of air in a particular pressure range in a particular pressure range to the second electronic pressure regulator **166**. In some implementations, the first electronic pressure regulator **162** is configured to regulate the air supply **160** in order to provide a stream of pressurized air to the second electronic pressure regulator **166** that is in a range of 1 to 5 psi.

[0192] The second electronic pressure regulator **166** has tighter tolerances compared to the first electronic pressure regulator **162** and is configured to provide a pressurized air stream within a particular pressure range to the air inlet **230** of the pulse generator **136**. In some implementations, the second pressure sensor **166** is configured to generate an air stream with a pressure of 5 psi. The pressure of the air stream generated by the second electronic pressure regulator **166** is controlled based on pressure signals received from the second pressure sensor **168** of the air supply system **138**. In addition, as will be described in further detail herein, the second electronic pressure regulator **166** is controlled based on signals received from a pressure sensor **141** and a flow sensor **142** positioned along the organ line **144** that fluidly couples the pulse generator **136** to the brain **110** inside the brain housing **102**. The use of two electronic pressure regulators **162**, **166** to regulate the pressure of the air stream provided to the pulse generator **136** results in a smoother and more controlled supply of pressurized air to the pulse generator **136** compared to using a single electronic pressure regulator. In some implementations, the electronic pressure regulators **162**, **166** are QPV electronic pressure regulators manufactured by EQUILIBAR®.

[0193] The air supply system **138** also includes an air calibration line **163** that fluidly couples the air supply line **161** to the organ line **144** downstream of the pressure sensors **140**, **141** positioned along the organ line **144**. As will be described in further detail, the pressure sensors **140**, **141** along the organ line **144** are calibrated by providing pressurized air to the organ line **144** along the air calibration line **163**. A filter **165** is positioned along the air calibration line **163** in order to filter out and prevent any particulates in the pressurized air stream flowing along the air calibration line **163** from entering the organ line **144**. In some implementations, the filter **165** is a 0.22 μm filter.

[0194] In order to restore and maintain the cellular and metabolic activity of the brain **110**, the pulse generator **136** and the air supply system **138** are controlled to provide a pulsatile flow of fresh perfusate to the brain **110** within predetermined boundaries of pressure and flow rate that are selected to effectively perfuse the brain while minimizing damage to the vasculature and other tissues of the brain **110**. Referring to FIGS. **1** and 2D, a sensor block inlet line **146** is coupled to the sensor block outlet **206** of the pulse generator **136** to fluidly couple the pulse generator **136** to the sensor block **124**. The organ line **144** is coupled to the organ line outlet **208** of the pulse generator **136** to fluidly couple the pulse generator **136** to the brain **110**. The sensor block inlet line **146** has a larger diameter than the organ line **144**. In addition, as can be seen in FIGS. **13-15**, the organ line **144** and the pulse generator **136** are each positioned on a respective platform **1402**, **1404** that maintains the brain housing **102**, the organ line **144**, and the pulse generator **136** at substantially equal heights above the height of the sensor block inlet line **146**.

[0195] As depicted in FIG. **15**, in some implementations, the sensor block inlet line **146** is a y-splitter line with a single line flowing out of the pulse generator **136** that splits into two fluid lines flowing into the sensor block **124**. The sensor block inlet line **146** extends downwards relative to the organ line **144**. As a result of the relative positioning of the pulse generator **136**, the brain housing **102**, the organ line **144**, and the sensor block **124**, there is no passive flow of fluid from the pulse generator **136** to the brain **110** inside the brain housing **102**, and all of the perfusate contained within the interior **212** of the central housing **202** of the pulse generator

136 passively flows through the sensor block inlet line 146 when the diaphragm 214 of the pulse generator 136 is in an unflexed ("neutral") position. Therefore, in order to provide fresh perfusate from the pulse generator 136 to the brain 110 via the organ line 144, the diaphragm 214 of the pulse generator 136 must be flexed downwards towards the fluid chamber 236 to cover the openings 210 in the fluid chamber 236 and increase the pressure within the interior 212 of the central housing 202. As result of the increased pressure caused by flexing the diaphragm 214, at least a portion of the fresh perfusate contained within the interior 212 of the pulse generator is forced through the organ line outlet 208 and along the organ line 144 to the brain 110.

[0196] In addition, the sensor block inlet line 146 has a larger diameter than the organ line 144. In some implementations, the sensor block inlet line 146 includes a Y-connector and two fluid lines extending from the Y-connector to fluidly couple the pulse generator 136 to the sensor block 124, further decreasing the resistance to flow between the pulse generator 136 and the sensor block 124. As a result of the decreased resistance to flow between the pulse generator 136 and the sensor block 124 compared to the resistance to flow between the pulse generator 136 and brain 110 inside the brain housing 102 due to the structural differences in the sensor block inlet line 146 and the organ line 144, at least a portion of the perfusate flowing through the pulse generator 136 is provided to the sensor block 124 from the pulse generator 136, even when the diaphragm 214 of the pulse generator 136 is in a flexed position.

[0197] In order to maintain an optimal pulsatile flow of perfusate along the organ line that is responsive to the internal resistance of the brain 110, the air supply system 138 coupled to the pulse generator 136 is controlled based on signals generated by a pressure sensor 141 and a flow sensor 142 positioned along the organ line 144. In some implementation, the pulse generation system 1302 is controlled to maintain the pressure along the organ line 144, as measured by the pressure sensor 141, between 5 mmHg and 100 mmHg. In some implementation, the pulse generation system 1302 is controlled to maintain the flow rate along the organ line 144, as measured by the flow sensor 142, between 0 mL/min and 750 mL/minute. The flow rate measured by the flow sensor 142 corresponds to the average flow rate of the pulse of fluid provided by the pulse generator 136.

[0198] The second electronic pressure regulator 166 of the air supply system 138 is controlled to provide pressurized air to the air inlet 230 of the pulse generator 136 at particular intervals and durations based on the signals generated the second pressure sensor 141 and the flow sensor 142 along the organ line 144 in order to achieve a particular pressure and/or flow rate of perfusate along the organ line 144. As the second electronic pressure regulator 166 provides pressurized air to the air inlet 230 of the pulse generator 136, the diaphragm 214 of the pulse generator 136 flexes downwards towards the fluid chamber 236 in response to the increased positive pressure in the air inlet 230 and covers the openings 210 in the fluid chamber 236. As result of the increased pressure within the interior 212 of the pulse generator 136 caused by the flexing of the diaphragm 214, at least a portion of the fresh perfusate contained within the interior 212 of the pulse generator is forced through the organ line outlet 208 and along the organ line 144 to the brain 110. By adjusting the frequency and/or duration of the pressurized air streams provided by the second electronic pressure regulator 166 to the pulse generator 136 in response to pressure and flow rate signals received from sensors 141, 142, the amplitude and base of the pulse of perfusate provided by the pulse generator 136 to the organ line 144 can be optimized based on the internal resistance of the brain detected by the sensor 141, 142.

[0199] In addition, as depicted in FIG. 1, the perfusion system 100 includes a resistance valve 152 positioned along the organ line 144 and that can be operated to further provide a particular flow resistance along the organ line 144. In particular, the resistance valve 152 is controlled to prevent passive flow of perfusate along the organ line 144 due to the driven flow provided by the flow pump 114.

[0200] The resistance valve 152 is positioned along the organ line 144 downstream of the first pressure sensor 140 and the flow sensor 142 and is positioned upstream of the second pressure sensor 141. During calibration of the system 100, the flow pump 114 is operated to flow perfusate along the organ line 144 toward the brain 110 at a predetermined rate without operating the pulse generation system 1302. As the flow pump flows perfusate along the organ line 144 during calibration, the pressure along the organ line 144 upstream of the resistance valve 152 is measured by the first pressure sensor 140 and the pressure along the organ line 144 downstream of the resistance valve 152 is measured by the second pressure sensor 141. Based on the pressure measured by the pressure sensors 140, 141, the resistance valve 152 can be adjusted (e.g., partially or fully opened or partially or fully closed) in order to predetermined internal resistance along the organ line 144. In some implementations, the resistance valve 152 is controlled, based on the data generated by the pressure sensors 140, 141, to provide an internal resistance of 1 mmHg along the organ line 144.

[0201] During the post-mortem interval prior to beginning the perfusion process, the brain 110 undergoes processes resulting in an increased internal resistance to flow within the vasculature of the brain 110. For example, during the post-mortem interval prior to beginning the perfusion process, water leaks into the brain tissue and collapses the capillaries of the brain 110, resulting in the formation of internal Starling resistors in the brain and an increased internal resistance to flow within the vasculature of the brain 110. Therefore, in order to condition the vasculature of the brain 110 to sufficiently uptake the fresh perfusate and perfuse the brain, the pulse generator 136 is controlled at the beginning of the perfusion process to gradually increase the pressure and flow rate of the fresh perfusate provided along the organ line 144 in a stepwise manner based on signals received from the second pressure sensor 141 and the flow sensor 142 positioned along the organ line 144. For example, at the beginning of the perfusion process, the electronic pressure regulators 162, 166 of the air supply system 138 are controlled to provide pressurized air to the pulse generator 136 and cause the pulse generator 136 to increase the pressure of the pulsatile flow of the fresh perfusate along the organ line 144 until an initial pressure threshold is detected by the second pressure sensor 141. In some implementation, the initial pressure threshold along the organ line is 16 mmHg. Once the initial pressure threshold along the organ line 144 has been maintained for a predetermined period, the electronic pressure regulator 166 of the air supply system 138 is controlled to cause the pulse generator 136 to increase the pulsatile flow along the organ line 144 until a second pressure threshold is detected along the organ line 144 by the second pressure sensor **141**. In some implementations, the air supply system **138** and the pulse generator **136** are controlled based on signals received from the pressure sensor **141** to increase the pressure along the organ line **144** at a rate of 0.0333 mmHg/minute. By controlling the pulse generation system **1302** to gradually increase the pressure of perfusate provided to the brain **110**, the internal resistance of the brain **110** is correspondingly gradually reduced, which enables the vasculature of the brain **110** to more effectively receive the perfusate while minimizing injury or other damage to the brain **110**.

[0202] As the pressure along the organ line **144** is increased, the flow rate of perfusate along the organ line **144** is simultaneously monitored using the flow sensor **142**, and the pulse generation system **1302** is controlled to continually increase the pressure along the organ line **144** until a predetermined target flow rate is detected along the organ line **144**. In some implementations, the target flow rate is 10 mL/minute. Once the target flow rate is detected by the flow sensor **142**, the pulse generation system **1302** is controlled based on signals generated by the flow sensor **142** in order to provide pulsatile flow along the organ line **144** sufficient to maintain the predetermined flow rate along the organ line **144** as detected by the flow sensor **142**.

[0203] The pulse generation system **1302** is controlled throughout the perfusion period (e.g., 24 hours or longer, 48 hours or longer, etc.) based on the signals generated by the pressure sensor **141** and the flow sensor **142** in order to maintain the predetermined flow rate of perfusate along the organ line **144** throughout the perfusion process while also maintaining the pressure along the organ line **144** within the predetermined pressure boundaries. For example, while perfusate is provided to the brain **110** in pulses generated by the pulse generation system **1302** to maintain a predetermined flow rate, the pressure along the organ line **144** continues to be measured by the pressure sensor **141**. If the pressure detected by the second pressure sensor **141** falls outside of predetermined pressure boundaries, the pulse generation system **1302** is controlled to adjust the pulsatile flow along the organ line **144** is adjusted so that the pressure measured by the second pressure sensor **141** is within the pressure boundaries. Because the pulsatile flow provided by the pulse generator **136** to the brain **110** is controlled in real-time based on the pressure and the flow rate detected by the second pressure sensor **141** and the flow sensor **142**, respectively, the flow of perfusate to the brain **110** is controlled dynamically in response to the real-time changes in internal resistance experienced by the brain **110** throughout the perfusion process. As used herein, a real-time operation may describe an operation that is performed with minimal delay, taking into account the limitations of the computing system (s) performing the operation. For example, in some implementations, the flow sensor **142** is sampled at a rate of 100 Hz, the pressure sensors **140**, **141** are sampled at a rate of 80 Hz, and the pulse generation system **1302** is controlled adjust the flow of perfusate to the brain every one second based on the data received from the flow sensor **142** and the pressure sensor **141**. In some implementations, the flow sensor **130** is sampled at a rate of 100 Hz and the flow pump **114** is controlled every 0.5 seconds based on the data generated by the flow sensor **130**. In some implementations, the sensor probes **128** in the sensor block **124** are each sampled at a rate of 1 Hz, and the operations of the heat exchanger **118** and the gas mixer **120** are each updated every one second based on the data generated by the sensor probes **128**. In some implementations, the resistance valve **152** is controlled at a 10 Hz frequency. The dynamic nature of the perfusate flow provided to the brain **110** by the pulse generation system **1302** ensures that the brain **110** is sufficiently perfused while simultaneously preventing injury to the brain **110**.

[0204] As previously discussed, at least a portion of the fluid flowing through the pulse generator **136** is provided to the sensor block **124** along the sensor block inlet line **146**. The sensor block **124** includes a housing **126** configured to receive and contain fresh perfusate flowing from the pulse generator **136**. The housing **126** of the sensor block **124** is positioned on a sensor block scale **137** that is configured to measure the weight of perfusate contained in the housing **126** of the sensor block **124** throughout the perfusion procedure. The flow rate of perfusate into and out of the sensor block **124**, the volume of perfusate flowing into and out of the sensor block **124**, and the times at which perfusate is added to and drained from the sensor block **124** can be monitored and recorded based on the weight measured by the sensor block scale **137**. In addition, the sensor block scale **137** can be used to validate the time at which samples of perfusate are collected from the sensor block **124** by the automatic sampler system **105**.

[0205] The sensor block **124** also includes a plurality of sensor probes **128** positioned within the housing **126** of the sensor block **124**. The sensor probes **128** are configured to measure certain characteristics of the fresh perfusate inside the housing **126** including, but not limited to, pH, dissolved oxygen concentration, dissolved nitrogen concentration, dissolved carbon dioxide concentration, viscosity, redox potential, and temperature. As previously described, the oxygenator **116**, the heat exchanger **118**, and the gas mixer **120** are controlled based on the characteristics measured by the sensor probes **128** in order to optimize the temperature, O_2 level, and gas composition of the perfusate. In some implementations, an alpha stat approach is used to control the concentration of gasses with the fresh perfusate and the pH of the fresh perfusate.

[0206] The housing **126** of the sensor block **124** is fluidly coupled to the perfusate reservoir **104** along a sensor block outlet line **148** and an arterial sample line **150**, and fresh perfusate flowing through the sensor block **124** is provided to the perfusate reservoir **104** along the sensor block outlet line **148** and/or the arterial sample line **150**. A stopcock **153** is positioned along the arterial sample line **150**, and can be used to sample the perfusate flowing out of the sensor block **124**, for example, to conduct tests on the perfusate exiting the sensor block **124**. Solenoid valves **149**, **151** are positioned along the sensor block outlet line **148** and the arterial sample line **150**, respectively, and are configured to control flow of fluid from the sensor block **124** to the perfusate reservoir **104**. For example, the sensor block **124** includes a level sensor **125** that is configured to detect the level of fluid (e.g., fresh perfusate) inside the housing **126** of the sensor block **124**, and the solenoid valve **149** along the sensor block outlet line **148** is controlled to open or close based on signals generated by the level sensor **125**. In some implementations, the level sensor **125** is an optical level sensor, such as an infrared level sensor. For example, the solenoid valve **149** is controlled to remain in a closed state and prevent flow of fresh perfusate from the sensor block **124** into the perfusate reservoir **104** until the level sensor **125** generates a signal indicating that the fluid level inside the housing **126** of the sensor block **124** has exceeded a threshold fluid level. In response to the level sensor **125** generating a signal indicating that the fluid level inside the housing **126** of the sensor block **124** has exceeded a threshold fluid level, the solenoid valve **149** is controlled to open to allow perfusate to flow from the sensor block **124** along the sensor block outlet line **148** to the perfusate reservoir **104**. Once the level sensor **125** detects that the fluid level inside the housing **126** of the sensor block **124** has fallen below the threshold level, the solenoid valve **149** is controlled to close, which prevents perfusate from flowing from the sensor block **124** to the perfusate reservoir **104** along the sensor block outlet line **148**.

[0207] Referring to FIGS. **1** and **3**, in addition to fluidly coupling the sensor block **124** to the perfusate reservoir **104**, the arterial sample line **150** fluidly couples the sensor block **124** to an automatic sampler system **105**. The automatic sampler system **105** includes a pump **184**, a solenoid manifold **185**, a sample collection line **186**, and a purge line **187**. The automatic sampler system **105** can be operated to automatically collect a sample of perfusate from the sensor block **124**, for example, for testing one or more characteristics of the fresh perfusate contained inside the sensor block **124**. A solenoid valve **171** is positioned along the arterial sample line **150** between the sensor block **124** and the automatic sampler system **105**. In order to automatically collect a sample of perfusate from the arterial sample line **150**, the solenoid valve **171** along the arterial sample line **150** between the sensor block **124** and the automatic sampler system **105** is opened, and the pump **184** of the automatic sampler system **105** is operated to draw perfusate out of the sensor block **124** along the arterial sample line **150**, and the solenoid manifold **185** is controlled to direct the perfusate through the sample collection line **186** and into a sample collection vessel. The automatic sampler system **105** can be controlled to collect samples of fresh perfusate **302** from the sensor block **124** at predetermined intervals during the perfusion process. For example, in some implementations, the automatic sampler system **105** is configured to collect samples of fresh perfusate **302** from the sensor block **124** at the start of a perfusion process ("0 hour"), 6 hours after the start of a perfusion process, 12 hours after the start of a perfusion process, and 24 hours after the start of a perfusion process. Samples can alternatively be collected at any other desired frequency.

[0208] The purge line **187** of the automatic sampler system **105** is used to purge the contents of the arterial sample line **150** and the contents of a corresponding venous sample line **183** prior to collecting a fluid sample from the arterial sample line **150** or the venous sample line **183**, respectively. For example, prior to sampling fluid from the arterial sample line **150**, the solenoid valve **171** along the arterial sample line **150** is opened, the pump **184** and the solenoid manifold **185** of the automated sampler system **105** are controlled to draw the fluid contained inside the arterial sample line **150** through the purge line **187** to a drain or waste container. After a predetermined amount of time has elapsed, the solenoid manifold **185** is operated to direct the fluid flowing along the arterial sample line **150** through the sample collection line **186** and into a sample collection vessel. As a result of this purging process, the perfusate samples **402** collected by the automatic sampler system **105** from the arterial sample line **150** corresponds to fresh perfusate that has exited the sensor block **124** at the time the sample collection is performed, rather than perfusate that was present in the arterial sample line **150** prior to the time of sample collection.

[0209] Still referring to FIG. **1**, a first stopcock **154** is positioned along the organ line **144** upstream of the resistance valve **152** and a second stopcock **155** is positioned along the organ line **144** downstream of the resistance valve **152**. An operator of the system **100** can use the stopcocks **154**, **155** to sample fluid from the organ line **144** upstream or downstream, respectively of the resistance valve **152**, for example, to test one of more properties of the perfusate being provided to the brain **110**. In addition, the stopcocks **154**, **155** can be operated in order to control the flow of air along the organ line during initial set up and calibration of the system **100**. For example, during initial calibration of the pressure sensor **141**, pressurized air is provided to the organ line along the air calibration line **163** through stopcock **155**, and the stopcocks **154**, **155** are both closed to generate a stable air pressure along the organ line **144** between the stopcocks **154**, **155**, and the pressure sensor **141** can be calibrated based on the measured pressure.

[0210] A solenoid valve **159** is positioned along the organ line **144** downstream of the second pressure sensor **141** and the second stopcock **155**. The solenoid **159** is configured to control fluid flow from the organ line **144** into a brain interface **170** inside the brain housing **102**. For example, during perfusion of the brain **110**, the solenoid **159** is opened to allow fluid within the organ line **144** flow to the brain interface **170** inside the brain housing **102**. As will be described in further detail herein, the solenoid **159** is closed during calibration of the pressure sensors **140**, **141** and the flow sensors **142** to prevent fluid along the organ line **144** from flowing into the brain interface **170**.

[0211] A brain housing shunt line **156** extends from the organ line **144** to the brain housing **102** and fluidly couples the organ line **144** to the interior of the brain housing **102**. Referring to FIGS. **10A-10C**, the brain housing **102** includes a fluid coupling **1046** that fluidly connects the shunt line **156** to the interior of the brain chamber **102**. In some implementations, the coupling **1046** is a quick-turn tube coupling. The brain housing shunt line **156** is positioned downstream of the resistance valve **152** and upstream of the second pressure sensor **141**. A solenoid valve **158** is positioned along the brain housing shunt line **156** and can be operated to fluidly couple the organ line **144** to the brain housing **102** along the brain housing shunt line **156**. In some implementations, the solenoid valve **158** along the brain shunt line **156** is opened to allow perfusate to flow from the organ line **144** to the brain housing **102** along the brain housing shunt line **156** during initial calibration of the pressure sensors **140**, **141** and flow sensor **142** along the organ line **144**.

[0212] In some implementations, the organ line **144** is fluidly coupled to the brain **110** positioned in the housing through a brain interface **170** inside the brain housing **102**. The brain interface **170** is a 3D-printed mechanical interface that includes a base plate configured to stabilize and support the brain **110** inside the housing, two fluid inlets, and two corresponding fluid outlets. A y-connector is coupled to an end of the organ line **144**, and each of the fluid inlets of the brain interface **170** are fluidly coupled to the organ line **144** using the y-connector and each inlet of the brain interface **170** receives fresh perfusate flowing out of the organ line **144** into the brain interface **170**. The brain interface **170**

defines flow paths fluidly coupling each of the fluid inlets of the brain interface **170** to a respective outlet of the brain interface **170**. Prior to placing the brain **110** in the brain housing **102** and beginning perfusion of the brain **110**, each of the carotid arteries of the brain **110** is sutured to a respective fluid outlet of the brain interface **170**. Once the carotid arteries of the brain **110** are sutured to the fluid outlets of the brain interface **170**, the inlets of the brain interface **170** are coupled to the organ line **144** using a y-connector and the brain **110** and brain interface **170** are enclosed in the brain housing **102** throughout the perfusion process. As fresh perfusate flows along the organ line **144**, the perfusate enters the fluid inlets of the brain interface **170** and flows through the fluid pathways defined by the brain interface **170**, through the fluid outlets of the brain interface **170**, and into the carotid arteries of the brain **110**.

[0213] The pressure and flow rate of the perfusate provided to the brain **110** are controlled to enable the perfusate entering the carotid arteries of the brain **110** to perfuse the entire vasculature of the brain **110**, including the penetrating arterioles, pre-capillary arterioles, and capillaries of the brain **110**.

[0214] The brain **110** is contained inside the brain housing **102** throughout the perfusion process. FIGS. **10**A-F depict an example brain housing **102**. Referring to FIGS. **10**A and **10**B, the brain housing **102** includes a basin **1022**, a hemispherical brain chamber **1024**, and an upper lid **1026**.

[0215] The lid **1026** is configured to releasably couple to the basin **1022** of the brain housing **102** and can be separated from the basin **1022** to allow a brain **110** to be positioned inside the brain housing **102** on the hemispherical brain chamber **1024**. The lid **1026** includes a pair of handles **1028**, **1030** to enable removal of the lid **1026** from the basin **1022** When the lid **1026** is coupled to the basin **1022**, the basin **1022** and lid **1026** form slots for the tubing of the organ line **144** to enter inside the interior of the brain housing **102** with minimal air gaps between the lid **1026** and the basin **1022**.

[0216] Referring to FIGS. **10**B and **10**C, the hemispherical brain chamber **1024** is configured to be positioned within the basin **1022** and is covered by the upper lid **1026** when the upper lid **1026** is attached to the basin **1022**. During the perfusion process, the brain **110** is positioned on the hemispherical brain chamber **1024** with the carotid and basilar arteries of the brain **110** facing upwards toward the lid **1026**.

[0217] Referring to FIGS. **10**D-**10**F, the hemispherical brain chamber **1024** includes a chamber body **1032** that is configured to receive and support the brain **110** throughout a perfusion process. The chamber body **1032** is formed of a polymer material that provides a soft surface on which the brain **110** can be positioned and supported within the brain housing **102** throughout the perfusion process. The size, shape, and material of the chamber body **1032** are selected to maintain integrity of the brain **110** throughout the perfusion process. The chamber body **1032** defines a plurality of drain holes **1036** through the chamber body **1032**. The drain holes **1036** allow perfusate flowing out of the veins of the brain **110** to flow through the hemispherical brain chamber **1024** into the basin **1022** such that the perfusate flowing out of the brain **110** is collected in the basin **1022**.

[0218] As can be seen in FIGS. **10**C-**10**F, the hemispherical brain chamber **1024** includes an upper lip **1034** extending from the chamber body **1032**. When the hemispherical brain chamber **1024** is positioned within the basin **1022**, the upper lip **1034** of the hemispherical brain chamber **1023** rests on an upper edge of the basin **1022** and allows the hemispherical brain chamber **1024** to be suspended within the basin **1022** without the contacting the bottom surface of the basin **1022** or blocking a level sensor **180** of the brain housing **102**. The upper lip **1034** also defines a plurality of support slots **1040** configured to receive corresponding support structures of the basin **1022** that help support the weight of the brain **110** positioned on the hemispherical brain chamber **1024**.

[0219] Referring to FIGS. **10**D and **10**F, the hemispherical brain chamber **1024** includes loops **1038**A-**1038**D that are coupled to the upper lip **1034** and that can be used to facilitate removal of the hemispherical brain chamber **1024** from the basin **1022**. For example, an operator of the system **100** can grasp one or more of the loops **1038**A-**1038**D with her fingers to lift the hemispherical brain chamber **1024** out of the basin **1022**.

[0220] The brain housing **102** is configured to maintain the brain **110** under normothermic conditions. For example, when the basin **1022** and lid **1026** of the brain housing **102** are securely coupled together, a warm, humid environment is maintained within the brain housing **102**. In some implementations, the brain housing **102** includes temperature and humidity controls configured maintain a particular temperature and humidity level to prevent drying or other injury to the brain **110**.

[0221] After the perfusate flows through the capillaries of the brain **110**, the perfusate flows out of the brain **110** via veins, through the drain holes **1036** in the hemispherical brain chamber **1024**, and into the basin **1022** of the brain housing **102**. The basin **1022** of the brain housing **102** collects and contains the perfusate that flows out of the brain **110** ("used perfusate").

[0222] The brain housing **102** is positioned on a brain housing scale **174** that measures the weight of the brain **110** and the contents inside the housing (e.g., the brain **110** and used perfusate) throughout the perfusion process. Based on the weight measurements generated by the brain housing scale **174**, the amount of swelling of the brain **110** due to fluid retention can be estimated in real-time throughout the perfusion process. In addition to monitoring swelling, the brain housing scale **174** can be used also to calculate and validate the flow measurements generated by the flow sensor **142** along the organ line **144**. For example, the amount of perfusate provided to the brain housing **102** during a particular period of time as determined by the weight measured by the brain housing scale **174** can be compared to the amount of perfusate flow measured by the flow sensor **142** during the particular time period in order to validate the measurements of the flow sensor **142**. In addition, the brain housing scale **174** can be used to validate the time at which samples of perfusate is collected from the brain housing **102** by the automatic sampler system **105**.

[0223] A housing outlet line **176** extends from the brain housing **102** and fluidly couples the brain housing **102** to the perfusate reservoir **104**. A solenoid valve **178** is positioned along the housing outlet line **176** to control the flow of used perfusate from the brain housing **102** into the perfusate reservoir **104**. For example, when the solenoid valve **178** is open, used perfusate collected in the brain housing **102** flows along the housing outlet line **176** into the perfusate reservoir **104**.

[0224] Referring to FIGS. **10**B and **10**C, the brain housing **102** includes a coupling **1042** connected to the side of the basin **1022**. The coupling **1042** fluidly connects the interior of the basin **1022** of the brain chamber **102** to the housing outlet line **176**. In some implementations, the coupling **1042** is a quick-turn tube coupling. In some implementations, the brain housing **102** includes a second coupling **1044** that can be coupled to the outlet line **176** or to another outlet line in order to increase the rate of drainage of fluid out of the brain housing **102** to the perfusate reservoir **104**.

[0225] The brain housing **102** includes a level sensor **180** to detect the level of fluid (e.g., used perfusate) inside the basin **1022** of the brain housing **102**, and the solenoid valve **178** along the housing outlet line **176** is controlled to open or close based on signals generated by the level sensor **180**. As depicted in FIG. **10**C, the level sensor **180** is coupled to the basin **1022** of the brain chamber **102**. In some implementations, the level sensor **180** is coupled to the basin **1022** at the same or similar height along the basin **1022** as the fluid coupling **1046** that fluidly connects the shunt line **156** to the interior of the brain chamber **102**. In some implementations, the level sensor **180** is an optical level sensor, such as an infrared level sensor. For example, the solenoid valve **178** is controlled to remain in a closed state and prevent flow of used perfusate from the brain housing **102** into the perfusate reservoir **104** until the level sensor **180** generates a signal indicating that the fluid level inside the brain housing **102** has exceeded a threshold fluid level. In response to the level sensor **180** generating a signal indicating that the fluid level inside the brain housing **102** has exceeded a threshold fluid level, the solenoid valve **178** is opened to allow used perfusate to flow from the brain housing **102** along the housing outlet line **176** to the perfusate reservoir **104**. Once the level sensor **180** detects that the fluid level inside the brain housing **102** has fallen below the threshold level, the solenoid valve **178** is closed, which prevents fluid from flowing from the brain housing **102** to the perfusate reservoir **104**.

[0226] Two stopcocks **181**, **182** are positioned along the housing outlet line **176** between the brain housing **102** and the perfusate reservoir **104**. An operator of the system **100** can use the stopcocks **181**, **182** to sample the used perfusate flowing out of the brain housing **102**, for example, to test one of more properties of the used perfusate discharged from the brain **110**.

[0227] The stopcock **181** along the housing outlet line **176** fluidly couples the housing outlet line **176** to a venous sample line **183**, which fluidly couples the brain housing **102** to the automatic sampler system **105**. The automatic sampler system **105** can be operated to automatically collect a sample of used perfusate from the brain housing **102**, for example, to test one or more characteristics of the used perfusate. A solenoid valve **173** is positioned along the venous sample line **183** between the brain housing **102** and the automatic sampler system **105**. In order to automatically collect a sample of perfusate from the venous sample line **183**, the stopcock **181** is automatically controlled (e.g., using a servo motor) to fluidly couple the housing outlet line **176** to the venous sample line **183**, the solenoid valve **173** along the venous sample line **183** is opened, and the pump **184** of the automatic sampler system **105** is operated to draw used perfusate out of the brain housing **102** along the venous sample line **183**, and the solenoid manifold **185** is controlled to direct the used perfusate through the sample collection line **186** and into a sample collection vessel. Referring to FIGS. **1** and **3**, the automatic sampler system **105** can be controlled to collect samples of used perfusate **304** from the brain housing **102** at predetermined intervals during the perfusion process. For example, in some implementations, the automatic sampler system **105** is configured to collect samples of used perfusate **304** from the brain housing **102** at the start of a perfusion process ("0 hour"), 6 hours after the start of a perfusion process, 12 hours after the start of a perfusion process, and 24 hours after the start of a perfusion process. Samples can alternatively be collected at any other desired frequency.

[0228] The automatic sampler system **105** purges the contents of the venous sample line **183** prior to collecting a fluid sample from the venous sample line **183**. For example, prior to sampling fluid from the venous sample line **183**, the solenoid valve **173** along the venous sample line **183** is opened, the pump **184** and the solenoid manifold **185** of the automated sampler system **105** are controlled to draw the fluid contained inside the venous sample line **183** through the purge line **187** to a drain or waste container. After a predetermined amount of time has elapsed, the solenoid manifold **185** is operated to direct the fluid flowing along the venous sample line **183** through the sample collection line **186** and into a sample collection vessel. As a result of this purging process, the samples **404** collected by the automatic sampler system **105** from the venous sample line **183** correspond to used perfusate that exited the brain housing **102** at the time of the sample collection, rather than used perfusate that was present in the venous sample line **183** prior to the time of sample collection.

[0229] Still referring to FIG. **1**, the perfusate reservoir **104** includes a reservoir housing **103** that is configured to receive fresh perfusate from the sensor block **124** along the sensor block outlet line **148** and along the arterial sample line **150**, and to receive used perfusate from the brain housing **102** along the housing outlet line **176**. The reservoir housing **103** is a watertight container that can be formed of any suitable material including, but not limited to, plastic, metal, or glass. The perfusate reservoir **104** includes a filter bag **101** positioned within the reservoir housing **103** to filter the fluid flowing into the perfusate reservoir **104**. In some implementations, the filter bag **101** is formed of a 0.22 μm filter.

[0230] In addition to the filter bag **101** positioned inside the reservoir housing **103**, the perfusion system **100** includes an in-line mechanical filtration system **1001** that includes a peristaltic filtration pump **1002**, a 0.22 μm filter **1004** fluidly coupled to the filtration pump **1002** along a filtration line **1006**, and a pressure sensor **1008** positioned along the filtration line **1006** between the filtration pump **1002** and the filter **1004**. In order to further filter the perfusate contained in the perfusate reservoir **104** prior to providing the perfusate to the brain **110**, the filtration pump **1002** can be operated to draw fluid out of the perfusate reservoir **104** along a reservoir outlet line **179** extending from the perfusate reservoir **104** and along the filtration line **1006**, through the filter **1004**, and back to the perfusate reservoir **104**. The filtration pump **1002** is communicably coupled to the pressure sensor **1008**, and the rate of the filtration pump **1002** is controlled to maintain a predetermined range of pressure along the filtration line **1006**, as detected by the pressure sensor **1008**. In some implementations, the filtration pump **1002** is controlled based on signals from the pressure sensor **1008** to maintain the pressure along the filtration line **1006** within a range of 50 mmHg and 150 mmHg. In some implementations, the filtration pump **1002**

is controlled to generate a particular pressure along the filtration line **1006**, as measured by pressure sensor **1008**, based on the condition of the perfusate and the level of residue in the filter **1004**. As the perfusate flows through the filter **1004** of the perfusate filtration system **1001**, additional particulates and other potential contaminants are removed from the perfusate before returning the perfusate to the perfusate reservoir **104** and flow the perfusate to the brain **110**. The in-line mechanical filtration system **1001** can be operated throughout the process of perfusing the brain **110** to further filter the perfusate any bacteria or contaminates.

[0231] The perfusion system **100** also includes a dialysis filtration system **109** that is configured to further filter the perfusate flowing through the system **100** and remove metabolic waste products or other unwanted substances from the perfusate. The dialysis filtration system **109** includes a dialysis pump **190**, a dialyzer **191**, a dialyzer inlet line **192** coupled to the dialyzer **191**, and a dialyzer outlet line **193** coupled to the dialyzer **191**. The dialyzer **191** includes a membrane configured filter waste particles out of the perfusate. The dialysis pump **190** is a peristaltic pump that is controlled throughout the perfusion process to draw perfusate from the perfusate reservoir **104** along the dialyzer inlet line **192** and into the dialyzer **191** at a controlled pressure. As the perfusate flows through the dialyzer **191**, waste particles within the perfusate are filtered out of the perfusate by the membrane of the dialyzer **191** and the filtered perfusate exits the dialyzer **191** along the dialyzer outlet line **193**. A stopcock **1912** is positioned along the dialyzer outlet line **193**. The stopcock **1912** can be used to sample perfusate fluid flowing out of the dialyzer **191** to test one of more properties of the perfusate being provided to the perfusate reservoir **104**, for example, to test the effectiveness of the dialysis filtration system **109**.

[0232] The dialysis filtration system **109** includes a first pressure sensor **194** positioned upstream of the dialyzer **191** and a second pressure sensor **195** positioned downstream of the dialyzer **191**, and the rate of the dialysis pump **190** is controlled based on pressure signals generated by pressure sensors **194**, **195** in order to control the pressure within the dialyzer **191** to provide efficient filtration across the membrane of the dialyzer **191**. For example, the pressure drop across the dialyzer **191** can be determined based on the difference between the pressure upstream of the dialyzer **191** measured by the first sensor **194** and the pressure downstream of the dialyzer **191** measured by the second sensor **195**, and the pump rate of the dialysis pump **190** can be controlled in order to maintain a pressure drop across the dialyzer **191** within a predetermined range that is sufficient to enable efficient waste removal and nutrient exchange within the dialyzer **191**. The dialysis filtration system **109** also includes a resistance valve **196** positioned along the dialyzer outlet line **193** between the pressure sensor **195** and the perfusate reservoir **104**. The resistance valve **196** is a pinch valve that is controlled based on the pressure signals generated by pressure sensors **194**, **195** along the dialyzer inlet and outlet lines **192**, **193**, respectively. For example, the resistance valve **196** can be controlled to increase the pressure along the dialyzer outlet line **193** in response to determining, based on the difference between the pressure signals generated by pressure sensors **194**, **195**, that the pressure within the dialyzer **191** is below a threshold pressure.

[0233] As filtered perfusate flows back to the perfusate reservoir **104** along the dialyzer outlet line **193**, the waste filtered out of the perfusate flows along the waste line **197** to the waste fluid receptacle **198** positioned on a waste scale **199**. As the waste flows into the waste fluid receptacle **198**, the waste scale **199** measures, in real-time, the weight of waste contained within the waste fluid receptacle **198**.

[0234] A resistance valve **1902** is positioned along waste line **197** and is configured to control the rate of flow of waste between the dialyzer **191** and the waste fluid receptacle **198**. For example, the resistance valve **1902** can be a pinch valve and, based on signals received from the waste scale **199**, the resistance valve **1902** can be adjusted to increase or decrease the flow rate of waste into the waste fluid receptacle **198** in order to maintain a particular rate of waste entering the waste fluid receptacle **198**. In some implementations, the resistance valve **1902** is controlled to cause waste to exit the dialyzer **191** and enter the waste fluid receptacle **198** at a rate of **1** gram/minute, as detected by the waste scale **199**. In addition, the resistance valve **1902** can be controlled based on the pressure signals generated by pressure sensors **194**, **195** along the dialyzer inlet and outlet lines **192**, **193**, respectively. For example, the resistance valve **1902** can be controlled to increase the pressure along the waste line **197** in response to determining, based on the difference between the pressure signals generated by pressure sensors **194**, **195**, that the pressure within the dialyzer **191** is below a threshold pressure.

[0235] The dialysis filtration system **109** also includes a fresh perfusate reservoir **1904** that contains a supply of perfusate solution and a fresh perfusate pump **1906** that is configured to pump fresh perfusate fluid contained within the fresh perfusate reservoir **1904** into the perfusate reservoir **104** along a fresh perfusate line **1908**. The fresh perfusate reservoir **1904** is positioned on a fresh perfusate scale **1910** that is configured to measure the weight of perfusate contained within the fresh perfusate reservoir **1904**. The fresh perfusate pump **1906** is controlled to pump fresh perfusate from the fresh perfusate reservoir **1904** into the perfusate reservoir **104** based on signals received from waste scale **199** and signals received from the fresh perfusate scale **1910**. For example, based on the signals received from the waste scale **199** indicating an amount of fluid removed from the dialyzer **191** into the waste fluid receptacle **198**, the fresh perfusate pump **1906** can be controlled to provide a corresponding amount of fresh perfusate from the fresh perfusate reservoir **1904** into the perfusate reservoir **104**, as determined based on the signals received from the fresh perfusate scale **1910**. In some implementations, the fresh perfusate pump **1906** is controlled to pump fresh perfusate from the fresh perfusate reservoir **1904** to the perfusate reservoir **104** at the same rate that waste fluid is provided from the dialyzer **191** to the waste fluid receptacle **198**, as determined based on signals generated by the waste scale **199** and the fresh perfusate scale **1910**.

[0236] As the brain is perfused with the perfusate, homeostasis and cellular activity of the brain **110** are maintained throughout the length of the perfusion process. In some implementations, the perfusion of the brain can be used to maintain homeostasis and cellular activity of the brain 24 hours or longer (e.g., 24-48 hours, 48-72 hours, 48-96 hours, 96 hours or longer, etc.) post-mortem. As a result of the restored metabolic and cellular activity provided by the perfusion of the brain, chemical exchanges and reactions occur within the brain **110** throughout the perfusion process, and these reactions can be monitored in real-time using the perfusion system **100**. For example, as depicted in FIG. **1**, the perfusion system **100** includes an in-line Raman spectroscopy system **107** (also referred to herein as "spectroscopy system **107**") that is configured to perform Raman spectroscopy on perfusate flowing through the system **100** and a Raman spectroscopy probe **172** coupled to or near the brain housing **102** for performing Raman spectroscopy on the tissue of the brain **110** in real-time during perfusion of the brain **110**.

[0237] The in-line Raman spectroscopy system **107** includes a pump **111**, a Raman chamber **117**, a Raman sample line **123** fluidly coupling the pump **111** to the Raman chamber **117**, and Raman spectrometer **119** configured to perform Raman spectroscopy on perfusate contained inside the Raman chamber **117**. A stopcock **127** is positioned along the Raman sample line **123** upstream of the pump **111**. The stopcock **127** can be used to sample the perfusate flowing to the Raman chamber **117**, for example, in order to conduct tests on the perfusate flowing to the Raman chamber **117** to calibrate the Raman spectrometer **119**.

[0238] In order to accurately perform Raman spectroscopy, the perfusate in the Raman chamber is maintained within a predetermined temperature range that is suitable for Raman spectroscopy. As depicted in FIG. **1**, the Raman spectroscopy system **107** includes a heat exchanger **115** upstream of the Raman chamber **117**, a first temperature sensor **113** positioned upstream of the Raman chamber **117** and the heat exchanger **115**, and a second temperature sensor **121** positioned downstream of the Raman chamber **117** and the heat exchanger **115**. As perfusate flows along the Raman sample line **123**, the temperature of the perfusate is measured by the first temperature sensor **113**, the perfusate is provided to the heat exchanger **115**, and the heat exchanger **115** is controlled based on the temperature signals generated by the first temperature sensor **113** to adjust the temperature of the perfusate provided to the Raman chamber **117** to be within the predetermined range. For example, the heat exchanger **115** can be configured to continually adjust (e.g., raise or lower) the temperature of the perfusate flowing through the Raman sample line **123** until the temperature of the perfusate downstream of the Raman chamber **117** measured by the second temperature sensor **121** is within the predetermined temperature range. In some implementations, the heat exchanger **115** is configured to adjust the temperature of the perfusate provided to the Raman chamber **117** to be approximately 26° C.

[0239] The Raman spectroscopy system **107** can be used to perform spectroscopy on both the fresh perfusate contained within the perfusate reservoir **104** and used perfusate flowing out of the brain housing **102**. For example, as depicted in FIG. **1**, the reservoir outlet line **179** that extends from the perfusate reservoir **104** is fluidly coupled to the Raman sample line **123** and, in order to perform Raman spectroscopy on the perfusate contained within the perfusate reservoir **104**, a solenoid valve **177** along the Raman sample line **123** downstream of the Raman chamber **117** is closed, the stopcock **127** is controlled (e.g., using a servo-motor) to fluidly couple the reservoir outlet line **179** to the Raman chamber **117**, a solenoid valve **129** along the reservoir outlet line **179** between the perfusate reservoir **104** and the Raman chamber **117** is opened, and the pump **111** is operated to draw perfusate from the perfusate reservoir **104**, along the reservoir outlet line **179** and the Raman sample line **123**, through the heat exchanger **115**, and into the Raman chamber **117**. In order to perform Raman spectroscopy on the used perfusate flowing out of the brain **110**, the stopcock **182**, **127** are controlled (e.g., using a servo-motor) to fluidly couple the housing outlet line **176** to the Raman chamber **117**, a solenoid valve **139** along the housing outlet line **176** between the brain housing **102** and the Raman chamber **117** is opened, and the pump **111** is operated to draw used perfusate along the housing outlet line **175** and the Raman sample line **123**, through the heat exchanger **115**, and into the Raman chamber **117**. Once the Raman chamber **117** is filled with perfusate, the pump **111** is stopped and the Raman spectrometer **119** is controlled to perform Raman spectroscopy analysis on the perfusate to analyze the chemical composition of the perfusate and identify one or more molecules present in the perfusate. Once the Raman spectrometer **119** has completed performing spectroscopy on the perfusate sample inside the Raman chamber **117**, the solenoid valve **177** along the Raman sample line **123** downstream of the Raman chamber **117** is opened and the pump **111** is controlled to pump the perfusate inside the Raman chamber **117** along the Raman sample line **123** to the perfusate reservoir **104**.

[0240] By performing Raman spectroscopy on the fresh perfusate inside the perfusate reservoir **104**, the chemical composition and proportions of components of the perfusate can be confirmed and adjusted, if needed. In addition, by performing Raman spectroscopy on the used perfusate flowing out of the brain **110**, the uptake of chemical compounds by the brain **110** from the perfusate can be determined. For example, the chemical composition of the fresh perfusate provided to the brain **110** is predefined and can be confirmed by performing Raman spectroscopy on the fresh perfusate in the perfusate reservoir **104** using the spectroscopy system **107**, as described above. The spectroscopy system **107** can then be used to determine the chemical composition of the used perfusate flowing out of the brain **110**, as described above. The known composition of the fresh perfusate provided to the brain **110** can be compared to the chemical composition of the used perfusate flowing out of the brain **110**, as determined using the spectroscopy system **107**, in order to identify any differences in chemical composition in the fresh perfusate and used perfusate. The identified differences between the fresh perfusate and used perfusate can be used to determine various metabolic processes performed by the brain **110** during perfusion. As an example, the fresh perfusate provided by the brain **110** can include a particular concentration of a particular molecule, and the concentration of the same molecule in the used perfusate flowing out of the brain **110** can be determined using the Raman spectroscopy system **107**. In response to determining that the concentration of the particular molecule is higher in the fresh perfusate than in the used perfusate, it can be determined that the particular molecule crossed the blood-brain barrier and that there has been uptake of the particular molecule by the brain **110**. As depicted in FIG. **4**, the blood-brain barrier selectively regulates passage of molecules **402** from the bloodstream (or perfusate) to the brain **110** through the use of specific proteins, such as transporters and tight junction proteins. In addition, the percentage of uptake of the particular molecule by the brain **110** can be determined by comparing the concentration of the particular molecule in the fresh perfusate and the concentration of the particular molecule in the used perfusate determined using the Raman spectroscopy.

[0241] Further, by comparing the chemical composition of the fresh perfusate provided to the brain **110** with the chemical composition of the used perfusate flowing out of the brain **110**, as determined using the spectroscopy system **107**, metabolic activity of the brain **110** and the health of the brain **110** can be determined in real-time during the perfusion process. For example, based on Raman spectroscopy on the used perfusate using the spectroscopy system **107**, it can be determined that the used perfusate flowing out of the brain **110** includes one or more chemical compounds that are not present in the fresh perfusate used perfusate, which indicates that the brain **110** has reacted to one or more components of the perfusate. In addition, based on performing Raman spectroscopy on the used perfusate using the spectroscopy system **107**. the response of the brain **110** to the perfusate can be determined, and other related side effects of the response of the brain **110** can be determined.

[0242] In addition to determining the response of the brain **110** to perfusate solution in real-time during the perfusion process, the perfusion system **100** can also be used to test the pharmacodynamics of pharmaceutical compounds by the brain **110** in real-time, such as the rate of uptake of the pharmaceutical compounds by the brain and the chemical and metabolic reaction of the brain **110** to the pharmaceutical compounds. For example, referring to FIG. **1**, the perfusion system **100** includes a syringe pump manifold **167** that is configured to support one or more syringes and a stepper motor that is configured to control movement of a respective plunger of each of the syringes contained within the manifold **167**. The syringe pump manifold **167** is fluidly coupled to the perfusate reservoir **104** along a drug input line **169**. Each of the syringes contained within the syringe pump manifold are filled with a pharmaceutical compound, and the syringe pump manifold **167** is controlled throughout the perfusion process to add the pharmaceutical compound to the fresh perfusate in the perfusate reservoir **104** at a predetermined rate. For example, the syringe pump manifold **167** can be controlled to add a pharmaceutical compound into the perfusate at a rate that corresponds to the release profile for oral administration or intravenous administration of the pharmaceutical compound to a human.

[0243] The syringe pump manifold **167** is communicably coupled to the Raman spectrometer **119**, and the rate of release of the pharmaceutical compound by the syringe pump manifold into the perfusate in the perfusate reservoir **104** can be controlled based on Raman spectroscopy analysis of the perfusate in the reservoir **104** by the spectroscopy system **107**. For example, as previously discussed, the Raman spectroscopy system **107** can be used to perform Raman spectroscopy of the fresh perfusate in the perfusate reservoir **104**. As the syringe pump manifold **167** operates to deliver a pharmaceutical compound into the perfusate inside the perfusate reservoir **104**, the concentration of the pharmaceutical compound in the fresh perfusate can be determined in real-time by performing Raman spectroscopy using the Raman spectroscopy system **107** on perfusate sampled from the reservoir **104**. Based on the concentration of the pharmaceutical compound in the fresh perfusate detected by the Raman spectroscopy system **107**, the syringe pump manifold **167** can be controlled to increase, decrease, or maintain the rate the pharmaceutical compound is provided to the perfusate reservoir **104** in order to align the concentration of the pharmaceutical compound within the perfusate with the desired release profile in real-time.

[0244] In addition, the dialysis filtration system **109** can be used in conjunction with the Raman spectroscopy system **107** to provide a desired release profile of the pharmaceutical compound to the brain **110**. As previously discussed, the perfusate system **100** is a closed-loop system. Therefore, any pharmaceutical compound added to the perfusate that is not metabolized by the brain **110** is recirculated within the perfusate. However, the release profile for many pharmaceutical compounds increases to a peak concentration, and then subsequently decreases. Therefore, in order to simulate certain release profiles, the concentration of the pharmaceutical compound within the perfusate needs to be reduced following a peak concentration. The dialysis filtration system **109** can be operated to remove the pharmaceutical compound from the perfusate at a controlled rate in order to cause the concentration of the pharmaceutical compound within the perfusate to decrease corresponding to the desired release profile. For example, as previously discussed, the concentration of the pharmaceutical compound in perfusate contained in the perfusate reservoir **104** can be determined in real-time by performing Raman spectroscopy using the Raman spectroscopy system **107** on perfusate sampled from the reservoir **104**. Based on the concentration of the pharmaceutical compound in the fresh perfusate detected by the Raman spectroscopy system **107**, the dialysis filtration system **109** can be controlled to decrease the concentration of the pharmaceutical compound in the perfusate at a predetermined rate to align the concentration of the pharmaceutical compound with the desired release profile for the pharmaceutical compound. For example, based on the concentration of the pharmaceutical compound detected in the fresh perfusate by the Raman spectroscopy system **107**, the dialysis pump **190** can be controlled to pump perfusate from the perfusate reservoir **104** along the dialyzer inlet line **192** to the dialyzer **191** and the dialyzer **191** removes at least a portion of the pharmaceutical compound from the perfusate. The perfusate with a reduced concentration of the pharmaceutical compound exits the dialyzer **191** and is pumped back to the perfusate reservoir **104** along the dialyzer outlet line **193**. The dialysis pump **190** can be controlled to continue flowing perfusate through dialyzer **191** until the Raman spectroscopy system **107** detects that the concentration of the pharmaceutical compound in the perfusate within the perfusate reservoir **104** is below a concentration corresponding to the desired release profile. In some implementations, the flow rate of the dialysis pump **190** is controlled in real time based on the level of pharmaceutical compound within the perfusate detected by the Raman spectroscopy system **107**. In some implementations, the dialysis pump **190** is controlled to flow perfusate from the perfusate reservoir **104** at a predetermined rate and for a predetermined duration in order to remove the pharmaceutical compound at a rate corresponding to the desired release profile.

[0245] FIG. **5** depicts Raman spectroscopy data **500** generated during example perfusion procedures in which adenosine was added to the perfusate during the perfusion procedure. During a first perfusion procedure, the syringe pump manifold **167** and dialysis filtration system **109** were controlled to adjust the concentration of adenosine within the perfusate in the perfusate reservoir **104** to correspond to a typical release profile **502** for oral administration of adenosine. As can be seen in FIG. **5**, the oral administration release profile **502** defines an initial increase in concentration of adenosine until the concentration of adenosine reaches a peak concentration **504**, followed by a decrease in concentration of adenosine. In order to administer adenosine to a brain being perfused by the system **100** according to the oral release profile **502**, the syringe pump manifold **167** was initially controlled to provide adenosine to the perfusate in the perfusate reservoir **104**. As the syringe pump manifold **167** operated to add adenosine to the perfusate in the perfusate reservoir **104**, samples of the perfusate from the perfusate reservoir **104** were periodically tested using the Raman spectroscopy system **107**, as described above. Based on the Raman spectroscopy data **500**, the concentration **502** of adenosine within the fresh perfusate in the perfusate reservoir **104** was tracked in real-time and the rate of the syringe pump manifold **167** was periodically adjusted to reduce the rate at which the concentration of adenosine increased, in accordance with the portion of the release profile **502** before the peak concentration **504**. Once the concentration of adenosine in the perfusate in the perfusate reservoir **104** was identified as being equal to the peak concentration **504**, as determined based on the Raman spectroscopy data **500** generated by performing Raman spectroscopy system **107**, the dialysis filtration system **109** was operated to remove adenosine from the perfusate at a controlled rate to gradually decrease the concentration of adenosine in the perfusate, as depicted in the portion of the release profile **502** after the peak concentration **504**. As the dialysis filtration system **109** was operated to remove adenosine from the perfusate, samples of the perfusate from the perfusate reservoir **104** were periodically tested using the Raman spectroscopy system **107**, as described above. Based on the Raman spectroscopy data **500**, the concentration **502** of adenosine within the fresh perfusate in the perfusate reservoir **104** was tracked in real-time and the dialysis filtration system **109** was controlled to progressively reduce the rate at which the concentration of adenosine within the perfusate decreased, as depicted in the portion of the release profile **502** after the peak concentration **504**.

[0246] During a second perfusion procedure on a second brain, the syringe pump manifold **167** and dialysis filtration system **109** were controlled to cause the concentration of adenosine within the perfusate in the perfusate reservoir **104** to correspond to a typical release profile **506** for intravenous (IV) administration of adenosine. As can be seen in FIG. **5**, the IV administration release profile **506** defines a peak initial increase concentration **508** of adenosine that decrease exponentially. In order to administer adenosine according to the IV release profile **506** to a brain being perfused by the system **100**, the syringe pump manifold **167** was controlled to provide an initial bolus of adenosine into the perfusate in the perfusate reservoir **104** to achieve the initial peak concentration **508** of adenosine in the perfusate. The initial peak concentration **508** of adenosine in the perfusate was confirmed by performing Raman spectroscopy on a sample of perfusate from the perfusate reservoir **104**, as described above. Once the initial peak concentration **508** was provided to the perfusate by the syringe pump manifold **167**, as confirmed based on the Raman spectroscopy data **500** generated by the Raman spectroscopy system **107**, the dialysis filtration system **109** was operated to remove adenosine from the perfusate at a controlled rate to exponentially decrease the concentration of adenosine in the perfusate in accordance with the IV release profile **506**. As the dialysis filtration system **109** was operated to remove adenosine from the perfusate, samples of the perfusate from the perfusate reservoir **104** were periodically tested using the Raman spectroscopy system **107**, as described above. Based on the Raman spectroscopy data **500**, the concentration **506** of adenosine within the fresh perfusate in the perfusate reservoir **104** was tracked in real-time and the dialysis filtration system **109** was dynamically controlled to exponentially reduce the rate at which the concentration of adenosine within the perfusate in accordance with the IV release profile **506**.

[0247] Referring back to FIG. **1**, as the pharmaceutical compound flows from the one or more syringes in the syringe pump manifold **167** into the perfusate in the perfusate reservoir **104**, the perfusate with the pharmaceutical compound is simultaneously provided to the brain **110** from the perfusate reservoir **104** along the organ line **144** and the used perfusate exits the brain **110** into the basin **1022** of the brain housing **102**, as previously described. The Raman spectroscopy system **107** can be controlled to perform Raman spectroscopy on the used perfusate flowing out of the brain housing **102** in order to determine the effects of the pharmaceutical compound on the brain **110**. For example, at predetermined intervals throughout the perfusion process, the solenoid valve **139** along the housing outlet line **176** between the brain housing **102** and the Raman chamber **117** is opened, and the pump **111** is operated to draw used perfusate along the housing outlet line **175** and the Raman sample line **123**, through the heat exchanger **115**, and into the Raman chamber **117**. As used perfusate flows through the Raman chamber **117**, the Raman spectrometer **119** is controlled to perform Raman spectroscopy analysis on the used perfusate to analyze the chemical composition of the used perfusate. The effect of the pharmaceutical compound on the brain **110** can be determined based on the chemical composition of the used perfusate determined by the Raman spectroscopy system **107**. For example, the spectroscopy analysis of the used perfusate can be used to determine the concentration of the pharmaceutical compound present in the used perfusate. In response to detecting that the concentration of pharmaceutical compound in the used perfusate flowing out of the brain **110** is lower than the concentration of the pharmaceutical compound in the fresh perfusate in the perfusate reservoir **104** at a corresponding time, it can be determined that the brain has metabolized (e.g., "taken up") at least a portion of the pharmaceutical compound. As a result, the concentration of the pharmaceutical compound in the used perfusate detected by the Raman spectroscopy system **107** can be used to determine whether the pharmaceutical compound is capable of crossing the blood-brain barrier and the rate at which the brain **110** has metabolized the pharmaceutical compound.

[0248] The used perfusate flowing out of the brain **110** can also be analyzed by the Raman spectroscopy system **107** to determine one or more molecules generated by the brain **110** in response to the to the pharmaceutical compound. For example, prior to adding a pharmaceutical compound to the perfusate, the chemical composition of the used perfusate flowing out of the brain **110** can be analyzed by the Raman spectroscopy system **107** in order to form a baseline composition. After the pharmaceutical compound has been introduced to the perfusate and flowed to the brain **110**, the chemical composition of the used perfusate flowing out of the brain **110** can be analyzed by the Raman spectroscopy system **107** and compared to the baseline chemical composition for the used perfusate to determine one or more molecules that are present in the used perfusate flowing out of the brain **110** following the introduction of the pharmaceutical compound that were not present prior to the introduction of the pharmaceutical compound. The molecules produced by the brain **110** in response to the introduction of the pharmaceutical compound and detected in the used perfusate by the spectroscopy system **107** can be used to determine the reaction of the brain **110** to the pharmaceutical compound, the likely efficacy of the pharmaceutical compound on treating the targeted disease, and potential side effects that may result from the pharmaceutical compound. As a result, potential new drugs can be tested using the perfusate system **100** to accurately determine their likely effectiveness in treating the target disease in humans.

[0249] As previously discussed, the perfusion system **100** also includes a Raman spectroscopy probe **172** that is coupled to or positioned near the brain housing **102** for performing Raman spectroscopy on the tissue of the brain **110** in real-time during perfusion of the brain **110**. Once the brain **110** is fluidly coupled to the organ line **144** and is positioned inside the brain housing **102**, the Raman spectroscopy probe **172** is positioned on the surface of the brain **110** and can be operated periodically throughout the perfusion process to perform Raman spectroscopy on the tissue of the brain **110** in real-time as the brain is perfused. Referring to FIGS. **11A** and **15**, in some implementations, the brain housing **102** includes a lid **1106** removably coupled to a basin **1104** and, in order to position the Raman spectroscopy probe **172** on or near the surface of the brain **110**, a lid **1106** of the brain housing **102** is removed from the basin **1104**. In some implementations, the Raman spectroscopy probe **172** is incorporated into the brain housing **102** (e.g., into a lid **1106** of the housing **102**) and the housing **102** can remain closed while performing Raman spectroscopy on the brain **110** inside the housing **102**.

[0250] Similar to the Raman spectroscopy data generated for the perfusate by the Raman spectroscopy system **107**, the data generated by the Raman spectroscopy probe **172** can be used to determine the metabolic activity of the brain **110** and chemical compounds generated by the brain **110** in real-time throughout the perfusion process. For example, at the beginning of the perfusion process, the Raman spectroscopy probe **172** can be controlled to perform Raman spectroscopy on the brain **110** to identify molecules and chemical compounds present in the tissue of the brain **110** in order to form a baseline spectroscopy reading for the brain tissue. After the brain **110** has been perfused with the perfusate for a predetermined period, the Raman spectroscopy probe **172** can be controlled to perform Raman spectroscopy on the brain **110** and the results can be compared to the baseline spectroscopy reading for the brain tissue to determine changes in the types and concentrations of molecules present in the tissue of the brain **110** compared to the baseline. These changes in the molecular composition of the brain tissue in response to perfusion of the brain **110** detected by the Raman spectroscopy probe **172** can be used to determine the effectiveness of the perfusion process, as well as predict the length of time the brain **110** can be effectively perfused, for example, before the brain **110** undergoes significant swelling.

[0251] Similar to the Raman spectroscopy system **107**, the Raman spectroscopy probe **172** can also be used to determine the effect of a pharmaceutical compound on the brain **110**. As discussed above, the syringe pump manifold **167** and the dialysis filtration system **109** can be operated to provide a pharmaceutical compound to the brain **110** through the perfusate at a rate corresponding to a particular release profile. As the pharmaceutical compound is simultaneously provided to the brain **110** from the perfusate reservoir **104**, the Raman spectroscopy probe **172** is controlled to perform Raman spectroscopy on the tissue of the brain **110** at predetermined intervals as the pharmaceutical compound is provided to the brain **110**.

[0252] The data generated by the Raman spectroscopy probe **172** can be used to determine whether the pharmaceutical compound is capable of crossing the blood-brain barrier. The pharmaceutical compound can be determined as capable of crossing the blood-brain barrier in response to detecting that one or more chemical compounds corresponding to the pharmaceutical compound are present in the tissue of the brain **110** using the Raman spectroscopy probe **172**. In addition, the percentage of the pharmaceutical compound taken up and metabolized by the brain **110** can be determined based on the concentration of one or more chemical compounds corresponding to the pharmaceutical compound that is detected in the tissue of the brain **110** by the Raman spectroscopy probe **172**.

[0253] In addition, the Raman spectroscopy probe **172** can be used to determine one or more chemical compounds generated by the brain **110** in response to administration of the pharmaceutical compound. For example, the Raman spectroscopy probe **172** can be controlled to perform Raman spectroscopy on the tissue of the brain **110** at predetermined intervals as a pharmaceutical compound is provided to the brain **110** through the perfusate. The results can be compared to a spectroscopy analysis of the tissue of the brain **110** taken prior to introduction of the pharmaceutical compound to the perfusate in order to determine changes in the types and concentrations of molecules present in the tissue of the brain **110** resulting from the introduction of the pharmaceutical compound. These changes in the molecular composition of the brain tissue in response to introduction of the pharmaceutical compound that are detected by the Raman spectroscopy probe **172** can be used to determine the reaction of the brain **110** to the pharmaceutical compound. For example, the molecules produced by the brain **110** in response to the introduction of the pharmaceutical compound into the perfusate and detected in the brain tissue by the Raman spectroscopy probe **172** can be used to determine the reaction of the brain **110** to the pharmaceutical compound, the likely efficacy of the pharmaceutical compound on treating the targeted disease, and potential side effects that may result from the pharmaceutical compound. As a result, potential new drugs can be tested using the perfusate system **100** to accurately determine their likely effectiveness in treating the target disease in humans.

[0254] Using the Raman spectroscopy probe **172** and the Raman spectroscopy system **107** to take Raman spectroscopy measurements on the tissue of the brain **110** and the used perfusate flowing out of the brain **110**, respectively, in real-time throughout the perfusion process allows for real-time analysis of the chemical processes occurring in the brain **110** during perfusion without requiring samples of the brain tissue to be removed during the perfusion process. As a result, the sterility and functionality of the brain **110** can be more closely regulated and maintained, which improves and prolongs the perfusion process.

[0255] In addition, the data generated by performing Raman spectroscopy on the brain tissue using probe **172** and on the used perfusate using the Raman spectroscopy system **107** can be provided to a trained machine learning model in order to predict one or more characteristics about the brain **110**. For example, by aggregating the Raman spectroscopy data collected throughout numerous perfusion processes on respective brains, a machine learning model can be trained to identify Raman spectroscopy data corresponding to particular responses by a human brain to a particular compound. Once the machine learning model has been trained, the Raman spectroscopy data generated by the sensor probe **172** and/or spectroscopy system **107** following introduction of a pharmaceutical compound into the perfusate can be provided to the trained machine learning model to predict certain responses or reactions relating to the administration of the pharmaceutical compound, such as a metabolic pathway of the pharmaceutical compound in the brain, an activation or deactivation of certain gene pathways in the brain, and/or physiological changes to the brain (e.g., swelling, tissue damage, infection, etc.). For example, the Raman spectroscopy data generated by the sensor probe **172** and/or spectroscopy system **107** indicating the changes in chemical bonds within the brain can be provided to a trained machine learning model, which can utilize the Raman spectroscopy data to determine the likelihood that the brain is currently experiencing swelling or will experience swelling during the perfusion procedure.

[0256] The machine learning model can be implemented as any appropriate type of machine learning model, e.g., including one or more of: a neural network, or a random forest, or a support vector machine, or a linear regression model. Further, the machine learning model can have any appropriate architecture that enables the machine learning model to perform its described functions, e.g., processing a model input that is based on Raman spectroscopy of brain tissue or used perfusate that has passed through brain tissue to generate a prediction characterizing the brain. For example, for a machine learning model implemented as a neural network model, the architecture of the neural network can include any appropriate types of neural network layers (e.g., convolutional layers, attention layers, fully connected layers, recurrent layers, message passing layers, pooling layers, transformer layers, and so forth) in any appropriate number (e.g., 3 layers, or 10 layers, or 25 layers) and connected in any appropriate configuration (e.g., as a directed graph of layers).

[0257] The model input provided to the machine learning model can represent the Raman spectroscopy data as an ordered collection of numerical values, e.g., a vector, matrix, or other tensor of numerical values. For instance, the Raman spectroscopy data can include a wavelength axis, representing the energy shift from the laser light and an intensity axis representing the amount of scattered light at each energy level. The model input can include Raman spectroscopy data captured at one time point, or at a sequence of multiple time points, e.g., that are spaced at regular intervals in a time window. The model output generated by the machine learning model can be, e.g., a classification output (e.g., that classifies the model input into a category from a predefined set of categories), or a regression output (e.g., that predicts a continuous value based on the model input), or any other appropriate type of model output.

[0258] The machine learning model can be trained on a set of training examples by a machine learning training technique. Each training example can include: (i) a training input that includes Raman spectroscopy data generated by performing Raman spectroscopy on brain tissue of a brain or on used perfusate that has passed through the brain tissue of the brain, and (ii) a target output defining a model output that should be generated by the machine learning model by processing the training input. The target output can characterize one or more characteristics of the brain, e.g., a metabolic pathway of a pharmaceutical compound that has been provided to the brain (e.g., in perfusate passed through the brain), an activation or deactivation of certain gene pathways in the brain, or physiological changes to the brain (e.g., swelling, tissue damage, infection, etc.). The target outputs included in the training examples may be generated based on experimental measurements or observations of brains during previous perfusion experiments.

[0259] Training the machine learning model on the set of training examples can include, for each training example, training the machine learning model to reduce a discrepancy between: (i) a predicted output generated by processing the training input of the training example using the machine learning model, and (ii) the target output specified by the training example. More specifically, the machine learning model can be trained to optimize an objective function that, for each training example, measures an error between: (i) a predicted output generated by processing the training input of the training example using the machine learning model, and (ii) the target output specified by the training example. The objective function can measure the error, e.g., as a cross-entropy error, or as a squared error, or in any other appropriate way.

[0260] FIG. **6** depicts Raman spectroscopy data **500** generated during an example perfusion procedure in which the drug rapamycin was provided to the brain **110**. During perfusion of the brain **110** using the perfusion system **100**, the syringe pump manifold **167** was controlled to provide rapamycin to the perfusate in the perfusate reservoir **104** at a particular rate, which was then provided to the brain **110** along the organ line **144**. As the perfusate containing the rapamycin was provided to the brain **110**, Raman spectroscopy was performed on the tissue of the dorsomedial prefrontal cortex (dmPFC) of the brain **110**. As can be seen from the Raman spectroscopy data **600** in FIG. **6**, the concentration **602** of rapamycin present in the brain **110** continually increased until a stable concentration **604** was reached. Based on the Raman spectroscopy data **600** indicating the presence of rapamycin in the tissue of the dmPFC of the brain **110**, it was determined that the rapamycin in the perfusate was able to cross the blood-brain barrier of the brain **110**. In addition, based on the Raman spectroscopy data **600**, the rate of uptake and metabolism of rapamycin by the brain **110** for the particular release profile of rapamycin provided by the perfusate system **100** was determined.

[0261] Referring back to FIG. **1**, the perfusate system **100** also includes a drain line **143** fluidly coupled to a waste container **145** configured to receive perfusate from the venous circuit **108**. The waste container **145** can be, for example, a rigid, liquid tight container formed of glass or plastic. Alternatively, the waste container **145** can be a plastic bag. A solenoid valve **147** is positioned along the drain line **143** upstream of the waste container **145** and is configured to control flow of used perfusate to the waste container **145**. The drain line **143** and the waste container **145** can be used to drain used perfusate from the system **100** without sending the used perfusate back to the perfusate reservoir **104**. To drain perfusate from the system **100**, the solenoid valve **178** along the housing outlet line **176** between the brain housing **102** and the perfusate reservoir **104** is closed, the solenoid valve **139** along the housing outlet line **176** between the brain housing **102** and the Raman chamber **117** is opened, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is closed, the solenoid valve **147** along the drain line **143** is opened, and the pump **111** is operated to pump used perfusate along the housing outlet line **175**, the Raman sample line **123**, and the drain line **143** into the waste container **145**. The waste container **145** is positioned on a waste scale **157** configured to record the weight of perfusate contained in the waste container **145**. In some implementations, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** and the solenoid valve **147** along the drain line **143** are controlled based on signals generated by the waste scale **157**.

[0262] At the beginning of the perfusion process, an initial flush of the brain **110** with perfusate is typically performed in order to remove toxins and other unwanted chemical compounds that have built up in the brain **110** during the post-mortem period prior to perfusion During the initial flush of the brain **110**, the used perfusate flowing out of the brain **110** is sent to the waste container **145** rather than back to the perfusate reservoir **104**. In order to sufficiently remove toxins and other unwanted chemical compounds that have built up in the brain **110** during the post-mortem period prior to perfusion, the brain **110** is flushed with a predetermined amount of perfusate that is not recirculated through the perfusate reservoir **104**. During the initial flush of the perfusate process, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is closed and the solenoid valve **147** along the drain line **143** is open to direct used perfusate flowing out of the brain **110** along the housing outlet line **176** and the Raman sample line **123** to the waste container **145**. Throughout the initial flush of the brain **110**, the waste scale **157** measures the weight of used perfusate inside the waste container **145** in real-time. In response to the waste scale **157** detecting that a predetermined amount of used perfusate has been provided to the waste container **145** during the initial flush, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** and the solenoid valve **178** along the housing outlet line **176** between the brain housing **102** and the perfusate reservoir **104** are each opened and the solenoid valve **147** along the drain line **143** is closed, which directs used perfusate flowing along the housing outlet line **176** and the Raman sample line **123** back to the perfusate reservoir **104**. In some implementations, the brain **110** is initially flushed with 500 mL of perfusate, as detected by the waste scale **157**.

[0263] In some implementations, the drain line **143** and the waste container **145** are used to drain the venous circuit **108** at the end of a perfusion process. For example, at the end of the perfusion process, solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is closed, the solenoid valve **147** along the drain line **143** is opened, and the pump **111** is operated to draw perfusate from the brain housing **102** and the perfusate reservoir **104** along the Raman sample line **123** into the waste container **145** to drain the brain housing **102**, the perfusate reservoir **104**, and the fluid lines of the venous circuit **108** of used perfusate.

[0264] Referring to FIG. **1**, the perfusate system includes a computer system **188** that is communicably coupled to the electrical components of the perfusate system **100** including, but not limited to the pumps **111**, **114**, **190**, **1002**, **1906**, the heat exchangers **115**, **118**, the oxygenator **116**, the gas mixer **120**, the electronic pressure regulators **162**, **166**, the sensor probes **128**, the level sensor **125**, the pressure sensors **140**, **141**, **164**, **168**, **195**, **1008**, the flow sensors **130**, **142**, the temperature sensors **113**, **121**, the resistance valves **152**, **196**, **1902**, the syringe pump manifold **167**, the scales **137**, **157**, **174**, **199**, **1910**, the solenoid valves **129**, **133**, **139**, **147**, **149**, **151**, **158**, **159**, **171**, **173**, **177**, **178**, the Raman spectrometer **119**, and the Raman spectroscopy probe **172**. In some implementations, the computer system **188** can be in wireless or wired communication with the electronic components of the perfusate system **100**. In some implementations, a display **189** of the computing device **188** is configured to display data generated by the perfusate system **100**. For example, sensor data generated by the sensor probes **128** can be transmitted to the computer system **188** in real-time, and the display **189** can be configured to display sensor data received from the sensor probes **128**. In some implementations, the data generated by the perfusate system **100** (e.g., the data sensor data generated by the sensor block **124** and Raman spectroscopy data generated by the Raman spectroscopy system **107** and the Raman spectroscopy probe **172**) is transmitted in real-time to the computer system **188**, and the computer system **188** is configured to transmit the data generated by the perfusate system **100** to a remote database for storage. The computer system **188** is also configured to communicably couple and controls the electronic components of the perfusate system **100** to perform the perfusion process as described herein. For example, the computer system **188** is configured to receive data from the pressure sensor **141** and the flow sensor **142**, and based on the sensor data, control the second electronic pressure regulator **168** to adjust the pulsatile flow provided by the pulse generator **136** along the organ line **144**.

[0265] An example process of calibrating the perfusion system **100** and performing perfusion of a brain **110** using the system **100** will now be described with reference to FIGS. **1**-**2**D and **4**.

[0266] Prior to performing a perfusion procedure, reusable components of the perfusion system **100** that come into contact with the perfusate during a perfusion process are sterilized (e.g., using autoclave) including, but not limited to, the pulse generator **136**, the perfusate reservoir **104**, the sensor block **124** and sensor probes, the brain housing **102**, the Raman chamber **117**, the oxygenator **116**, the pressure sensor **140**, **141**, **194**, **195**, **1008**, and the flow sensors **130**, **142**. After sterilization of the relevant system components is complete, each of the components of the system **100** is positioned inside a clean room that houses the perfusion system **100** throughout the perfusion process. In some implementations, the clean room that houses the perfusion system **100** is provided with a positive pressure system that forces positive pressure into the clean room to prevent contamination of the perfusion system. Once the reusable components of the perfusion system **100** have been sterilized and positioned in the clean room, the various fluid lines **112**,

123, **131**, **144**, **143**, **146**, **148**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908** are attached to the system **100** as depicted in FIG. **1** to form the arterial circuit **106**, the venous circuit **108**, the in-line Raman spectroscopy system **107**, and the dialysis filtration system **109**.

[0267] Once components of the perfusion system **100** are positioned in the clean room and fluidly coupled as depicted in FIG. **1** and prior to adding perfusate to the system, the air supply system **138** and the pressure sensors **140**, **141** along the organ line **144** are calibrated. For example, the resistance valve **152** along the organ line **144** and the solenoid valves **158**, **159** along the brain housing shunt line **156** and the organ line **144**, respectively, are closed, and a flow of pressurized air is provided from the air source **160** to the first electronic pressure regulator **162**. The first electronic pressure regulator **162** is operated, based on pressure signals generated by the first pressure sensor **164**, to regulate the air flow to a pressurized air stream having a pressure of about 5 psi. The pressurized air stream is then provided from the first electronic pressure regulator **164** to the second electronic pressure regulator **166**. As previously discussed, the second pressure regulator **166** has tighter tolerances compared to the first electronic pressure regulator **162** and is configured to provide a pressurized air stream to the air inlet **230** of the pulse generator **136** at a particular pressure (e.g., 5 psi). The smoothed, pressurized air stream flows from the second electronic pressure regulator **166**, through the pulse generator **136**, along the organ line **144**, and to the brain housing **102**. As the pressurized air stream flows from the second electronic pressure regulator **166** along the air supply line **161** and the organ line **144**, the pressure of the air stream is measured by the second pressure sensor **168** along the air supply line **161** and the pressure sensors **140**, **141** along the organ line **144**. The pressure measured by the second pressure sensor **168** along the air supply line **161** is compared to the pressure measured by the pressure sensors **140**, **141** along the organ line **144**, and the pressure sensors **140**, **141** along the organ line **144** are calibrated based on any differences between the pressure measured by the second pressure sensor **168** along the air supply line **161** and the pressure measured by the respective pressure sensor **140**, **141** along the organ line **144**.

[0268] In some implementations, the pressure sensor **141** downstream of the resistance valve **152** is further calibrated using air provided from the second pressure regulator **166** along the air calibration line **163**. For example, pressurized air is provided from the second pressure regulator **166** to the organ line along the air calibration line **163** through stopcock **155**. The stopcock **154** upstream of the pressure sensor **141** and the stopcock **155** downstream of the pressure sensor **141** are each closed to generate a stable air pressure along the organ line **144** between the stopcocks **154**, **155** corresponding to the air pressure provided by the second pressure regulator **166**. Once the stopcocks **154**, **155** are closed, the pressure sensor **141** can be calibrated based on any difference in the pressure measured by the pressure sensor **141** can be compared to the pressure provided by the second pressure regulator **166**,

[0269] Once the air supply system **138** and the pressure sensors **140**, **141** along the organ line **144** are calibrated, fresh perfusate is added to the perfusate reservoir **104**. In some implementations, the perfusate is sterilized using ultraviolet irradiation prior to being provided to the perfusate reservoir **104**. Once the perfusate reservoir **104** is filled with fresh perfusate, the system **100** is controlled to flood the fluid lines **112**, **123**, **131**, **144**, **146**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908** of the arterial circuit **106**, the venous circuit **108**, the Raman spectroscopy system **107**, and the dialysis filtration system **109** with fresh perfusate in order to remove any air contained within the fluid lines. In order to "flood" the system **100** with perfusate and purge air from the fluid lines of the system **100**, the solenoid valves **133**, **158**, **159**, **177**, **129**, **139** of the arterial circuit **106**, the venous circuit **108**, the Raman spectroscopy system **107** are opened, the solenoid valves **149**, **151**, **178** between the sensor block **124** and the brain housing **102** and the perfusate reservoir **104** are closed, and the flow pump **114**, the dialysis pump **190**, and the pump **111** of the Raman spectroscopy system **107** and the pulse generator **136** are each controlled to flow perfusate from the perfusate reservoir **104** through fluid lines **112**, **123**, **131**, **144**, **146**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908** of the arterial circuit **106**, the venous circuit **108**, the Raman spectroscopy system **107**, and the dialysis filtration system **109**, as described in detail above. The perfusate flows through the fluid lines to the sensor block **124** and the brain housing **102**. Once the level sensor **180** in the brain housing **102** detects that the brain housing **102** is filled to a desired level with perfusate, the solenoid valve **178** between the brain housing **102** and the perfusate reservoir **104** opens to allow perfusate to flow from the brain housing **102** to the perfusate reservoir **104**. Similarly, once the level sensor **125** in the sensor block **124** detects that the sensor block **124** is filled to a desired level with perfusate, the solenoid valves **149**, **151** between the sensor block **124** and the perfusate reservoir **104** open to allow perfusate to flow from the sensor block **124** to the perfusate reservoir **104**. As the perfusate flows through the system **100**, the air contained within the fluid lines **112**, **123**, **131**, **144**, **146**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908** is vented out of the fluid lines into a head space at the top of the perfusate reservoir **104**, a head space at the top of the sensor block **124**, and a head space at the top of the brain housing **102**. The pumps **114**, **190**, **111** and pulse generator **136** continue to operate to pump perfusate throughout the system **100** until all of the air is vented out of the fluid lines **112**, **123**, **131**, **144**, **146**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908**. In addition, the air supply system **138** is configured to provide a pulsatile stream of pressurized air to the pulse generator **136** sufficient to cause the pulse generator **136** to force any air contained in the central housing **202** of the pulse generator **136** downstream to the sensor block **124** or the brain housing **102**.

[0270] Once the arterial circuit **106**, the venous circuit **108**, the Raman spectroscopy system **107**, and the dialysis filtration system **109** have been flood with perfusate, the solenoid valves **133**, **158** along the oxygenator shunt line **131** and the brain housing shunt line **156** are closed, and an initial sample of the perfusate is collected using one of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**, **1912**. The initial ("zero hour") sample of the perfusate can be tested using various techniques (e.g., spectroscopy, gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, metabolomic analysis) in order to determine baseline measurements for the perfusate, including the chemical composition of the perfusate. For example, a user can use other laboratory equipment to test the initial perfusate sample collected from the system **100**, for example, to perform gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, metabolomic analysis. In addition, an initial sample of the perfusate can be tested e using the Raman spectroscopy system **107**, as described herein.

[0271] Once the system **100** is flooded with perfusate, the sensor probes **128** in the sensor block **124** are controlled to take initial sensor readings, which are used to calibrate the sensor probes **128**. In some implementations, the sensor probes **128** are calibrated using electronic calibration devices, such as the i-STAT analyzer device manufactured by Abbott.

[0272] Once the system **100** is flooded with perfusate and the sensor probes **128** have been calibrated, the initial conditions for the gas mixer **120** and heat exchanger **118** are set, and the oxygenator **116**, heat exchanger **118**, and gas mixer **120** are operated to optimize the temperature and gas composition of the perfusate for the initial stage of perfusion of a brain. As previously discussed, the heat exchanger **118** and gas mixer **120** are controlled based on signals generated by one or more sensor probes **128** in the sensor block **124**. For example, the heat exchanger **118** is controlled to adjust the temperature of the perfusate to be within a predetermined range of temperature based on the temperature signals received from the sensor block **124**. In addition, the gas mixer **120** is controlled to adjust the amount of oxygen, nitrogen, and/or carbon dioxide in the perfusate within the oxygenator **116** based on one or more signals generated by the sensor block **124** indicating the dissolved oxygen concentration of the perfusate within the sensor block **124**, the dissolved nitrogen concentration of the perfusate within the sensor block **124**, and the dissolved carbon dioxide concentration of the perfusate within the sensor block **124**. In some implementations, the gas mixer **120** is configured to adjust the amount of oxygen, nitrogen, and/or carbon dioxide in the perfusate to be within a predetermined range, as detected by one or more sensor probes **128** in the sensor block.

[0273] Once the system **100** is flooded with perfusate and the temperature and gas composition of the perfusate have been adjusted to the correct initial settings, the system **100** enters a "zero flow stage" in which the pumps **114**, **190**, **111** are controlled to stop pumping the perfusate system **100** and the resistance valve **152** along the organ line **144** is closed for a predetermined amount of time. In some implementations, the system is controlled to be in the zero-flow stage for approximately 5 minutes. While the system is in the zero-flow stage, there is no flow through the flow sensors **130**, **142** along the arterial circuit **106** and the flow sensors **130**, **142** are calibrated.

[0274] Once the flow sensors **130**, **142** along the arterial circuit **106** have been calibrated, the pumps **114**, **190**, **111** are controlled to resume pumping perfusate through the system **100** and the pulse generation system **1302** is controlled to achieve a target pressure and flow rate along the organ line **144**, as described in detail above. While the pumps **114**, **190**, **111** and the pulse generator **136** are operated, initial measurements of one or more characteristics of perfusate can be performed before attaching the brain **110** to the system **100** in order to determine one or more baseline characteristics of the perfusate. These initial readings can include Raman spectroscopy analysis data generated by performing Raman spectroscopy on the perfusate using the Raman spectroscopy system **107**, as described herein. In some implementations, samples of the perfusate **302**, **304** are collected using one or more of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**, **1912**, and the samples of the perfusate **302**, **304** can be tested manually to determine one or more characteristics of the perfusate. In some implementations, the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**, **1912** are manually operated by a user of the system **100** to collect perfusate samples. The initial readings of the perfusate can also include measurements performed by the sensor probes **128** of the sensor block **124**.

[0275] Once the initial measurements of the perfusate characteristics have been performed, the system **100** is controlled to drain the perfusate from the brain housing **102**. In order to drain the brain housing **102**, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is closed and the solenoid valve **147** along the drain line **143** is open to direct perfusate inside the brain housing **102** along the housing outlet line **176** and the Raman sample line **123** into the waste container **145**. The waste scale **157** measures the weight of perfusate inside the waste container **145** in real-time as the brain housing **102** is drained. In response to the waste scale **157** detecting that a predetermined amount of perfusate has been provided to the waste container **145** from the brain housing **102** corresponding to the volume of the brain housing **102**, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is opened and the solenoid valve **147** along the drain line **143** is closed. Once the brain housing **102** is drained of perfusate, the brain housing scale is zeroed out to calibrate the brain housing scale **174**. Once the brain housing **102** is drained and the brain housing scale **174** is calibrated, the flow pump **114** and pulse generator **136** are controlled to pulse perfusate along the organ line **144** to provide a small amount of perfusate to the brain housing **102** prior to connecting the brain **110** to the system.

[0276] The brain **110** is then positioned in the brain housing **102** and fluidly coupled to the system **100** using the brain interface **170**. Prior to placing the brain **110** in the brain housing **102**, each of the carotid arteries of the brain **110** is sutured to a respective fluid outlet of the brain interface **170**. Once the carotid arteries of the brain **110** are sutured to the fluid outlets of the brain interface **170**, the inlet line of the brain interface **170** is coupled to the organ line **144**, and the brain **110** and the brain interface **170** are enclosed in the brain housing **102**.

[0277] Once the brain **110** is positioned in the brain housing **102** and fluidly coupled to the arterial circuit **106** through the brain interface **170**, the flow pump **114** is controlled to flow perfusate through the oxygenator **116** to the pulse generator **136**, and the air supply system **138** and the pulse generator **136** are controlled to pump perfusate to the sensor block **124** and to provide pulsatile flow of perfusate to the brain **110** along the organ line **144**.

[0278] Perfusate flows out of the oxygenator **116** along the arterial fluid line **112** to the pulse generator **136**. As previously described, due to the relative positioning of the pulse generator **136**, the brain housing **102**, and the sensor block **124**, and the structure of the pulse generator **136**, all of the perfusate provided to the pulse generator **136** from the arterial fluid line **112** passively flows through the sensor block inlet line **146** when the diaphragm **214** of the pulse generator **136** is in an unflexed ("neutral") position. As perfusate passes through the sensor block **124**, sensor probes **128** of the sensor block **124** measure one or more characteristic of the fresh perfusate inside the housing **126**, including, but not limited to, pH, dissolved oxygen concentration, dissolved nitrogen concentration, dissolved carbon dioxide concentration, viscosity, and temperature. Based on the characteristics measured by the sensor probes **128**, the oxygenator **116**, heat exchanger **118**, and gas mixer **120** are controlled in order to optimize the temperature, O_2 level, and gas composition of the perfusate.

[0279] As previously discussed, the heat exchanger **118** is configured to adjust the temperature of the perfusate flowing through the heat exchanger **118** based on the temperature of the perfusate detected by the sensor block **124**. At the beginning of a perfusion procedure, the temperature of the perfusate is gradually increased by the heat exchanger **118** until a target temperature for the perfusate is reached, as detected based on the sensor probes **128** in the sensor block **124**. In some implementations, the temperature of the perfusate is increased at a rate of 0.1° C./minute until a target temperature of the perfusate is reached, as detected by the sensor probes **128** of the sensor block **124**. In some implementations, the target temperature is 36° C., and the heat exchanger **118** is controlled to increase the temperature of the perfusate from is 20° C. to is 36° C. over the course of a three hour period.

[0280] In addition, at the beginning of a perfusion procedure, the O_2 concentration of the perfusate is gradually increased by the gas mixer **120** until a target partial pressure of O_2 in the perfusate (pO_2) is reached, as detected based on the sensor probes **128** in the sensor block **124**. In some implementations, the target partial pressure of O_2 in the perfusate is 450 mmHg. For example, at the beginning of the perfusion process, the partial pressure of O_2 in the perfusate is 200 mmHg and, based on data generated by the sensor probes **128**, the gas mixer **120** and oxygenator **116** are controlled to gradually increase the concentration of the O_2 in the perfusate until the partial pressure of O_2 in the perfusate is 450 mmHg. In some implementations, the O_2 concentration for the perfusate is increased at a rate of 1% increase in pO_2 every thirty minutes for the first 12 hours of perfusions and increases to a rate of 2% increase in pO_2 every thirty minutes until the end of the perfusion process. By gradually increasing the temperature and the O_2 concentration of the perfusate provided to the brain **110** at the beginning of the perfusion procedure, the internal resistance of the brain **110** is decreased, which results in improved perfusion of the brain **110** with reduced risk of injury to the brain **110**.

[0281] While the temperature and O_2 concentration of the perfusate are being ramped up by the heat exchanger **118** and the gas mixer **120**, the pulse generation system **1302** is simultaneously controlled to progressively increase the pressure of the perfusate provided to the brain **110** along the organ line **144** until a target flow rate of perfusate along the organ line **144** is reached. For example, as previously discussed, due to the relative positioning of the pulse generator **136**, the brain housing **102**, and the sensor block **124**, and the structure of the pulse generator **136**, none of the perfusate provided to the pulse generator **136** passively flows through the organ line **144** to the brain **110** when the diaphragm **214** of the pulse generator **136** is in an unflexed ("neutral") position. Therefore, in order to provide pulsatile flow of perfusate to the brain **110** along the organ line **144**, the air supply system **138** is controlled to provide pulsatile, pressurized air streams to the pulse generator **136** in order to cause the diaphragm **214** of the pulse generator **136** to flex downwards towards the fluid chamber **236**, which causes the diaphragm **214** to cover the openings **210** in the fluid chamber **236** of the pulse generator **136** and increase the pressure within the interior **212** of the pulse generator **136**. As result of the increased pressure caused by the flexing of the diaphragm **214**, at least a portion of the fresh perfusate contained within the interior **212** of the pulse generator is forced through the organ line outlet **208** of the pulse generator **136** and along the organ line **144** to the brain **110**.

[0282] At the beginning of the perfusion procedure, the pulse generation system **1302** is controlled to provide pulsatile flow of perfusate to the brain **110** along the organ line **144** at an initial pressure, as measured by the second pressure sensor **141** along the organ line **144**. For example, as previously discussed, the second electronic pressure regulator **166** is controlled based on pressure signals generated by second pressure sensor **141** along the organ line to adjust the frequency and duration which the second electronic pressure regulator **166** provides pressurized air to the pulse generator **136** in order to adjust amplitude and/or base of the pulsatile flow of perfusate to the brain **110** until the initial pressure threshold along the organ line **144** is reached, as detected by the second pressure sensor **141**. In some implementation, the initial pressure threshold along the organ line is 16 mmHg.

[0283] Once the initial pressure threshold along the organ line **144** has been maintained for a predetermined period, as determined based on the signals generated by the pressure sensor **141** along the organ line **144**, the second electronic pressure regulator **166** is controlled to cause the pulse generator **136** to progressively increase the pulsatile flow along the organ line **144** until a particular target flow rate along the organ line **144** is detected by the flow sensor **142** along the organ line **144**. In some implementations, the pulse generation system **1302** is controlled based on signals received from the pressure sensor **141** to increase the pressure along the organ line **144** at a rate of 0.0333 mmHg/minute. In some implementations, the pulse generation system **1302** is configured to increase the pressure in the organ line **144** from 16 mmHg to 20 mmHg, increasing the pressure along the organ line at a rate of 1 mmHg every 30 minutes. In some implementations, the pulse generation system **1302** in controlled to maintain a constant amplitude of pressure along the organ line **144**. In some implementations, the pulse generation system **1302** in controlled to maintain a constant amplitude of **10** mmHg along the organ line **144**.

[0284] Once the target flow rate is detected by the flow sensor **142**, the pulse generation system **1302** is controlled based on signals generated by the flow sensor **142** in order to provide pulsatile flow along the organ line **144** sufficient to maintain the predetermined flow rate along the organ line **144** as detected by the flow sensor **142**, while maintaining the pressure along the organ line **144** (as detected by pressure sensor **141**) within a predetermined range of pressure. In some implementations, the target flow rate is 10 mL/minute. By gradually increasing the pressure and the flow rate of the perfusate provided to the brain **110**, the internal resistance of the brain **110** is correspondingly gradually reduced, which enables the vasculature of the brain **110** to more effectively receive the perfusate while minimizing injury or other damage to the brain **110**.

[0285] Perfusate flowing along the organ line **144** is provided to the brain **110** and perfuses the entire vasculature of the brain **110**, including the penetrating arterioles, pre-capillary arterioles, and capillaries of the brain **110**. After flowing through the vasculature of the brain **110**, the used perfusate is collected in the basin **1022** of the brain housing **102** and exits the brain housing **102** along the housing outlet line **176**.

[0286] As previously discussed, at the beginning of the perfusion of the brain **110**, an initial "flush" of the brain **110** in order to removed toxins and other unwanted chemical compounds that have built up in the brain **110** during the post-mortem period prior to perfusion During the initial flush of the brain **110**, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** is closed and the solenoid valve **147** along the drain line **143** is open to direct used perfusate flowing out of the brain **110** to the waste container **145**. Once the waste scale **157** detects that a predetermined amount of used perfusate has been provided to the waste container **145** during the initial flush, the solenoid valve **177** along the Raman sample line **123** upstream of the waste container **145** and the solenoid valve **178** along the housing outlet line **176** between the brain housing **102** and the perfusate reservoir **104** are each controlled to open and the solenoid valve **147** along the drain line **143** is controlled to close, which directs used perfusate flowing along the housing outlet line **176** and the Raman sample line **123** back to the perfusate reservoir **104**. In some implementations, the brain **110** is initially flushed with 500 mL of perfusate.

[0287] Once the initial flush of the brain **110** is complete and the solenoid valve **147** along the drain line **143** is closed, the perfusion system **100** is a closed-loop system and the brain **110** is perfused with perfusate from the perfusate reservoir **104** throughout the duration of the perfusion procedure. As the brain **110** is being perfused, the dialysis filtration system **109** can be operated to remove metabolic waste products or other unwanted substances from the perfusate, as previously described herein. In some implementations, the brain **110** is perfused using the perfusion system **100** for 24 hours or longer (e.g., 24-48, 48-72 hours, 48-96 hours, 96 hours or longer).

[0288] During the perfusion process, the automatic sampler system **105** is controlled to collect samples of the fresh perfusate **302** in the arterial circuit **106** and the used perfusate in the venous circuit **108** at predetermined intervals. For example, the automatic sampler system **105** can be controlled to collect samples of fresh perfusate **302** from the perfusate reservoir **104** and used perfusate from the brain housing **102** at the start of a perfusion process ("0 hour"), 6 hours after the start of a perfusion process, 12 hours after the start of a perfusion process, and 24 hours after the start of a perfusion process. In order to collect a sample of fresh perfusate from the arterial circuit **106**, the solenoid valve **151** along the arterial sample line **150** between the sensor block **124** and the perfusate reservoir **104** is closed, the solenoid valve **171** along the arterial sample line **150** between the sensor block **124** and the automatic sampler system **105** is opened, and the pump **184** of the automatic sampler system **105** is operated to draw perfusate out of the sensor block **124** along the arterial sample line **150**, and the solenoid manifold **185** is controlled to direct the perfusate through the sample collection line **186** and into a sample collection vessel. In order to collect samples of the used perfusate **304** from the venous circuit **108**, the solenoid valve **173** along the venous sample line **183** is opened, and the pump **184** of the automatic sampler system **105** is operated to draw perfusate out of the brain housing **102** along the venous sample line **183**, and the solenoid manifold **185** is controlled to direct the perfusate through the sample collection line **186** and into a sample collection vessel.

[0289] During the perfusion process, the syringe pump manifold **167** is controlled to introduce one or more pharmaceutical compounds into the perfusate reservoir **104**, which are then provided to the brain **110** through the perfusate. The syringe pump manifold **167** is operated in conjunction with the dialysis filtration system **109** to introduce the pharmaceutical compound to the brain **110** in accordance with a particular predetermined release profile. For example, as previously described, the Raman spectroscopy system **107** can be used to determine the concentration of the pharmaceutical compound in the perfusate contained in the perfusate reservoir **104** in real-time throughout the perfusion procedure. Based on the concentration of the pharmaceutical compound in the fresh perfusate detected by the Raman spectroscopy system **107**, the syringe pump manifold **167** can be controlled to increase, decrease, or maintain the rate the pharmaceutical compound is provided to the perfusate reservoir **104** in order to align the concentration of the pharmaceutical compound with the desired release profile. In addition, based on the concentration of the pharmaceutical compound detected in the fresh perfusate by the Raman spectroscopy system **107**, the dialysis pump **190** can be controlled to pump perfusate from the perfusate reservoir **104** along the dialyzer inlet line **192** into the dialyzer **191**, which removes at least a portion of the pharmaceutical compound from the perfusate. The perfusate with a reduced concentration of the pharmaceutical compound exits the dialyzer **191** is pumped back to the perfusate reservoir **104** along the dialyzer outlet line **193**.

[0290] In addition, the Raman spectroscopy system **107** is controlled to perform Raman spectroscopy on the fresh perfusate in the perfusate reservoir **104** and on the used perfusate flowing out of the brain housing **102** at predetermined intervals throughout the perfusion process in order to determine one or more characteristics of the fresh perfusate and used perfusate in real-time throughout the perfusion process. As discussed above, the data generated by the Raman spectroscopy system **107** can be used to control the syringe pump manifold **167** and the dialysis filtration system **109** to produce a particular release profile of a pharmaceutical compound. In addition, the data generated by the Raman spectroscopy system **107** can be used to determine metabolic activity of the brain **110** and the health of the brain **110** in real-time during the perfusion process. For example, as previously discussed, the chemical composition of the fresh perfusate in the perfusate reservoir **104** and the chemical composition of the used perfusate flowing out of the brain **110** can each be determined using the spectroscopy system **107**, and various metabolic processes performed by the brain **110** during perfusion can be determined based on any differences in chemical composition in the fresh perfusate and used perfusate. In addition, while a pharmaceutical compound is being provided to the brain **110** through the perfusate, the Raman spectroscopy system **107** is controlled to perform Raman spectroscopy on samples of the used perfusate **304** flowing out of the brain **110** in order to determine whether the pharmaceutical compound has crossed the blood-brain barrier of the brain **110**, the reaction of the brain **110** to the pharmaceutical compound, the likely efficacy of the pharmaceutical compound on treating the targeted disease, and potential side effects that may result from the pharmaceutical compound.

[0291] The Raman spectroscopy probe **172** is controlled to perform Raman spectroscopy on the tissue of the brain **110** in real-time during the perfusion procedure. As previously discussed, the Raman spectroscopy data generated by the Raman spectroscopy probe **172** can be used to determine the effectiveness of the perfusion process, as well as predict the length the of time the brain **110** can be effectively perfused, for example, before the brain **110** undergoes significant swelling. In addition, the Raman spectroscopy probe **172** is controlled to perform Raman spectroscopy on the tissue of the brain **110** while a pharmaceutical compound is being provided to the brain **110** through the perfusate in order to determine whether the pharmaceutical compound has crossed the blood-brain barrier and to determine the effect of a pharmaceutical compound on the brain **110**.

[0292] The Raman spectroscopy data generated by the Raman spectroscopy system **107** and the Raman spectroscopy probe **172** is stored in a database. In some implementations, the Raman spectroscopy data generated by the Raman spectroscopy system **107** and the Raman spectroscopy probe **172** is provided to the trained machine learning model to predict, e.g., a metabolic pathway of the pharmaceutical compound in the brain, an activation or deactivation of certain gene pathways in the brain, physiological changes to the brain (e.g., swelling, tissue damage, infection, etc.), relating to the administration of the pharmaceutical compound.

[0293] Once the perfusion process is complete, for example, because the predetermined length of perfusion has elapsed, the pumps **111**, **114**, **190**, **1002**, **1906** are each controlled to stop pumping the perfusate through the system **100**, the brain **110** is removed from the brain housing **102**, and the perfusate contained within the perfusate reservoir **104** and the brain housing **102** is disposed of in a waste container or directly to a drain. The fluid lines **112**, **123**, **131**, **144**, **143**, **146**, **148**, **150**, **156**, **176**, **183**, **192**, **193**, **197**, **179**, **1908** of the arterial circuit **106**, the venous circuit **108**, the in-line Raman spectroscopy system **107**, and the dialysis filtration system **109** are disconnected from respective components of the system **100** and are replaced with new fluid lines prior to the start of another perfusion procedure.

[0294] FIG. **9** is a block diagram of an example computer system **900**. For example, referring to FIG. **1**, the computer system **188** of the perfusate system **100** could be an example of the system **900** described here. The system **900** includes a processor **910**, a memory **920**, a storage device **930**, and an input/output interface **940**. Each of the components **910**, **920**, **930**, and **940** can be interconnected, for example, using a system bus **950**. The processor **910** is capable of processing instructions for execution within the system **900**. The processor **910** can be a single-threaded processor, a multi-threaded processor, or a quantum computer. The processor **910** is capable of processing instructions stored in the memory **920** or on the storage device **930**. The processor **910** may execute operations such as receiving signals from one or more sensors (e.g., the pressure sensors **140**, **141**, **164**, **168**, **195**, **1008**, the flow sensors **130**, **142**, and the temperature sensors **113**, **121** shown in FIG. **1**) and controlling or more elements of the system (e.g.,. the solenoid valves **129**, **133**, **139**, **147**, **149**, **151**, **158**, **159**, **171**, **173**, **177**, **178** of FIG. **1**) based on the received signals.

[0295] The memory **920** stores information within the system **900**. In some implementations, the memory **920** is a computer-readable medium. The memory **920** can, for example, be a volatile memory unit or a non-volatile memory unit. In some implementations, the memory **920** stores a data structure. In some implementations, multiple data structures are used.

[0296] The storage device **930** is capable of providing mass storage for the system **900**. In some implementations, the storage device **930** is a non-transitory computer-readable medium. The storage device **930** can include, for example, a hard disk device, an optical disk device, a solid-date drive, a flash drive, magnetic tape, or some other large capacity storage device. The storage device **930** may alternatively be a cloud storage device, e.g., a logical storage device including multiple physical storage devices distributed on a network and accessed using a network.

[0297] The input/output interface **940** provides input/output operations for the system **900**. In some implementations, the input/output interface **940** includes one or more of network interface devices (e.g., an Ethernet card), a serial communication device (e.g., an RS-232 10 port), and/or a wireless interface device (e.g., an 802.11 card, a 3G wireless modem, or a 4G wireless modem). In some implementations, the input/output device includes driver devices configured to receive input data and send output data to other input/output devices, e.g., keyboard, printer and display devices (e.g., display device **189** of FIG. **1**). In some implementations, mobile computing devices, mobile communication devices, and other devices are used.

[0298] In some implementations, the system **900** is a microcontroller. A microcontroller is a device that contains multiple elements of a computer system in a single electronics package. For example, the single electronics package could contain the processor **910**, the memory **920**, the storage device **930**, and input/output interfaces **940**.

[0299] Although an example processing system has been described in FIG. **9**, implementations of the subject matter and the functional operations described above can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this specification can be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a tangible program carrier, for example a computer-readable medium, for execution by, or to control the operation of, a processing system. The computer readable medium can be a machine readable storage device, a machine readable storage substrate, a memory device, a composition of matter effecting a machine readable propagated signal, or a combination of one or more of them.

[0300] The term "computer system" may encompass all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. A processing system can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them.

[0301] A computer program (also known as a program, software, software application, script, executable logic, or code) can be written in any form of programming language, including compiled or interpreted languages, or declarative or procedural languages, and it can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

[0302] Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile or volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks or magnetic tapes; magneto optical disks and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), e.g., the Internet.

[0303] A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure.

[0304] For example, while the perfusate system **100** has been described as including both a filter bag **101** inside the perfusate reservoir **104** and an in-line mechanical filtration system **1001** has been described as including a filter bag **101**, in some implementations, the perfusate system **100** includes only the filter bag **101** inside the perfusate the reservoir **104** to filter the perfusate entering the reservoir **104** without the in-line mechanical filtration system **1001**. In other implementations, the perfusate system **100** includes only the in-line mechanical filtration system **1001** to filter the perfusate entering the reservoir **104** without the filter bag **101** inside the perfusate the reservoir **104**.

[0305] In addition, while the pumps **111**, **114**, **190**, **1002**, **1906** have been described as being peristaltic pumps, in some implementations, other types of pumps can be used. In some implementations, one or more of the pumps **111**, **114**, **190**, **1002**, **1906** are roller pumps or centrifugal pumps.

[0306] While the brain interface **170** has been described as including two fluid outlets with each of the carotid arteries of the brain **110** being sutured to a respective fluid outlet of the brain interface **170**, in some implementations, the brain interface **170** includes other numbers of fluid outlets (e.g., 1, 3, 4, etc.). For example, the brain interface **170** can include a single fluid outlet and a single carotid artery of the brain **110** can be coupled to the fluid outlet of the brain interface **170**.

[0307] In addition, while the brain interface **170** has been described as including two fluid inlets, in some implementations, the brain interface **170** includes a single fluid inlet that can be coupled to the organ line **144** and that is fluidly coupled to the two fluid outlets of the brain interface **170**.

[0308] Further, in some implementations, the brain **110** is fluidly coupled to the organ line **144** without the use of a brain interface **170**. For example, in some implementations, samples of pig vasculature are sutured to the carotid arteries and basilar artery of the human brain **110** at a first end and are fluidically connected to the organ line (e.g., through suturing) at a second end. Perfusate flowing through the organ line **144** flows through the pig vasculature sutured to the brain **110**, through the carotid and basilar arteries of the brain **110**, and perfuses the brain **110**.

[0309] While the level sensors **125**, **180** have been described as being infrared level sensor, other types of level sensors can be used to detect the level of perfusate in the sensor block **124** and the brain housing **102**, respectively. For example, the level sensors **125**, **180** are ultrasonic sensors, capacitance sensors, or other types of optical sensors.

[0310] While the sensor block inlet line **146** has been described as a y-splitter line with a single line flowing out of the pulse generator **136** that splits into two fluid lines flowing into the sensor block **124**, in other implementations, the sensor block inlet line can be a single, unsplit line.

[0311] While the dialysis filtration system **109** has been described as filtering the perfusate without the use of an exchange solution, in some implementations, the dialysis filtration system **109** includes a source of exchange solution and the perfusate is filtered using the exchange solution. For example, in some implementations, the dialyzer **191** is a capillary dialyzer, and perfusate and exchange solution are pumped through the dialyzer **191** at the same time. The exchange solution flowing through the dialyzer **191** interacts with the perfusate flowing through the dialyzer **191** to remove waste products from the perfusate and add nutrients to the perfusate. For example, while the perfusate flows within porous microtubes of the dialyzer **191**, the exchange solution can flow outside the porous microtubes such that waste products pass from the perfusate in the interior of the porous microtubes to the exchange solution surrounding the porous microtubes and such that nutrients pass from the exchange solution surrounding the porous microtubes into the persuade inside the microtubes. The filtered perfusate exits the dialyzer **191** and flows back to the perfusate reservoir **104** along the dialyzer outlet line **193**, while the exchange solution exits the dialyzer **191** and flows to the waste fluid receptacle **198** via the spent exchange solution line **197**. In some embodiments, the exchange solution is replaced or replenished over time to prevent the depletion of nutrients or the accumulation of toxins.

[0312] While the perfusion system **100** has been described as including a syringe pump manifold **167** with two or more syringes filled with a pharmaceutical compound, in some implementations, the perfusion system **100** includes a single syringe pump with a single syringe filled with a pharmaceutical compound.

[0313] While the lid **218**, the central housing **202**, and the base **220** of the pulse generator **136** have been described as being attached to one another using a set of bolts **224** and nuts **228**, other types of mechanical fasteners can be used to couple the lid **218**, central housing **202**, and base **220**. For example, in some implementations, the lid **218**, the central housing **202**, and the base **220** are attached to one another using one or more of bolts, rivets, pins, or adhesive.

[0314] While the drain line **143** of the perfusate system **100** has been described as being coupled to a waste container **145** to collect the perfusate drained from the system **100** along the drain line **143**, in some implementations, the drain line **143** is directly coupled a drain and perfusate flowing along the drain line **143** is directed into the drain coupled to the drain line **143**.

[0315] While the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182** of the system have been described as being manually operated by a user of the system **100** to collect perfusate samples, in some implementations, one or more of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**, **1912** can be automatically controlled to collect samples of the perfusate. For example, one or more of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**, **1912** can be automatically controlled to collect a sample of perfusate from the system **100** at one or more particular points during the perfusion process. For example, one or more of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182** can be fluidly coupled to an autosampler or a sensor (e.g., pressure sensor) in order to automatically collect fluid sample or other data using the respective stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182**. In addition, the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182** have previously been described as automatically being controlled using servo-motor to control and change flow paths throughout the system **100**. However, in some implementations, an operator can manually operate one or more of the stopcocks **127**, **132**, **135**, **153**, **154**, **155**, **181**, **182** to alter the flow paths of the system.

[0316] While the computer system **188** has been depicted as including a single computing device, in some implementations, the computing system **188** includes multiple computing devices or processors (2, 3, 4, etc.) in communication with the various electronic components of the perfusion system **100** and configured to control operation of one or more electronic components of the perfusion system **100**. Each of the computing devices of the computer system **188** may be in wired or wireless communication with one or more electronic components of the perfusion system **100**. In addition, one or more of the computing devices of the computer system **188** can be a remote computing device that is located at a different, remote location from the perfusion system **188**.

[0317] In addition, while the process of performing an initial flush of the brain **110** using the perfusate system **100** has been described as being stopped automatically in response to the waste scale **157** detecting that a threshold weight of perfusate has been received by the waste container **145**, in some implementations, the initial flush procedure is manually stopped by a user. For example, in some implementations, the waste container **145** includes one or more markings indicating the volume of fluid inside the waste container **145**, and as perfusate flows along the drain line **143** into the waste container **145**, an operator of the system **100** monitors the volume of perfusate collected by the waste container **145**. In response to the operator observing that the waste container **145** has received a threshold volume of perfusate, the operator closes the solenoid valve **147** along the drain line **143** and opens the solenoid valve **177** along the along the Raman sample line **123**. In some implementations, the operator adjusts the solenoid valves **147**, **177** to stop the initial flush of the brain **110** in response to observing that 500 mL of perfusate has been collected in the waste container **145**. In some implementations, the operator of the system **100** adjusts the solenoid valves **147**, **177** to stop the initial flush of the brain **110** based on a predetermined amount of time elapsing since the start of the initial flush procedure. In some implementations, for example, an operator of the system **100** manually stops the initial flush procedure 30 minutes to 1 hour after starting the flush procedure.

[0318] While the syringe pump manifold **167** and the dialysis filtration system **109** have been described as being controlled in order to provide concentrations of a drug according to a predetermined release profile, in some implementations, the uptake and metabolism of a particular drug is tracked in real time using the Raman spectroscopy system **107** and/or the Raman spectroscopy probe **172**, as described above, and the syringe pump manifold **167** and dialysis filtration system **109** are dynamically controlled to increase or decrease the concentration of the drug in the perfusate based on the brain's uptake and metabolism of the drug, as detected using the Raman spectroscopy system **107** and/or the Raman spectroscopy probe **172**. By dynamically adjusting the concentration of the drug in the perfusate using the syringe pump manifold **167** and the dialysis filtration system **109** in response to real-time Raman spectroscopy data indicating the uptake and metabolism of the drug by the brain, an optimized release profile for the drug that provides increased uptake and optimized metabolism of the drug by the brain can be determined.

[0319] While the perfusion system **100** has been described as having both an in-line Raman spectroscopy system **107** for performing Raman spectroscopy on perfusate and a Raman spectroscopy probe **172** for performing Raman spectroscopy on the tissue of the brain **110** in the brain housing **102**, in some implementations, the perfusion system **100** does not include the Raman spectroscopy probe **172**. In such implementations, the uptake and metabolism of drugs and other components of the perfusate by the brain **110** can be determined based on performing Raman spectroscopy of the used perfusate exiting the brain using the Raman spectroscopy system **107** alone.

[0320] While the spectroscopy system **107** and the spectroscopy probe **172** of the perfusion system **100** have been described as being configured to perform Raman spectroscopy, it should be understood that such spectroscopy can include surface-enhance Raman spectroscopy (SERS). In addition, while the spectroscopy system **107** and the spectroscopy probe **172** of the perfusion system **100** have been described as being configured to perform Raman spectroscopy, in some implementations, the spectroscopy system **107** and spectroscopy probe **172** can be configured to other types of spectroscopy. For example, one or both of the spectroscopy system **107** and the spectroscopy probe **172** can be configured to perform fluorescence spectroscopy, UV spectroscopy, or infrared spectroscopy.

[0321] Further, as an alternative to or in additional to using spectroscopy of the type described above, the perfusion systems described herein can use transcriptomics (including but not limited to qPCR, RNAseq, scRNAseq, snRNAseq, RNAscope, etc), proteomics, mass spectrometry, ELISA, histology, genotyping, etc. to analyze the perfusion fluid and/or brain tissue samples.

[0322] In addition, referring to FIG. **3**, while the cellular and metabolic activity of the brain have been described as being determined based on performing Raman spectroscopy using the Raman spectroscopy system and/or the Raman spectroscopy probe **172**, in some implementations, tissue samples **406** of the brain **110** and/or samples of the perfusate exiting the brain **110** are alternatively or additionally collected during and/or after perfusion of the brain **110** by the system **100**, and one or more tests or assays are performed on the collected brain tissue samples **406** and/or collected perfusate samples to determine cellular and metabolic activity of the brain **110** during the perfusion procedures. Various testing procedures can be performed on the collected brain tissue samples **406** and the collected perfusate samples including, but limited to, spectroscopy, gene sequencing, cytokine analysis, anti-body analysis, genomic analysis, transcriptomic analysis, proteomic analysis, metabolomic analysis.

[0323] While the perfusion system **100** has been described as being used to determine the uptake and efficacy of a pharmaceutical compound, the perfusion system **100** can similarly be used to determine the uptake and efficacy of any of various other therapeutic agents, including, but not limited to, antibodies, biologics, viral vectors, lipid nanoparticles, peptides, etc.

[0324] In some examples, a particular gene therapy for treating a particular disease can be tested using the perfusion system **100**. For example, a single vector or a collection of vectors carrying respective DNA segments can be added to the perfusate in the perfusate reservoir using similar procedures described herein for introducing a pharmaceutical compound to the perfusate, and the brain **110** can be perfused with the perfusate containing the vector(s). The used perfusate exiting the brain and the brain tissue can be tested using techniques described herein to determine whether the vector(s) have crossed the blood-brain barrier of the brain **110**, to determine the vector transcytosis rates through the blood-brain barrier, and to determine an optimized release profile for the vector(s) to be most effectively taken up by the brain **110**.

[0325] FIG. **16** is an RNAScope image showing that GFP and mCherry were successfully delivered to the frontal cortex of the brain using viral vectors, namely adeno-associated virus (AAV). Box **16**A in FIG. **16** is an enlarged view of region **1602** showing evidence of GFP, while box **16**B is an enlarged view of region **1604** showing evidence of mCherry. In this example, rather than introducing a chemical compound into the perfusate of the perfusion system **100**, the viral vectors carrying the GFP and mCherry are introduced into the perfusate. The presence of the GFP and mCherry in the frontal cortex indicates that the viral vectors used were able to cross the blood-brain barrier and reach the frontal cortex of the brain.

[0326] While methods of using the perfusion system **100** to test whether therapeutic agents can cross the blood-brain barrier, in some cases, the perfusion system **100** is alternatively or additionally used to determine whether the blood-brain barrier of a brain is intact. In such cases, for example, one or more agents known to be incapable of crossing the blood-brain barrier (e.g., certain large and/or water soluble molecules) can be introduced into the perfusate and then any of the various testing techniques described herein can be used to determine whether the agent or agents has/have crossed into the brain. The presence of any of those agents in the brain would be evidence that the blood-brain barrier of the brain is no longer intact. This may help researchers to determine whether test data related other therapeutic agents tested on that brain (particularly test data showing whether those therapeutic agents were able to cross the blood-brain barrier) are reliable.

[0327] In addition, while the perfusion system **100** has been described as being used to perfuse and test brains that have not otherwise been mechanically or chemically altered, the perfusion system **100** can be used to perfuse and tests brains that have been mechanically or chemically altered (e.g., to simulate vascular and inflammatory injuries) prior to perfusing the brain with the system **100**. For example, referring to FIGS. **7** and **8**, an experiment was conducted in which three brains **702**, **704**, **706** were each perfused using the perfusion system **100** and the rate of lactate released by each of the brains **702**, **704**, **706** was monitored throughout the perfusion process. The first brain **702** ("baseline brain **702**") was not mechanically or chemically altered prior to being perfused by the perfusion system **100**. The second brain **704** was mechanically modified to mimic vascular disruptions prior to perfusion of the second brain **704** using the perfusion system **100**, effectively replicating the impact of vascular injury on brain tissue. The third brain **706** was treated with lipopolysaccharide (LPS) during perfusion using the perfusion system **100** to mimic pathophysiological conditions associated with vascular impairments and inflammation in the brain. The rate of lactate release by each of the three brains **702**, **704**, **706** was tested throughout the **24** hour perfusion process (e.g., by performing Raman spectroscopy on the used perfusate exiting the brains **702**, **704**, **706** using the Raman spectroscopy system **107**). As can be seen in FIG. **8**, the rate of lactate release **802** for the untreated brain **702** was higher than both the rate of lactate release **804** for the mechanically altered brain **704** and the rate of lactate release **806** for the brain **706** perfused with lipopolysaccharide (LPS). By utilizing the perfusion system to perfuse and test brains that are mechanically or chemically altered to simulate vascular and inflammatory injuries, the perfusion system **100** can be used to effectively study mechanisms and potential treatments for brain injuries.

[0328] While the perfusate system **100** has been described as being used to perfuse a human brain **110**, the perfusion system **100** can be used to perfuse other mammalian brains, such as pig brains. In some implementations, the perfusate used in the perfusate system **100** may be altered in order to more effectively perfuse non-human mammalian brains. In addition, an alternate brain housing **102** may be used to better accommodate and support a pig brain. For example, in some implementations, when the perfusion system **100** is being used to perfuse a pig brain, the brain housing **102** is replaced with a pig brain housing that is smaller than the brain housing **102** used to house the human brain **110**.

[0329] FIGS. **11**A-**11**D depict an example brain housing **1102** that can be used to support a pig brain during a perfusion process performed using the system **100** of FIG. **1**. As can be seen in FIGS. **11**A-**11**D, the brain housing **1102** includes a basin **1104** and lid **1106**. The lid **1106** is configured to releasably couple to the basin **1104** and can be separated from the basin **1104** to allow a pig brain to be positioned inside the interior of the brain housing **1102**.

[0330] When the brain housing **1102** is fluidly connected to the system **100** of FIG. **1**, the housing outlet line **176** extends from the brain housing **1102** and fluidly couples the brain housing **1102** to the perfusate reservoir **104**. Referring to FIGS. **11**A, **11**B, and **11**D, the brain housing **1102** includes a coupling **1142** connected to the side of the basin **1104**. The coupling **1142** fluidly connects the interior of the basin **1104** of the brain chamber **1102** to the housing outlet line **176**. In some implementations, the coupling **1142** is a quick-turn tube coupling. In some implementations, the brain housing **1102** includes a second coupling **1144** that can be coupled to the outlet line **176** or to another outlet line in order to increase the rate of drainage of fluid out of the brain housing **1102** to the perfusate reservoir **104**.

[0331] The brain housing **1102** also includes a fluid coupling **1146** that fluidly connects the shunt line **156** to the interior of the brain chamber **1102**. In some implementations, the coupling **1146** is a quick-turn tube coupling.

[0332] The brain housing **1102** also includes a three arm knob **1108** releasably coupled to the lid **1106** of the brain housing **1102**. The knob **1108** serves as an ergonomic handle for opening the lid **1106** of the brain housing **1102**.

[0333] The brain housing **1102** includes a level sensor **180** to detect the level of fluid (e.g., used perfusate) inside the basin **1104** of the brain housing **1102**, and the solenoid valve **178** along the housing outlet line **176** is controlled to open or close based on signals generated by the level sensor **180**. As depicted in FIG. **11**D, the level sensor **180** is coupled to the basin **1104** of the brain chamber **1102**. In some implementation, the level sensor **180** is coupled to the basin **1104** at the same or similar height along the basin **1104** as the fluid coupling **1146** that fluidly connects the shunt line **156** to the interior of the brain chamber **1102**. In some implementations, the level sensor **180** is an optical level sensor, such as an infrared level sensor. For example, the solenoid valve **178** is controlled to remain in a closed state and prevent flow of used perfusate from the brain housing **1102** into the perfusate reservoir **104** until the level sensor **180** generates a signal indicating that the fluid level inside the brain housing **1102** has exceeded a threshold fluid level. In response to the level sensor **180** generating a signal indicating that the fluid level inside the brain housing **1102** has exceeded a threshold fluid level, the solenoid valve **178** is opened to allow used perfusate to flow from the brain housing **1102** along the housing outlet line **176** to the perfusate reservoir **104**. Once the level sensor **180** detects that the fluid level inside the brain housing **1102** has fallen below the threshold level, the solenoid valve **178** is closed, which prevents fluid from flowing from the brain housing **1102** to the perfusate reservoir **104**.

[0334] The brain housing **1102** is configured to maintain the pig brain under normothermic conditions. For example, when the basin **1104** and lid **1106** of the brain housing **1102** are securely coupled together, a warm, humid environment is maintained within the brain housing **1102**. In some implementations, the brain housing **1102** includes temperature and humidity controls configured maintain a particular temperature and humidity level to prevent drying or other injury to the brain.

[0335] Prior to placement of the pig brain in the brain housing **1102** for perfusion, the pig brain is fluidly coupled to a brain interface **1170** that is configured to fluidly couple the vasculature of the pig brain to the organ line **144** of the system **100**. FIGS. **12**A-**12**C depict an example brain interface **1170** for fluidly coupling a pig brain to the organ line **144**. The brain interface **1170** includes a base plate **1172**, two fluid input ports **1174**, **1176**, and two fluid output ports **1178**, **1180**.

[0336] In some implementations, the base plate **1172** is formed of a titanium alloy. In some implementations, the base plate **1172** is formed through laser sintering a titanium alloy. In some implementations, the base plate **1172** is formed through 3D printing. Referring to FIG. **12**A, the base plate **1172** defines a plurality of drain holes **1182** therethrough that allow perfusate flowing out of the pig brain to flow through the base plate **1172** into the basin **1104** of the brain housing **1102** such that the perfusate flowing out of the pig brain is collected in the basin **1104**.

[0337] Each of the fluid inlet ports **1174**, **1176** of the brain interface **1170** are fluidly coupled to the organ line **144** using a y-connector and each inlet port **1174**, **1176** of the brain interface **1170** receives fresh perfusate flowing out of the organ line **144** into the brain interface **1170**. Referring to FIG. **12**C, the brain interface **1170** defines flow paths **1184** fluidly coupling each of the fluid inlets ports **1174**, **1176** of the brain interface **170** to a respective outlet port **1178**, **1180** of the brain interface **170**.

[0338] Prior to placing a pig brain in the brain housing **1102** and beginning perfusion of the pig brain, each of the carotid arteries of the pig brain is sutured to a respective fluid outlet port **1178**, **1180** of the brain interface **1170**. Once the carotid arteries of the pig brain are sutured to the fluid outlet ports **1178**, **1180** of the brain interface **1170** and the inlet ports **1174**, **1176** of the brain interface **1170** are fluidly coupled to the organ line **144**, the pig brain and the brain interface **1170** can be positioned within the basin **1104** of the brain housing **1102**. The lid **1106** of the brain housing **1102** is coupled to the basin **1102** to enclose the pig brain and brain interface **1170** within the brain housing **1102** throughout the perfusion process. Referring to FIG. **11**B, the basin **1104** of the brain housing **1102** defines a slot **1110** that allows a portion of the organ line **144** to enter the basin **1104** with minimal air gaps when the lid **1106** is coupled to the basin **1104**.

[0339] As fresh perfusate flows along the organ line **144**, the perfusate enters the fluid inlet ports **1174**, **1176** of the brain interface **1170** and flows through the flow paths **1184** defined by the brain interface **1170**, through the fluid outlet ports **1178**, **1180** of the brain interface **1170**, and into the carotid arteries of the pig brain coupled to the brain interface **1170**. As previously discussed, the pressure and flow rate of the perfusate provided to the pig brain are controlled to enable the perfusate entering the carotid arteries of the brain to perfuse the entire vasculature of the brain, including the penetrating arterioles, pre-capillary arterioles, and capillaries of the brain.

[0340] While the brain interface **1170** has been depicted as having two fluid outputs, in some implementations, in order to perfuse a pig brain using the perfusion system **100**, a brain interface having a single fluid output is provided, and a single artery of the pig brain is sutured to the single fluid output of the brain interface to fluidly couple the pig brain to the brain interface.

[0341] In addition, while the perfusion systems described herein have been described as being used to perfuse brains, in some implementations, the perfusion system can be used to perfuse other mammalian (e.g., human) organs. For example, the perfusion system can be used to perfuse kidneys, hearts, livers, lungs, pancreases, etc. In some implementations, an alternate interface is used to fluidly couple the non-brain mammalian organ to the perfusions system **100**. Further, in some implementations, an alternate housing device is used to contain and collect perfusate expelled from the non-brain mammalian organ perfused by the system **100**. In addition, the perfusate used in the perfusate system **100** may be altered in order to more effectively perfuse non-brain mammalian organs. For example, in some implementations, perfusate used to perfuse non-brain mammalian organs has a higher concentration of oncotically active particles compared to perfusate of brains in order to accommodate the higher capillary filtration rates that occur in non-brain tissue.

[0342] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any invention or on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular inventions. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially be claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0343] Similarly, while operations are depicted in the drawings and recited in the claims in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system modules and components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0344] Particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In some cases, multitasking and parallel processing may be advantageous.

What is claimed is:

1. An ex-vivo brain perfusion system for perfusion of a brain, the system comprising:

a reservoir configured to receive perfusate;

a housing configured to contain the brain;

an arterial circuit configured to fluidly couple the reservoir to the brain;

a venous circuit fluidly coupling the housing to the reservoir; and

a syringe pump fluidly coupled to the reservoir and configured to introduce a therapeutic agent to the perfusate in the reservoir,

wherein the ex-vivo brain perfusion system is configured to perfuse the brain with the therapeutic agent to test at least one property of the therapeutic agent.

2. The system of claim **1**, wherein the arterial circuit comprises:

a fluid line fluidly coupling the reservoir to the brain;

a pressure sensor configured to measure pressure along the fluid line;

a flow sensor configured to measure a flow rate along the fluid line; and

a pulse generation system configured to generate pulsatile flow of perfusate from the reservoir to the brain along the fluid line based on one or more signals generated by at least one of the pressure sensor or the flow sensor.

3. The system of claim **2**, wherein the pulse generation system comprises:

a pulse generator comprising:

a housing;

a housing inlet; and

a flexible diaphragm; and

an air supply system fluidly coupled to the housing inlet of the pulse generator,

wherein the air supply system is configured to provide pressurized air into the housing inlet of the pulse generator based on one or more signals generated by at least one of the pressure sensor or the flow sensor.

4. The system of claim **3**, wherein the pulse generation system is controlled based on an internal resistance of the brain detected by the pressure sensor.

5. The system of claim **1**, wherein the syringe pump is configured to introduce the therapeutic agent into the reservoir according to a predefined release profile for the therapeutic agent.

6. The system of claim **1**, further comprising a filtration system configured to filter the perfusate, wherein the filtration system comprises:

a dialyzer comprising an inlet port and an outlet port, the dialyzer configured to filter the perfusate;

an inlet line fluidly coupling the reservoir to the inlet port;

an outlet line fluidly coupling the outlet port to the reservoir; and

a pump configured to pump perfusate from the reservoir through the dialyzer.

7. The system of claim **1**, further comprising an in-line spectroscopy system configured to perform spectroscopy on the perfusate in real-time during a perfusion process.

8. The system of claim **7**, wherein the in-line spectroscopy system comprises:

a spectroscopy chamber configured to receive perfusate from the reservoir or the venous circuit;

a heat exchanger fluidly coupled to the spectroscopy chamber and configured to control a temperature of perfusate provided to the spectroscopy chamber; and

a spectrometer configured to perform spectroscopy on perfusate in the spectroscopy chamber.

9. The system of claim **7**, wherein:

the in-line spectroscopy system is configured to perform spectroscopy on one or more perfusate samples from the reservoir; and

the syringe pump is configured to introduce the therapeutic agent into the reservoir according to a predefined release profile for the therapeutic agent based on spectroscopy data generated by performing spectroscopy on one or more perfusate samples from the reservoir using the in-line spectroscopy system.

10. The system of claim **7**, wherein:

the system further comprises a filtration system configured to filter the therapeutic agent out of the perfusate; and

the filtration system is configured to remove the therapeutic agent from the perfusate according to a predefined release profile for the therapeutic agent based on spectroscopy data generated by performing spectroscopy on one or more perfusate samples from the reservoir using the in-line spectroscopy system.

11. The system of claim **7**, wherein the in-line spectroscopy system is configured to perform spectroscopy on one or more perfusate samples collected from the venous circuit to test the at least one property of the therapeutic agent.

12. The system of claim **11**, wherein the at least one property of the therapeutic agent comprises whether the therapeutic agent passes through a blood-brain barrier of the brain.

13. The system of claim **7**, wherein the in-line spectroscopy system is configured to perform Raman spectroscopy on the perfusate.

14. The system of claim **1**, further comprising a spectroscopy probe coupled to the housing, wherein the spectroscopy probe is configured to perform spectroscopy on tissue of the brain in the housing.

15. The system of claim **14**, wherein the at least one property of the therapeutic agent is determined based on spectroscopy data generated by performing spectroscopy on the brain using the spectroscopy probe.

16. The system of claim **14**, wherein the spectroscopy probe is configured to perform Raman spectroscopy on the tissue of the brain.

17. The system of claim **1**, wherein the at least one property of the therapeutic agent is a pharmacodynamic property of the therapeutic agent or a pharmacokinetic property of the therapeutic agent.

18. The system of claim **1**, wherein the at least one property of the therapeutic agent is an ability of the therapeutic agent to pass through a blood-brain barrier of the brain.

19. A method of testing at least one property of a therapeutic agent, the method comprising:

perfusing a brain with a perfusate using an ex-vivo perfusion system;

introducing the therapeutic agent into the perfusate;

while perfusing the brain, performing spectroscopy on tissue of the brain in real-time using the ex-vivo perfusion system to generate spectroscopy data; and

determining, based on spectroscopy data, the at least one property of the therapeutic agent.

20. A method of testing at least one property of a therapeutic agent, the method comprising:

perfusing a brain with a perfusate using an ex-vivo perfusion system;

introducing the therapeutic agent into the perfusate;

while perfusing the brain, performing spectroscopy on the perfusate using an in-line spectroscopy system of the ex-vivo perfusion system to generate spectroscopy data; and

determining, based on the spectroscopy data, the at least one property of the therapeutic agent.

* * * * *